Boyce-Codd Normal Form with Bounded Dependency Loss

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ABSTRACT

State-of-the-art schema design computes a lossless, dependency-preserving decomposition into Third Normal Form (3NF) that is in Boyce-Codd Normal Form (BCNF) whenever possible. SQL systems, however, enforce keys natively, and non-key functional dependencies (FDs) only at a high cost or not at all, so a 3NF schema outside BCNF leaves its non-key FDs unenforced, as a BCNF decomposition does with the FDs it does not preserve. We introduce BCNF designs, which enforce FDs through keys only and whose losslessness rests on these keys, and characterize when these keys also keep independently stored relations consistent. The dependency loss of a design, the number of minimal FDs that its keys do not imply, is singled out by three natural requirements and under violation weights bounds the probability that an update violates the FDs unnoticed. A relation schema is in k-loss BCNF if it has a BCNF design with loss at most k. Loss zero recovers the dependency-preserving BCNF decompositions, deciding membership in k-loss BCNF is NP-hard for every fixed k under arbitrary covers and coNP-hard for every fixed $k \geq 1$ given all minimal FDs, and the minimum loss cannot be approximated within a factor below two. We develop exact algorithms, one certifying lower bounds by integer programming, a local search that lowers the loss of any design, and an algorithm that repairs the critical schemata of 3NF synthesis. Experiments on discovered and meaningful FDs quantify how many FDs classical designs leave unenforced, how much our designs reduce this loss, and what they cost in insertion, foreign-key and query time and in integrity risk, on the universal relation and on the stored tables. Counting the FDs that keys leave unenforced turns the choice between 3NF and BCNF into a matter of degree.

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The source code, data, and/or other artifacts have been made available at https://github.com/zhuoxingzhang/bounded-loss-bcnf.

1 INTRODUCTION

The logical design of relational database schemata is a classical topic of research and education in the database field [6, 8, 39, 55]. BCNF guarantees that no relation exhibits redundant data values caused by FDs [62], but a lossless, dependency-preserving decomposition into BCNF does not always exist [5, 6]. A lossless, dependency-preserving decomposition into 3NF always exists [8, 12], but may tolerate unbounded data redundancy [37]. State-of-the-art normalization therefore computes a lossless, dependency-preserving decomposition into 3NF that is in BCNF whenever possible [27, 46, 68, 71].

Dependency preservation serves integrity maintenance, since an FD embedded in a schema can be validated on a single relation without joins [12, 37]. Validation, however, is left to the database system, which treats keys and non-key FDs differently. SQL systems enforce a key by a unique index, which validates an update with one index probe and stays correct under concurrent transactions. For non-key FDs, the assertions of the SQL standard have only just appeared in Oracle [32, 45] and remain unsupported in other major systems [42, 44, 57]. Among table constraints, PostgreSQL alone can state an FD, as an exclusion constraint at the price of an index per FD [58]. Otherwise, a non-key FD is maintained by triggers or application code. Both slow down updates [71], are unsafe under weak isolation levels [4, 20], and hide business rules in procedural code that is hard to maintain [13]. Consequently, integrity logic lives mostly outside the database, and even foreign keys are absent from many open-source schemata [4, 16, 54, 61, 66]. Alibaba's Java coding guidelines, for example, make unique indexes mandatory wherever the business logic calls for them, but forbid foreign keys and leave them to the application layer [2]. Microsoft's guidance on normalization likewise keeps $Zip \rightarrow City$ in a customer table and lets the application verify the related fields [41]. In practice, a 3NF schema outside BCNF thus enforces its keys only. Violations of its non-key FDs go undetected, and the values they determine are stored redundantly, while classical theory counts them as preserved.

This changes the classical trade-off between 3NF and BCNF. A BCNF decomposition leaves the FDs it does not preserve unenforced, and a 3NF schema outside BCNF leaves its non-key FDs unenforced. Previous work, however, counts the first as a loss of dependencies and the second as their preservation [29, 37, 71]. Both guarantee less than the given FDs promise. Hence, the question is no longer whether to give up FDs but how few a lossless decomposition into BCNF must give up. This is illustrated as follows.

Example 1.1 (Running example). Our running example is the schema $R = \{C, S, T, Z\}$ with the attributes C(ustomer), S(treet), T(own) and Z(ip), which records customers with their postal address. The set Σ of FDs consists of $C \rightarrow ST$, $ST \rightarrow Z$ and $Z \rightarrow T$. The only minimal key is C, so $ST \rightarrow Z$ and $Z \rightarrow T$ are non-key FDs, and (R, Σ) is not even in 3NF. No lossless, dependency-preserving decomposition of (R, Σ) into BCNF exists. A design is a set of relation schemata with the keys they declare, and it enforces exactly the FDs that these keys imply. A designer may consider four designs.

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Table 1: Running Example: Designs and Unenforced FDs

Design	Schemata (keys)	Unenforced FDs	Sound
$\mathcal{D}_1$	$CSTZ$ (C)	$ST \to Z, Z \to T$	yes
$\mathcal{D}_2$	ZT (Z), CSZ (C)	$ST \to Z$	yes
$\mathcal{D}_3$	ZT (Z), SZ (SZ), CST (C)	$ST \to Z, C \to Z$	no
$\mathcal{D}_4$	STZ (ST, SZ), CST (C)	$Z \to T$	yes

Design $\mathcal{D}_1$ keeps R with key C, the key-only design of the guidance above. Designs $\mathcal{D}_2$ and $\mathcal{D}_3$ are the outputs of the classical BCNF decomposition algorithm when it decomposes $Z \to T$ or $ST \to Z$ first, respectively. Design $\mathcal{D}_4$ leaves $Z \to T$ unenforced but declares both keys ST and SZ on STZ. Tab. 1 lists the four designs with the FDs they leave unenforced. □

Ex. 1.1 exposes three shortcomings of the state of the art, which raise four questions. Firstly, the key-only design $\mathcal{D}_1$ leaves two FDs unenforced and $\mathcal{D}_2$ and $\mathcal{D}_4$ leave one. Classical theory cannot see this difference, since it counts every FD of a single schema as preserved, whether a key enforces it or not. Hence, we ask (Q1) *Which measure of dependency loss counts the FDs that keys do not enforce, independently of how the FDs are represented?* Secondly, $\mathcal{D}_2$ and $\mathcal{D}_3$ are both lossless for Σ, but the join of $\mathcal{D}_3$ is lossless only because $ST \to Z$ holds, the very FD that $\mathcal{D}_3$ cannot enforce. Two customers with the same street and town but different zip codes satisfy the keys of $\mathcal{D}_3$, and the join of their projections returns four records instead of two. This leads to (Q2) *What must a design guarantee when the FDs it cannot enforce through keys are not enforced at all?* Thirdly, the classical algorithm returns $\mathcal{D}_2$ or $\mathcal{D}_3$ by an arbitrary choice and never $\mathcal{D}_4$, the output of 3NF synthesis with all minimal keys declared. 3NF synthesis, in turn, never decomposes its critical schemata, and on FD sets mined from real-world data it leaves many more FDs unenforced than necessary, as our experiments show. This raises (Q3) *How can designs with minimum dependency loss be computed, and how hard is it to compute them?* Finally, FDs go unenforced to avoid the costs and risks of enforcing them, so we ask (Q4) *How does minimizing the loss at the logical level translate into costs and integrity at the operational level?*

Fig. 1 previews the answer to (Q4) on the running example, together with the design $\mathcal{D}_5$ that adds to $\mathcal{D}_4$ an exclusion constraint for $Z \to T$. In PostgreSQL, we store 100,000 customers in 200 towns with ten zip codes each, insert 10,000 new customers, half of whom move to an existing address, and apply 10,000 random updates to the stored tables, each copying the value of an attribute from another customer. The key-only design $\mathcal{D}_1$ inserts fastest but misses three in four of the harmful updates, those that violate an FD, while $\mathcal{D}_5$ enforces every FD at 5× the insertion time of $\mathcal{D}_1$. The designs $\mathcal{D}_2$ and $\mathcal{D}_4$ lose one FD each, but only $\mathcal{D}_2$ leaves violations in the data, those of $ST \to Z$. The other harmful updates they let pass change a row shared by several customers, and with it all their records, an update anomaly that comes with decomposition rather than with the loss.

Our **contributions** answer the four questions.

(1) Fundamentally, we introduce BCNF designs, which declare keys on the schemata of a decomposition and are sound, that is, lossless for the FDs these keys imply, and locally enforced, that is, every FD they enforce on a schema follows from the keys of that schema (Q2). Soundness is necessary, since classical BCNF decompositions can produce lossy joins on states that satisfy their keys, and a polynomial-time test decides whether the keys of a design also keep independently stored relations globally consistent. The dependency loss of a design counts the minimal FDs that its keys do not imply, which three requirements on measures of unenforced rules single out, and under violation weights it bounds the probability that an update violates Σ unnoticed (Q1). The BCNF designs with loss zero are the classical lossless, dependency-preserving BCNF decompositions, the bottom of the hierarchy k-loss BCNF.

(2) Computationally, we lift the NP-hardness of the classical case [5] to every fixed k under arbitrary covers, which rules out algorithms that are fixed-parameter tractable in k unless P equals NP. When Σ is given by its atomic cover, deciding whether the loss is at most one is coNP-hard, already for two obstructions, so no algorithm approximates the minimum within a factor below two or is fixed-parameter tractable in the number of obstructions, unless P equals NP (Q3).

(3) Algorithmically, we characterize the FD sets that BCNF designs can enforce, bound the loss by the obstructions of this characterization, and show that optimal designs may need weakened FDs and undeclared keys. The algorithm Drop computes a design with minimum loss by iterative deepening, Hit casts MinLoss as an implicit hitting-set problem whose integer programs certify increasing lower bounds, Ascend lowers the loss of any BCNF design, and Repair repairs the critical schemata of 3NF synthesis for FD sets with thousands of minimal FDs (Q3).

(4) Operationally, we quantify on FD sets mined from real-world data and on meaningful FDs how many FDs 3NF synthesis, classical BCNF decomposition and the key-only design leave unenforced, how much smaller the loss of our designs is, what the designs cost in insertion, foreign-key and query time in PostgreSQL, and how much integrity risk they carry on the universal relation and on the stored tables, also under violation weights (Q4).

The Supplementary Material (Supp.) contains further results and all proofs.

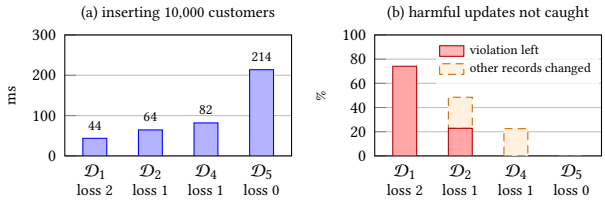


Figure 1: Insertion Time and Integrity Risk of the Designs of the Running Example

2 FUNDAMENTALS AND PREVIOUS WORK

We fix notions and notation from previous work, following [39, 68, 71]. Ex. 1.1 illustrates them.

FDs, keys, and normal forms. A *relation schema* is a finite set R of attributes A with domains $dom(A)$, a *relation* over R is a finite set r of tuples from $\prod_{A \in R} dom(A)$, and $t[S]$ and $r[S] = \{t[S] \mid t \in r\}$

denote the projections of a tuple t and of r onto $S \subseteq R$. An *FD* is an expression $X \to Y$ with $X, Y \subseteq R$, which r *satisfies* iff every pair of its tuples with matching values on X has matching values on Y, and which is *trivial* iff $Y \subseteq X$. An FD set Σ *implies* an FD φ if every relation that satisfies Σ satisfies φ, the *closure* Σ^+ of Σ is the set of all FDs implied by Σ, computable with Armstrong's axioms [3], FD sets with the same closure are *covers* of one another, and $X_\Sigma^+ = \{A \in R \mid X \to A \in \Sigma^+\}$ is the attribute closure of $X \subseteq R$. An FD $X \to A \in \Sigma^+$ with $A \notin X$ is *minimal* if $Y \to A \notin \Sigma^+$ for every proper subset $Y \subset X$, and the *atomic cover* Σ_a of Σ is the set of all minimal FDs in Σ^+ [27, 46], which is a cover of Σ determined by Σ^+ alone. We assume throughout that Σ^+ contains no FD $\emptyset \to A$, that is, no attribute is constant under Σ. For the running example, $\Sigma_a = \{C \to S, C \to T, C \to Z, ST \to Z, Z \to T\}$.

X is a *key* of R for Σ iff $X \to R \in \Sigma^+$, and a *minimal key* if no proper subset of X is a key. An FD $X \to Y$ is *non-key* if X is not a key. An attribute is *prime* if it belongs to some minimal key. For an FD set Σ over R, (R, Σ) is in *BCNF* iff for every non-trivial FD $X \to A \in \Sigma^+$, X is a key of R, and it is in *3NF* iff for every non-trivial FD $X \to A \in \Sigma^+$, X is a key of R or A is prime.

Decompositions. A *decomposition* of R is a set $\mathcal{D}$ of subsets $S \subseteq R$ such that $\bigcup_{S \in \mathcal{D}} S = R$. For an FD set $\mathcal{T}$ over R, $\mathcal{D}$ is *lossless* for $\mathcal{T}$ iff every relation r over R that satisfies $\mathcal{T}$ is the join $\bowtie_{S \in \mathcal{D}} r[S]$ of its projections. Losslessness for $\mathcal{T}$ can be decided in polynomial time with the chase [1]. The *projection* of $\mathcal{T}$ onto S is $\mathcal{T}[S] = \{X \to Y \in \mathcal{T}^+ \mid XY \subseteq S\}$. $\mathcal{D}$ is *dependency-preserving* for $\mathcal{T}$ iff $\mathcal{T}^+ = (\bigcup_{S \in \mathcal{D}} \mathcal{T}[S])^+$, and $\mathcal{D}$ is in BCNF (3NF) for $\mathcal{T}$ iff $(S, \mathcal{T}[S])$ is in BCNF (3NF) for every $S \in \mathcal{D}$. A lossless, dependency-preserving decomposition into BCNF for $\mathcal{T}$ is *faithful*, and $(R, \mathcal{T})$ is *faithfully decomposable* if it has one. A schema $(S, \mathcal{T}[S])$ of a 3NF decomposition that is not in BCNF is called *critical* [68]. In the running example, the decomposition $\{STZ, CST\}$ is lossless and dependency-preserving for Σ and in 3NF, and $(STZ, \{ST \to Z, Z \to T\})$ is its only critical schema.

Complexity results. Deciding whether (R, Σ) is in BCNF takes polynomial time, deciding whether $(S, \Sigma[S])$ is in BCNF for a subset $S \subseteq R$ is coNP-complete, and deciding whether a lossless, dependency-preserving decomposition into BCNF exists is NP-hard [5]. Lossless decompositions into BCNF can be computed in polynomial time [60].

Previous work on dependency loss. Osborn [46] and Köhler [27, 28] decide whether a lossless, dependency-preserving decomposition into BCNF exists by searching the atomic cover for a cover of Σ whose FDs $X \to A$ all have schemata XA in BCNF, and return a 3NF decomposition otherwise. Zhang et al. [68] minimize the number of critical schemata of such a decomposition, and Zhang et al. [71] optimize it for the numbers of minimal keys and non-key FDs. Link et al. [37] and Wei et al. [65] bound the cardinality of redundant value occurrences by a level ℓ and lose FDs in exchange for a smaller level, choosing the lost FDs by the priority of ℓ. All of them measure the redundancy and the non-key FDs that a design keeps. Measuring and minimizing the FDs that a design gives up when only its keys are enforced, while these keys guarantee its losslessness, is the contribution of this work.

3 BCNF DESIGNS AND DEPENDENCY LOSS

The only FDs that every SQL system validates natively are the keys declared on the schemata of a design. We therefore define key designs and the FDs their keys enforce, require designs to be sound and locally enforced, and measure their dependency loss by the minimal FDs of Σ they do not enforce. Tab. 4 in Supp. A summarizes the notation.

3.1 Key Designs and Enforced FDs

A design is a decomposition together with the keys declared on each of its schemata. This motivates the following definition.

Definition 3.1 (Key design). Let Σ be an FD set over relation schema R. A *key design* of (R, Σ) is a pair $(\mathcal{D}, \kappa)$ where $\mathcal{D}$ is a decomposition of R and κ assigns to every $S \in \mathcal{D}$ a set $\kappa(S)$ of subsets $K \subseteq S$ such that $K \to S \in \Sigma^+$. The set of *enforced FDs* of $(\mathcal{D}, \kappa)$ is $\mathcal{E}(\mathcal{D}, \kappa) = \{K \to S \mid S \in \mathcal{D}, K \in \kappa(S)\}^+$. □

That is, a key design declares only keys that Σ permits, so $\mathcal{E}(\mathcal{D}, \kappa) \subseteq \Sigma^+$, and we write $\mathcal{E}$ when the design is clear. A *state* of a key design assigns to every $S \in \mathcal{D}$ a relation r_S over S, and it *satisfies the keys* iff every r_S satisfies $K \to S$ for every $K \in \kappa(S)$. As in classical theory, losslessness refers to states of projections $r_S = r[S]$ of one relation r over R [39], which satisfy the keys iff r satisfies $\mathcal{E}$. We illustrate the definition on our running example.

Example 3.2 (Key designs). The four designs of Tab. 1 are key designs of (R, Σ) from Ex. 1.1. For instance, $\mathcal{D}_4 = \{STZ, CST\}$ with $\kappa(STZ) = \{ST, SZ\}$ and $\kappa(CST) = \{C\}$ is a key design because $ST \to STZ$, $SZ \to STZ$ and $C \to CST$ are all implied by Σ. Its enforced FDs form the closure $\mathcal{E}_4 = \{ST \to Z, SZ \to T, C \to ST\}^+$. The FD $Z \to T$ is implied by Σ but not by $\mathcal{E}_4$. Indeed, the two records (s, t, z) and (s', t', z) with $s \neq s'$ and $t \neq t'$ satisfy both keys of STZ but violate $Z \to T$. □

The following lemma shows that the minimal FDs of $\mathcal{E}$ are weakenings of the minimal FDs of Σ.

Lemma 3.3 (Weakened FDs). *Let $(\mathcal{D}, \kappa)$ be a key design of (R, Σ) with enforced FDs $\mathcal{E}$. For every minimal FD $X \to A$ of $\mathcal{E}$ there is some minimal FD $Y \to A$ of Σ with $Y \subseteq X$.*

That is, a design enforces a minimal FD $Y \to A$ of Σ either itself or in a weakened form $X \to A$ with $Y \subset X$, as $\mathcal{D}_4$ enforces $SZ \to T$ instead of $Z \to T$.

3.2 Sound and Locally Enforced Designs

A decomposition that is lossless for Σ may have lossy joins on states that satisfy its keys, as $\mathcal{D}_3$ in Ex. 1.1 shows, so losslessness must rest on the enforced FDs. This motivates the following definition.

Definition 3.4 (Sound design). A key design $(\mathcal{D}, \kappa)$ of (R, Σ) is *sound* iff $\mathcal{D}$ is lossless for $\mathcal{E}(\mathcal{D}, \kappa)$. □

The join of a sound design is therefore lossless on every relation that satisfies the declared keys. Soundness implies losslessness for Σ, but not vice versa. We illustrate the definition on our running example.

Example 3.5 (Soundness). Design $\mathcal{D}_3 = \{ZT, SZ, CST\}$ declares the keys Z of ZT, SZ of SZ and C of CST, so $\mathcal{E}_3 = \{Z \to T, C \to$

$ST\}^+$ since the key of SZ is trivial. The relation with the two tuples (c, s, t, z) and (c', s, t, z'), where $c \neq c'$ and $z \neq z'$, satisfies $\mathcal{E}_3$ but violates $ST \to Z$, and the join of its projections onto ZT, SZ and CST contains all four combinations of c, c' with z, z'. Hence, $\mathcal{D}_3$ is not sound. In contrast, $\mathcal{D}_2$ is sound since $ZT \cap CSZ = Z$ and $Z \to T \in \mathcal{E}_2$, and $\mathcal{D}_4$ is sound since $STZ \cap CST = ST$ and $ST \to Z \in \mathcal{E}_4$. □

The enforced FDs may also imply on a schema S an FD that only the keys of other schemata imply, which causes redundant values that no key of S checks. This motivates the following definition.

Definition 3.6 (BCNF design). A key design $(\mathcal{D}, \kappa)$ of (R, Σ) is *locally enforced* iff for every $S \in \mathcal{D}$, $\mathcal{E}(\mathcal{D}, \kappa)[S] = \{K \to S \mid K \in \kappa(S)\}^+[S]$ holds. A key design is a *BCNF design* iff it is sound and locally enforced. □

That is, every FD that the design implies on a schema is already implied by the keys declared on that schema. Equivalently, every non-trivial FD $X \to Y \in \mathcal{E}[S]$ satisfies $K \subseteq X$ for some $K \in \kappa(S)$. We illustrate the definition on our running example.

Example 3.7 (Local enforcement). Design $\mathcal{D}_4$ is locally enforced, since the non-trivial FDs in $\mathcal{E}_4[STZ]$ have a left-hand side that contains ST or SZ, and those in $\mathcal{E}_4[CST]$ one that contains C. Hence, $\mathcal{D}_4$ is a BCNF design, and so are $\mathcal{D}_1$ and $\mathcal{D}_2$. Soundness does not imply local enforcement: for $\Sigma' = \{A \to B, B \to C\}$ over $ABCD$, the sound design $\{AB, BC, ACD\}$ with keys A, B and AD implies $A \to C$ on ACD, which no key of ACD enforces, so the relation over ACD with the tuples (a, c, d) and (a, c', d') satisfies the key AD but violates $A \to C$. □

The name of BCNF designs is justified by the following lemma, which relates them to the classical notions.

Lemma 3.8 (BCNF designs and BCNF decompositions). *Let $(\mathcal{D}, \kappa)$ be a key design of (R, Σ) with enforced FDs $\mathcal{E}$. Then $\mathcal{D}$ is dependency-preserving for $\mathcal{E}$, and if $(\mathcal{D}, \kappa)$ is locally enforced, then $(S, \mathcal{E}[S])$ is in BCNF for every $S \in \mathcal{D}$. Conversely, if $(S, \mathcal{E}[S])$ is in BCNF and every minimal key of $(S, \mathcal{E}[S])$ belongs to $\kappa(S)$ for every $S \in \mathcal{D}$, then $(\mathcal{D}, \kappa)$ is locally enforced.*

Consequently, a BCNF design is a faithful decomposition for the FDs it enforces, which may be a proper subset of the given FDs. The design $\mathcal{D}_4$ is such a decomposition of $(R, \mathcal{E}_4)$ but not of (R, Σ). The following proposition states what a BCNF design guarantees and what it does not.

Proposition 3.9 (Semantics of BCNF designs). *Let $(\mathcal{D}, \kappa)$ be a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$. Then the following hold for every state $(r_S)_{S \in \mathcal{D}}$ that satisfies the keys.*

1. *Every relation r_S satisfies $\mathcal{E}[S]$ and contains no data value that is redundant for $\mathcal{E}[S]$.*
2. *The join $\bowtie_{S \in \mathcal{D}} r_S$ satisfies $\mathcal{E}$.*
3. *If $r_S = r[S]$ for every $S \in \mathcal{D}$ and a relation r over R that satisfies $\mathcal{E}$, then $r = \bowtie_{S \in \mathcal{D}} r_S$.*

An FD $\varphi \in \Sigma^+$ is not in $\mathcal{E}$ iff the join of some state that satisfies the keys violates φ.

Prop. 3.9 tells us that a BCNF design protects the enforced FDs on every relation and on the join, and every other FD in Σ^+ not at all. Only statement 3 rests on the assumption after Def. 3.1, as losslessness does for every decomposition, and Sec. 3.7 characterizes when keys alone keep the stored relations consistent, while statement 1 fails without local enforcement (Ex. 3.7). Statement 1 excludes redundancy for the enforced FDs only, so an unenforced FD inside a schema, such as $Z \to T$ in STZ of $\mathcal{D}_4$, still causes redundant values whenever the data satisfies it. The unenforced FDs are the business rules that the application must protect, which motivates measuring the loss of a design by them.

3.3 Dependency Loss

The closure Σ^+ may contain exponentially many FDs, and the number of FDs in a cover depends on the cover [38]. The atomic cover is the set of all minimal FDs, so counting its unenforced elements is independent of how Σ is given. This motivates the following definition.

Definition 3.10 (Dependency loss). Let $(\mathcal{D}, \kappa)$ be a key design of (R, Σ) with enforced FDs $\mathcal{E}$, and let $w : \Sigma_a \to \mathbb{R}_{>0}$ assign weights to the minimal FDs of Σ. The *dependency loss* of $(\mathcal{D}, \kappa)$ is $\lambda_w(\mathcal{D}, \kappa) = \sum_{\varphi \in \Sigma_a - \mathcal{E}} w(\varphi)$, and $\lambda(\mathcal{D}, \kappa) = |\Sigma_a - \mathcal{E}|$ for unit weights. The *dependency loss* $\lambda_w(R, \Sigma)$ of (R, Σ) is the minimum of $\lambda_w(\mathcal{D}, \kappa)$ over all BCNF designs $(\mathcal{D}, \kappa)$ of (R, Σ). □

Every key design has a loss, but only BCNF designs count towards $\lambda_w(R, \Sigma)$. We illustrate the definition on our running example.

Example 3.11 (Dependency loss). The enforced FDs $\mathcal{E}_4$ imply the minimal FD $C \to Z$ of Σ through $C \to ST$ and $ST \to Z$, and $C \to S$, $C \to T$ and $ST \to Z$ directly, but not $Z \to T$. Hence, $\lambda(\mathcal{D}_4) = 1$, and similarly $\lambda(\mathcal{D}_2) = 1$ and $\lambda(\mathcal{D}_1) = 2$. The key design $\mathcal{D}_3$ has loss two, but it is not a BCNF design and does not count towards $\lambda(R, \Sigma)$. Consequently, $\lambda(R, \Sigma) \leq 1$, and Ex. 3.20 will show that $\lambda(R, \Sigma) = 1$. □

The following proposition shows that the dependency loss exists for every (R, Σ), what the design without decomposition costs, and that no such design can do better.

Proposition 3.12 (Key-only designs). *Let $\mathcal{K}$ denote the set of minimal keys of (R, Σ), and let $\nu(R, \Sigma)$ denote the number of non-key minimal FDs of Σ. Then $(\{R\}, \kappa)$ with $\kappa(R) = \mathcal{K}$ is a BCNF design of (R, Σ) with $\lambda(\{R\}, \kappa) = \nu(R, \Sigma)$. Every BCNF design $(\{R\}, \kappa')$ of (R, Σ) satisfies $\lambda(\{R\}, \kappa') \geq \nu(R, \Sigma)$.*

Hence, decomposition is the only means of enforcing a non-key FD through keys. In the running example, $\lambda(\mathcal{D}_1) = \nu(R, \Sigma) = 2$.

Why minimal FDs. A measure of the FDs that a design leaves unenforced should satisfy three requirements. (R1) It depends only on Σ^+, not on the cover. (R2) It adds up fixed costs of the unenforced FDs, since each of them is a rule at risk (Prop. 3.9). (R3) It charges nothing for an FD $X \to A$ with $Y \to A \in \Sigma^+$ for some $Y \subset X$, since every relation that violates $X \to A$ also violates $Y \to A$. Exactly the losses λ_w of Def. 3.10 satisfy these requirements, and unit costs single out λ (Prop. A.15 in Supp. A). Natural alternatives violate one of the requirements. Counting the unenforced FDs of the given cover violates (R1): $\mathcal{D}_3$ leaves only $ST \to Z$ of Σ unenforced, but also $C \to Z$, which the cover $\Sigma \cup \{C \to Z\}$ counts. Counting all unenforced FDs of Σ^+ violates (R3), and this count may be exponential in $|R|$ when the loss is one. The *repair distance*,

the fewest FDs whose addition to $\mathcal{E}$ restores Σ^+, satisfies (R1) but violates (R2): it never exceeds the loss, but it may be smaller by an unbounded factor while every unenforced minimal FD remains at risk (Prop. A.12 in Supp. A).

The weights also connect the loss to integrity. For a key design with enforced FDs $\mathcal{E}$ and a random relation r' over R, such as the result of a random update, let $p(\varphi)$ be the probability that r' violates $\varphi \in \Sigma_a$, and let λ_p denote the loss of Def. 3.10 with these non-negative weights.

Proposition 3.13 (Risk bound). *The probability that r' violates Σ but satisfies every declared key dependency is at most $\lambda_p(\mathcal{D}, \kappa)$, and the expected number of minimal FDs of Σ that r' violates and $\mathcal{E}$ does not contain is $\lambda_p(\mathcal{D}, \kappa)$. The bound is attained when the design is plain and r' never violates two minimal FDs of Σ.*

Hence, minimizing the loss under violation weights minimizes an upper bound on the updates that violate Σ unnoticed, the integrity risk of Sec. 6, and unit weights are the special case of equally likely violations. Moreover, weights such as FD rankings and certainty degrees attach to minimal FDs, and computing the loss takes one implication test per minimal FD.

3.4 Bounded Dependency Loss

The dependency loss of a schema separates the faithfully decomposable schemata from the others, and it grades the latter. This gives rise to a hierarchy of normal forms.

Definition 3.14 (Bounded dependency loss). Let Σ be an FD set over relation schema R, and let k be a non-negative integer. Then (R, Σ) is in *Boyce-Codd Normal Form with loss k* (k-loss BCNF) iff $\lambda(R, \Sigma) \leq k$. □

That is, a schema in k-loss BCNF has a BCNF design that leaves at most k minimal FDs unenforced. Unlike BCNF and 3NF, the normal form thus constrains the best design of a schema, not the schema itself, so a schema in 0-loss BCNF need not be in BCNF. Every schema is in $\nu(R, \Sigma)$-loss BCNF by Prop. 3.12. The next theorem shows that the bottom of the hierarchy is the classical notion.

Theorem 3.15 (Loss zero). *For all relation schemata R and FD sets Σ over R, (R, Σ) is in 0-loss BCNF if and only if (R, Σ) is faithfully decomposable.*

The classical theory thus emerges as the special case $k = 0$.

Above level zero, the gap between the optimum and both the key-only design and classical BCNF decomposition is unbounded, as the following proposition shows.

Proposition 3.16 (Unbounded gap). *For every $n \geq 1$ there is a schema (R_n, Σ_n) with $\lambda(R_n, \Sigma_n) = 1$ such that the design $(\{R_n\}, \kappa)$ of Prop. 3.12 has loss $n + 1$ and every BCNF decomposition of (R_n, Σ_n) that the classical algorithm obtains has loss $n + 1$.*

The schemata of the proof add attributes that depend on the zip code to STZ, and their optimal designs keep STZ with key ST and move these attributes into a separate schema. Neither 3NF synthesis nor classical BCNF decomposition, which loses $ST \to Z$ and every FD derived through it, produces such a design.

3.5 Enforceable FD Sets

The loss of a design depends only on its enforced FDs. Thm. 3.18 below reduces the loss to faithful decomposability:

$$\lambda_w(R, \Sigma) = \min\{\, w(\Sigma_a - \mathcal{E}) \mid \mathcal{E} \subseteq \Sigma^+ \text{ is closed and } (R, \mathcal{E}) \text{ is faithfully decomposable}\}. \tag{1}$$

Definition 3.17 (Enforceable FD set). Let Σ be an FD set over relation schema R. A closed FD set $\mathcal{E} \subseteq \Sigma^+$ is *enforceable* for (R, Σ) iff $\mathcal{E} = \mathcal{E}(\mathcal{D}, \kappa)$ for some BCNF design $(\mathcal{D}, \kappa)$ of (R, Σ). A minimal FD $Y \to B$ of $\mathcal{E}$ *violates* a schema $S \subseteq R$ iff $YB \subseteq S$ and $S \not\subseteq Y^+_{\mathcal{E}}$, and a minimal FD $X \to A$ of $\mathcal{E}$ is *critical* for $\mathcal{E}$ iff some minimal FD of $\mathcal{E}$ violates XA, and *uncritical* otherwise. □

Since a non-key FD $Z \to B$ of $(XA, \mathcal{E}[XA])$ yields a violating minimal FD $Y \to B$ of $\mathcal{E}$ with $Y \subseteq Z$, $X \to A$ is critical iff $(XA, \mathcal{E}[XA])$ is not in BCNF, the obstacle that Osborn [46] and Köhler [27] identified. This schema is in 3NF, so $X \to A$ is critical exactly when its schema is critical in the sense of Sec. 2. The following theorem characterizes enforceable FD sets.

Theorem 3.18 (Enforceable FD sets). *Let Σ be an FD set over relation schema R, and let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set. Then the following statements are equivalent.*

(1) $\mathcal{E}$ is enforceable for (R, Σ).
(2) $(R, \mathcal{E})$ is faithfully decomposable.
(3) The uncritical minimal FDs of $\mathcal{E}$ form a cover of $\mathcal{E}$.

Eq. (1) is the equivalence of statements 1 and 2. The equivalence of statements 2 and 3 is the characterization of Köhler [27, 28] applied to $\mathcal{E}$ instead of Σ. The proof also provides a design for every set $\Psi \subseteq \Sigma^+$ of non-trivial FDs that is *uncritical*, that is, $(XA, \Psi^+[XA])$ is in BCNF for every $X \to A \in \Psi$.

Corollary 3.19 (Canonical designs). *Let $\Psi \subseteq \Sigma^+$ be an uncritical set. Then Ψ^+ is enforceable for (R, Σ), and the schemata XA of the elements $X \to A$ of Ψ together with a minimal key of (R, Ψ^+), each declaring its minimal keys for Ψ^+, form a BCNF design of (R, Σ) with enforced FDs Ψ^+.*

Hence, the uncritical sets are exactly the generators of the enforceable FD sets, and every enforceable FD set has a BCNF design with at most one schema per uncritical minimal FD and one key schema. We illustrate the theorem on our running example.

Example 3.20 (Enforceable FD sets). The enforced FDs $\mathcal{E}_4 = \{ST \to Z, SZ \to T, C \to ST\}^+$ of $\mathcal{D}_4$ have the minimal FDs $ST \to Z$, $SZ \to T$, $C \to S$, $C \to T$ and $C \to Z$. All of them are uncritical, since $(STZ, \mathcal{E}_4[STZ])$ has the key dependencies of ST and SZ only, so $\mathcal{E}_4$ is enforceable. In contrast, $ST \to Z$ is critical for Σ^+ since $Z \to T$ is a non-key FD of $(STZ, \Sigma[STZ])$, and the uncritical minimal FDs $Z \to T$, $C \to S$, $C \to T$ and $C \to Z$ of Σ^+ do not imply $ST \to Z$. Hence, Σ^+ is not enforceable, which confirms $\lambda(R, \Sigma) = 1$. □

The uncritical minimal FDs of Σ also bound the loss from above, which motivates the following definition.

Definition 3.21 (Obstructions). Let Σ be an FD set over relation schema R, and let $\mathcal{U}$ be the set of uncritical minimal FDs of Σ. The *obstructions* of (R, Σ) are the elements of $\mathcal{O} = \Sigma_a - \mathcal{U}^+$. □

That is, the obstructions are the minimal FDs of Σ that the uncritical minimal FDs do not imply, so Σ^+ is enforceable iff (R, Σ) has no obstruction by Thm. 3.18, and every obstruction is critical. In the running example, $ST \to Z$ is the only obstruction. An uncritical minimal FD of Σ stays uncritical in every closed set $\mathcal{E} \subseteq \Sigma^+$ that contains all uncritical minimal FDs of Σ (Lem. A.2 in Supp. A). Consequently, the closure of the uncritical minimal FDs passes the test of Thm. 3.18.

Theorem 3.22 (Obstruction bound). *Let $\mathcal{U}$ be the set of uncritical minimal FDs of Σ, and let $\mathcal{O}$ be the set of obstructions of (R, Σ). Then $\mathcal{U}^+$ is enforceable for (R, Σ), and $\lambda_w(R, \Sigma) \leq w(\mathcal{O})$ for every weight function w.*

By Lem. A.2 in Supp. A, $\mathcal{U}$ is an uncritical set, and Cor. 3.19 turns it into a BCNF design with loss $w(\mathcal{O})$, which we call the *uncritical design* of (R, Σ). Hence, (R, Σ) is in k-loss BCNF whenever it has at most k obstructions, and the criterion of Köhler [27] for level zero is the case $\mathcal{O} = \emptyset$. The bound is not tight in general (Prop. A.5 in Supp. A), but Prop. 5.4 will bound the loss from below by the same obstructions.

3.6 The Price of Soundness

The classical decomposition algorithm can return unsound designs such as $\mathcal{D}_3$ (Prop. A.1 in Supp. A). Soundness, however, is free in terms of loss, since every decomposition into BCNF schemata can be completed to a sound design that enforces the same FDs.

Proposition 3.23 (Soundness is free). *Let $(\mathcal{D}, \kappa)$ be a locally enforced key design of (R, Σ) with enforced FDs $\mathcal{E}$, and let K be a minimal key of $(R, \mathcal{E})$. Adding the schema K with the key K yields a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$. In particular, every decomposition $\mathcal{D}$ with $(S, \Sigma[S])$ in BCNF for all $S \in \mathcal{D}$, declaring the minimal keys of these schemata, extends to a BCNF design that enforces the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$.*

Consequently, soundness never increases the minimum loss. The schemata of a BCNF design, however, need not be in BCNF for Σ, as in $\mathcal{D}_4$, and Thm. 5.2 shows that this freedom can lower the loss.

3.7 Independent Designs

A database stores the relations r_S independently, and only their keys are checked. Such a state need not consist of the projections of a single relation, so we adopt the weak-instance semantics of independent schemata [22, 53]. A state $(r_S)_{S \in \mathcal{D}}$ of a key design is *globally consistent* iff some relation r over R satisfies $\mathcal{E}$ and $r_S \subseteq r[S]$ for every $S \in \mathcal{D}$, and the key design is *independent* iff every state that satisfies the keys is globally consistent. In an independent design, the keys protect the enforced FDs not only on every relation and on the join, as in Prop. 3.9, but also across the stored relations: every stored tuple extends to a tuple over R that agrees with all other stored tuples on every FD of $\mathcal{E}$. For $S, T \in \mathcal{D}$, let $S^+_{\setminus T}$ denote the closure of S under the declared key dependencies of the schemata in $\mathcal{D} - \{T\}$.

Theorem 3.24 (Independence). *A key design $(\mathcal{D}, \kappa)$ is independent iff for all distinct $S, T \in \mathcal{D}$ and every $K \in \kappa(T)$ with $K \subseteq S^+_{\setminus T}$, the sets $T - K$ and $S^+_{\setminus T}$ are disjoint. Independence can therefore be decided in polynomial time.*

That is, a stored tuple of S, extended by key lookups in the other relations, reaches every attribute through at most one relation; otherwise a state with one tuple per relation satisfies all keys but assigns some attribute two values. The designs $\mathcal{D}_1$, $\mathcal{D}_2$ and $\mathcal{D}_4$ are independent, but classical decompositions need not be, and independence may cost FDs.

Example 3.25 (Price of independence). For $\Sigma = \{A \to BD, B \to C, D \to C\}$ over $ABCD$, the decomposition $\{ABD, BC, CD\}$ with keys A, B and D is lossless, dependency-preserving and in BCNF, so it is a BCNF design with loss zero. It is not independent: the state with the tuples (a, b, d), (b, c) and (d, c') for $c \neq c'$ satisfies all keys, but no relation over $ABCD$ that satisfies Σ contains them, since a would need both c and c'. Indeed, every BCNF design with loss zero contains the schemata BC with key B and CD with key D, and the lookups from a schema with A reach C through both. The BCNF design $\{ABD, BC\}$ with keys A and B is independent and leaves only $D \to C$ unenforced, so the minimum loss of the independent BCNF designs is one. □

The algorithms of Sec. 5 can test independence on their output. On the eleven FD sets of Sec. 6 with records and positive loss, the key-only design and the classical BCNF decomposition are always independent, but the designs of Repair only on three of the four sets of meaningful FDs and the keys of iConf only on two (Tab. 6 in Supp. A), since their schemata overlap, so that a stored tuple reaches some attribute through two relations. Independence is thus a guarantee beyond minimum loss, and its price in loss is open.

4 COMPUTATIONAL COMPLEXITY

The hardness of the classical case transfers to every level of the hierarchy, and to the problem MinLoss of Sec. 5 under the atomic cover.

Theorem 4.1 (NP-hardness). *For every fixed $k \geq 0$, deciding whether (R, Σ) is in k-loss BCNF is NP-hard under arbitrary covers.*

The case $k = 0$ is NP-hard by Thm. 3.15 and [5]. The proof lifts it to every $k > 0$ by adding k attribute-disjoint copies of the schema $(\{S, T, Z\}, \{ST \to Z, Z \to T\})$ with loss one each, whose losses add up by the following lemma.

Lemma 4.2 (Additivity). *Let Σ_1 and Σ_2 be FD sets over disjoint schemata R_1 and R_2. Then $\lambda(R_1 \cup R_2, \Sigma_1 \cup \Sigma_2) = \lambda(R_1, \Sigma_1) + \lambda(R_2, \Sigma_2)$.*

Hence, the hierarchy is strict, and since level zero is already NP-hard, no algorithm for the loss under arbitrary covers is fixed-parameter tractable in the loss itself unless P equals NP. This hardness rests on the succinctness of Σ: when Σ_a is given, as in our experiments, membership in 0-loss BCNF takes polynomial time by Thm. 3.18, but MinLoss remains hard.

Theorem 4.3 (NP-hardness with explicit atomic cover). *Deciding whether $\lambda(R, \Sigma) \leq k$ for a given schema (R, Σ) and integer k is NP-hard, even when Σ is its own atomic cover.*

The proof maps a graph with n vertices and m edges to a schema with $n + 6m$ minimal FDs whose minimum loss is $2m + n - \alpha$, where α is the size of a maximum independent set of the graph [21]. The hardness thus stems from the choice of the FDs to leave unenforced, the choice that our algorithms face on the atomic covers of Sec. 6, and it extends to approximation.

Corollary 4.4 (Inapproximability). *Unless P equals NP, there is no polynomial-time algorithm that computes for every schema (R, Σ), given by an arbitrary cover, a BCNF design whose loss is at most $\rho \cdot \lambda(R, \Sigma)$ for some function ρ of the input size, or at most $\lambda(R, \Sigma) + c$ for some constant c. Even when $\Sigma = \Sigma_a$, the same holds for a loss below $2 \cdot \lambda(R, \Sigma)$ and for a loss of at most $\lambda(R, \Sigma) + c$.*

Indeed, an algorithm of the first kind decides membership in 0-loss BCNF, and an algorithm of the second kind decides it on $c+1$ attribute-disjoint copies of the input, whose loss is $c + 1$ times the loss of the input. For the last statement, telling loss one from loss two is coNP-hard under the atomic cover (Thm. A.9 in Supp. A), while the loss of a computed design takes polynomial time when Σ_a is given.

Theorem 4.5 (Verification). *Given (R, Σ) and a key design $(\mathcal{D}, \kappa)$ represented by its declared keys, deciding whether $(\mathcal{D}, \kappa)$ is sound is possible in polynomial time, and deciding whether $(\mathcal{D}, \kappa)$ is locally enforced is coNP-complete. The hardness holds even when Σ consists of the declared key dependencies, for sound designs in which every schema but two has two attributes and declares one key.*

Soundness is decided by the chase, while local enforcement is hard for the same reason as BCNF of projections [5], so a design given by its keys is no certificate of membership in k-loss BCNF that can be checked in polynomial time unless P equals NP. With the atomic cover given, deciding whether $\lambda(R, \Sigma) \leq k$ is coNP-hard for every fixed $k \geq 1$, in contrast to $k = 0$, even for two obstructions, and DP-hard for k as part of the input, while MinLoss for plain designs is in Σ_2^p (Thms. A.8 to A.11 in Supp. A).

5 COMPUTING DESIGNS WITH MINIMUM LOSS

By Thm. 3.18, the search for a design with minimum loss is a search for an enforceable FD set, and two natural restrictions of this search lose optimality. The algorithms Drop and Hit compute a design with minimum loss, Ascend lowers the loss of any BCNF design, and Repair repairs the critical schemata of 3NF synthesis locally, trading optimality for scalability.

Problem 1 (MinLoss). Given a relation schema R, an FD set Σ over R, and weights w on Σ_a, compute a BCNF design $(\mathcal{D}, \kappa)$ of (R, Σ) such that $\lambda_w(\mathcal{D}, \kappa) = \lambda_w(R, \Sigma)$. □

5.1 Plain and Key-Complete Designs

A design may enforce a minimal FD of Σ in weakened form only (Lem. 3.3), and it may declare fewer keys than Σ permits. We name the designs that avoid these complications.

Definition 5.1 (Plain and key-complete designs). Let $(\mathcal{D}, \kappa)$ be a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$. The design is *plain* iff $\mathcal{E} = (\Sigma_a \cap \mathcal{E})^+$, and it is *key-complete* iff $\kappa(S)$ contains every minimal key of $(S, \Sigma[S])$ for every $S \in \mathcal{D}$. □

By Lem. 3.3, every minimal FD of $\mathcal{E}$ is a minimal FD of Σ or a weakening of one, and a design is plain iff its weakenings are implied by the minimal FDs of Σ it enforces, while a key-complete design declares every key that Σ permits. Every classical faithful decomposition is plain and key-complete, and so are the designs $\mathcal{D}_1$ and $\mathcal{D}_2$ of the running example. The design $\mathcal{D}_4$ is key-complete but not plain, since $SZ \to T \in \mathcal{E}_4$ is not implied by $\Sigma_a \cap \mathcal{E}_4 = \{C \to S, C \to T, C \to Z, ST \to Z\}$. The design $\mathcal{D}_4'$ that declares only the key ST on STZ is plain but not key-complete, and it has the same loss as $\mathcal{D}_4$. Plain designs reduce MinLoss to a search over subsets of the atomic cover, and key-complete designs to a search over decompositions. The following theorem shows that neither restriction preserves optimality.

Theorem 5.2 (Plain and key-complete designs). *There is a schema (R, Σ) such that every plain BCNF design of (R, Σ) has a loss greater than $\lambda(R, \Sigma)$, and there is a schema (R, Σ) such that every key-complete BCNF design of (R, Σ) has a loss greater than $\lambda(R, \Sigma)$.*

Consequently, the search for an optimal design can be confined neither to the atomic cover nor to decompositions. Confining the search to plain designs can multiply the minimum loss by the number of obstructions, which may be exponential in the number of attributes (Prop. A.7 in Supp. A).

5.2 The Algorithm Drop

Thm. 3.18 and Lem. 3.3 bound the search space of MinLoss. Every enforceable FD set is generated by the minimal FDs of Σ it contains together with weakenings of the minimal FDs of Σ it does not contain. For a set $\Delta \subseteq \Sigma_a$ we write $\mathcal{W}(\Delta) = \{X' \to A \mid X \to A \in \Delta, X \subset X' \subseteq R - A\}$ for the set of weakenings of its elements. Violation and criticality refer to $\mathcal{E} = \Sigma^+$ unless stated otherwise. Let $\mathcal{U}$ denote the set of uncritical minimal FDs of Σ, and for $\Delta \subseteq \Sigma_a$ let $C(\Delta)$ denote the set of critical minimal FDs $X \to A$ of Σ such that every minimal FD of Σ that violates XA belongs to Δ. The following lemma restricts the sets Δ of unenforced minimal FDs.

Lemma 5.3 (Cover condition). *Let $\mathcal{E}$ be the enforced FDs of a BCNF design of (R, Σ), and let $\Delta = \Sigma_a - \mathcal{E}$. Then $\mathcal{U} \cup C(\Delta) \cup \Delta$ is a cover of Σ. In particular, if Σ^+ is not enforceable, then Δ contains a critical minimal FD of Σ or a minimal FD of Σ that violates the schema XA of a critical minimal FD $X \to A$ of Σ.*

The lemma says that the unenforced minimal FDs, together with the uncritical minimal FDs and the critical minimal FDs whose schemata they relieve of every violation, must still imply Σ. The condition is monotone: if Δ satisfies it, so does every superset of Δ, as $C(\Delta)$ grows with Δ. Hence, if Δ fails it, every BCNF design leaves some minimal FD outside Δ unenforced, and the requirement sets below identify such FDs for the designs that leave all of Δ unenforced. The test is polynomial in the size of Σ_a, and Alg. 1 applies it in line 3. The cover condition also yields lower bounds on the loss, which prune the enumeration and certify a design when the budget of the enumeration is exhausted. For $\Delta \subseteq \Sigma_a$ and a minimal FD $g = Y \to B$ of Σ that $\mathcal{U} \cup C(\Delta) \cup \Delta$ does not imply, the *requirement set* $Q(g, \Delta)$ consists of the minimal FDs in $\Sigma_a - \Delta$ outside $\mathcal{U} \cup C(\Delta)$ with left-hand side in Y_Σ^+, and of the minimal FDs in $\Sigma_a - \Delta$ that violate the schema XA of some critical minimal FD $X \to A \notin C(\Delta)$ with $X \subseteq Y_\Sigma^+$. Intuitively, $Q(g, \Delta)$ contains the FDs whose loss could still open a derivation of g from FDs that the cover condition admits.

Proposition 5.4 (Packing bound). *Let $\Delta \subseteq \Sigma_a$, and let $g_1, \ldots, g_m$ be minimal FDs of Σ not implied by $\mathcal{U} \cup C(\Delta) \cup \Delta$ whose*

Algorithm 1 Drop(R, Σ, w)

Require: Relation schema R, FD set Σ over R, weights w on Σ_a
Ensure: A BCNF design of (R, Σ) with minimum loss
1: compute the atomic cover Σ_a of Σ and its uncritical minimal FDs $\mathcal{U}$
2: **for** $b \leftarrow 0, w_{\min}, \ldots$ in increasing order of total weights of subsets of Σ_a **do**
3: **for all** $\Delta \subseteq \Sigma_a$ with $w(\Delta) = b$ and $(\mathcal{U} \cup C(\Delta) \cup \Delta)^+ = \Sigma^+$ **do**
4: $\Psi \leftarrow \Sigma_a - \Delta$
5: **for all** $W \subseteq \mathcal{W}(\Delta)$ in increasing order of $|W|$ **do**
6: $\mathcal{E} \leftarrow (\Psi \cup W)^+$
7: **if** $\mathcal{E} \cap \Delta \neq \emptyset$ **then**
8: **continue** ▷ *$\mathcal{E}$ has a smaller loss and was examined before*
9: $\mathcal{D} \leftarrow$ FaithfulBCNF$(R, \Psi \cup W)$ ▷ *faithful decomposition of $(R, \mathcal{E})$, or $\perp$*
10: **if** $\mathcal{D} \neq \perp$ **then**
11: $\kappa(S) \leftarrow$ minimal keys of $(S, \mathcal{E}[S])$ for every $S \in \mathcal{D}$
12: **return** $(\mathcal{D}, \kappa)$

requirement sets $Q(g_1, \Delta), \ldots, Q(g_m, \Delta)$ are pairwise disjoint. Then every BCNF design of (R, Σ) that leaves every element of Δ unenforced leaves some element of every $Q(g_i, \Delta)$ unenforced. Hence, its loss is at least $w(\Delta) + \sum_{i=1}^{m} \min_{f \in Q(g_i, \Delta)} w(f)$, and no such design exists if some $Q(g_i, \Delta)$ is empty.

For $\Delta = \emptyset$, the FDs g_i are obstructions of (R, Σ), and the bound is a lower bound on $\lambda_w(R, \Sigma)$ that any pairwise disjoint family of requirement sets provides in time polynomial in the size of Σ_a. In the running example, the only obstruction $ST \to Z$ has the requirement set $Q(ST \to Z, \emptyset) = \{ST \to Z, Z \to T\}$, since $Z \to T$ violates STZ, so every BCNF design leaves $ST \to Z$ or $Z \to T$ unenforced. Together with Thm. 3.22, the obstructions enclose the loss from both sides, and Alg. 1 terminates at a budget of at most $w(O)$, since $\Delta = O$ with $W = \emptyset$ succeeds. Our implementation prunes line 3 with this bound and certifies as a lower bound the smallest weight of a set Δ that its time or a cap left unexplored.

Alg. 1 enumerates the sets Δ of minimal FDs to leave unenforced in increasing order of their weight, discards those that fail the cover condition, and for every Δ enumerates the sets W of weakened FDs to enforce instead in increasing order of their size, so that plain designs are found first. Lines 7 and 8 skip the sets $\mathcal{E}$ that imply an element of Δ, which were examined at a smaller budget. Line 9 applies the test of Thm. 3.18 and, in time polynomial in the size of the atomic cover of $\mathcal{E}$, returns a faithful decomposition of $(R, \mathcal{E})$, which exists by Cor. 3.19. The following theorem states the guarantee of the algorithm.

Theorem 5.5 (Correctness of Drop). *On input (R, Σ, w), Alg. 1 returns a BCNF design of (R, Σ) with loss $\lambda_w(R, \Sigma)$. If line 5 is restricted to $W = \emptyset$, then the algorithm returns a plain BCNF design of (R, Σ) whose loss is minimum among all plain BCNF designs of (R, Σ).*

Thm. 4.3 rules out polynomial time unless P equals NP, even when Σ_a is given, and the runtime of Alg. 1 is worst-case doubly exponential in $|R|$, since line 5 enumerates subsets of $\mathcal{W}(\Delta)$. This enumeration is the price of Thm. 5.2. Our implementation grows W by a goal-directed search, which branches only over weakenings within the closure of the left-hand side of a critical FD and which Lem. A.16 in Supp. A makes complete.

5.3 The Algorithm Hit

Alg. 1 enumerates the sets Δ in increasing order of their weight, and the certified lower bounds of its implementation rest on the disjoint requirement sets of Prop. 5.4. The requirement sets suggest a different view. Every BCNF design must leave some element of each of them unenforced, so MinLoss is a weighted hitting-set problem over the atomic cover whose constraints can be generated on demand [17, 43]. The weakened FDs do not enter the objective, since the loss of a design depends only on the set Δ of minimal FDs that it leaves unenforced.

We encode a set $\Delta \subseteq \Sigma_a$ by $y \in \{0,1\}^{\Sigma_a}$ with $y_f = 1$ iff $f \in \Delta$. A *cut* is a pair (P, Q) of subsets of Σ_a, which Δ *satisfies* iff $P \not\subseteq \Delta$ or $Q \cap \Delta \neq \emptyset$, that is, iff $\sum_{f \in Q} y_f + \sum_{f \in P} (1 - y_f) \geq 1$. A cut is *valid* for (R, Σ) if the set $\Sigma_a - \mathcal{E}$ of every BCNF design with enforced FDs $\mathcal{E}$ satisfies it. For a set $\mathcal{K}$ of cuts, the *master problem* asks for a set $\Delta \subseteq \Sigma_a$ that satisfies every cut in $\mathcal{K}$ and minimizes $w(\Delta) = \sum_{f \in \Sigma_a} w(f)\, y_f$, an integer linear program. If every cut in $\mathcal{K}$ is valid, the optimum of the master problem is a lower bound on $\lambda_w(R, \Sigma)$. The results of Sec. 5 provide four kinds of valid cuts.

Lemma 5.6 (Valid cuts). *Let $\Delta \subseteq \Sigma_a$.*

(1) For every minimal FD g of Σ not implied by $\mathcal{U} \cup C(\Delta) \cup \Delta$, the cut $(\Delta, Q(g, \Delta))$ is valid.
(2) For every $f \in \Sigma_a$ and every $D \subseteq \Sigma_a$ that implies f, the cut $(\{f\}, D)$ is valid.
(3) If no BCNF design leaves exactly Δ unenforced, the cut $(\Delta, \Sigma_a - \Delta)$ is valid.
(4) For every $g = Y \to B$ as in statement 1, the cut $(\Delta \cap \mathcal{N}(g), Q(g, \Delta))$ is valid, where $\mathcal{N}(g)$ consists of the critical minimal FDs of Σ with left-hand side contained in Y_Σ^+ and of the minimal FDs that violate their schemata.

The first kind restates Prop. 5.4, the second the closure of enforced FD sets, and the third excludes a single set. For $\Delta = \emptyset$, the cuts of the first kind are the hitting-set constraints $\sum_{f \in Q(g, \emptyset)} y_f \geq 1$ for the obstructions g, and already the linear relaxation of the master problem with these cuts dominates the bound of Prop. 5.4, which is the special case of a pairwise disjoint family. The fourth kind strengthens the first, since it excludes every superset of $\Delta \cap \mathcal{N}(g)$ that misses $Q(g, \Delta)$, not only the supersets of Δ.

Alg. 3 in Supp. A alternates between the master problem and the test whether its solution Δ is the set of unenforced minimal FDs of some BCNF design. A solution Δ whose complement implies one of its elements is excluded by cuts of the second kind. Otherwise, the test searches the weakened FDs of Δ by the goal-directed search of Lem. A.16 in Supp. A, as lines 5 to 9 of Alg. 1 do, and if it fails, cuts of the third and fourth kind exclude Δ.

Theorem 5.7 (Correctness of Hit). *On input (R, Σ, w), Alg. 3 in Supp. A terminates and returns a BCNF design of (R, Σ) with loss $\lambda_w(R, \Sigma)$, and the weight $w(\Delta)$ of every solution Δ of a master problem is a lower bound on $\lambda_w(R, \Sigma)$. If the test of Δ is restricted to $\mathcal{E} = (\Sigma_a - \Delta)^+$, then the algorithm returns a plain BCNF design whose loss is minimum among all plain BCNF designs, with lower bounds on this minimum.*

On the running example, the only initial cut is $y_{ST \to Z} + y_{Z \to T} \geq 1$, the master problem returns one of the two FDs, and the first test succeeds, which certifies $\lambda(R, \Sigma) = 1$. Hence, Hit replaces the enumeration of Alg. 1 by integer programs for off-the-shelf solvers, and with the upper bounds of Ascend, an interrupted run reports a design and its gap to the optimum.

5.4 The Algorithm Ascend

Thm. 3.22 bounds the loss by the weight of the obstructions, and the uncritical design leaves every obstruction unenforced. Many obstructions can be enforced after all, at the price of the uncritical FDs that violate their schemata, as $ST \to Z$ can be enforced instead of $Z \to T$ in the running example. The algorithm Ascend climbs from the uncritical design, or from any BCNF design, towards smaller loss by such exchanges. It searches the uncritical sets of Cor. 3.19, so it never leaves the space of BCNF designs.

Alg. 4 in Supp. A starts from the set Ψ of uncritical minimal FDs of the given enforceable set and climbs by examining the unenforced minimal FDs g of Σ in decreasing order of their weight. It adds g to Ψ when the set stays uncritical, which decreases the loss. Otherwise, it exchanges the elements of Ψ that violate the schema of g for g when the exchange leaves the set uncritical and the minimal FDs of Σ that g brings in weigh at least as much as those that the violators take along. Exchanges of equal loss cross plateaus, with their violators tabu until the loss decreases: for $\Sigma'_n = \{S_iT \to Z \mid 1 \le i \le n\} \cup \{Z \to T\}$, exchanging $Z \to T$ for one $S_iT \to Z$ keeps the loss n of the uncritical design, and only then can the other $S_jT \to Z$ be added, down to loss one. When a climb ends, the algorithm keeps the set Ψ^* of least loss and kicks the search out of local optima at random, with the removed FDs tabu for the next climb. A move computes no atomic cover and at most 2^m closures of Ψ per element, where m is the size of the largest schema. The following proposition states the guarantee of the ascent.

Proposition 5.8 (Ascent). *Let $\mathcal{E} \subseteq \Sigma^+$ be enforceable for (R, Σ). Every set Ψ that Alg. 4 in Supp. A maintains on input $(R, \Sigma, w, \mathcal{E})$ is uncritical, and $w(\Sigma_a - \Psi^{*+})$ never increases. Hence, the algorithm terminates and returns a BCNF design of (R, Σ) with loss at most $w(\Sigma_a - \mathcal{E})$, and so does its interruption at any time.*

5.5 The Algorithm Repair

Alg. 1 searches the whole atomic cover. In practice, most schemata of a 3NF decomposition are already in BCNF, and only the critical schemata need treatment [69]. Repair exploits this observation.

Alg. 2 in Supp. A first computes with iConf [71] a lossless, dependency-preserving 3NF decomposition $\mathcal{D}_0$ that is in BCNF whenever possible. Every schema of $\mathcal{D}_0$ in BCNF contributes the key dependencies of its minimal keys, and every critical schema $(S, \Sigma[S])$ those of a design with minimum loss for $(S, \Sigma[S])$, which Alg. 1 computes on the small schema, where w applies since every minimal FD of $\Sigma[S]$ is a minimal FD of Σ. Let Γ denote the set of these key dependencies. The local designs interact through the FDs they enforce, so Repair tests whether Γ^+ is enforceable for (R, Σ). If so, it returns the lighter of the ascents of Alg. 4 in Supp. A from Γ^+ and from the uncritical design of Thm. 3.22, and otherwise it falls back to Drop on (R, Σ). The following theorem states the guarantee of Repair.

Theorem 5.9 (Correctness of Repair). *On input (R, Σ, w), Alg. 2 in Supp. A returns a BCNF design of (R, Σ). If Γ^+ is enforceable for (R, Σ), then the loss of the design is at most the minimum of $w(\Sigma_a - \Gamma^+)$ and $w(O)$, and otherwise it is $\lambda_w(R, \Sigma)$.*

If every schema of $\mathcal{D}_0$ is in BCNF, then Γ is a cover of Σ by dependency preservation, $\Gamma^+ = \Sigma^+$ is enforceable, and the loss is zero. In general, the local optima do not add up, since an FD that a critical schema leaves unenforced may be needed to derive minimal FDs embedded in no schema of $\mathcal{D}_0$. In the running example, if Drop leaves $ST \to Z$ unenforced on STZ, the assembled design also loses $C \to Z$, which the ascent adds back. Repair is optimal when the critical schemata of $\mathcal{D}_0$ share no attribute with each other or with any other schema except a minimal key of (R, Σ) (Prop. A.6 in Supp. A), but on the FD sets of Sec. 6 they share attributes with almost all others.

Parts of a schema that share only attributes that no FD determines can be designed separately, which makes MinLoss fixed-parameter tractable in the size of the largest part for plain designs (Thm. A.13 in Supp. A). On the FD sets of Sec. 6, every discovered FD set and TPC-DS form a single part, while the meaningful FDs of food, price paid and hospital split into 8 to 12 parts of at most ten attributes, on which Cor. A.14 in Supp. A computes the minimum loss of plain designs part by part.

Certainty degrees [36], errors of approximate FDs [33], rankings of discovered FDs [64] and violation counts serve as weights.

6 EXPERIMENTS

We evaluate our framework along four research questions. RQ1: How large is the dependency loss of the designs that practice and the state of the art produce, and how much smaller is the minimum loss? RQ2: How efficient are Drop, Hit and Repair on discovered FD sets, and how far is Repair from the optimum? RQ3: How do designs with minimum loss affect insertion, foreign-key and query costs? RQ4: How much integrity risk do the unenforced FDs carry on real data, on the universal relation and on the stored tables?

Set up. All algorithms are implemented in Python 3.14, Hit solves its master problems with HiGHS [25], the database experiments run on PostgreSQL 18.4, and every run is single-threaded on a machine with an AMD Ryzen 7 9700X processor and 64 GB of main memory. The FD sets are from ten data sets of the FD discovery benchmark [48][1] and the lineitem table of TPC-H. Tab. 2 lists them, lineitem and uniprot without records, and the artifact repository holds the code, all records and the full protocol.

Budgets. Every FD set had four hours. The exact searches of Drop and Repair had budgets of five minutes to an hour, with caps on the weakened FDs. Hit had an hour per FD set and weighting, and it stops when its bound reaches the least loss of the designs of Repair and Drop. A search that hits its budget or a cap reports its design as unproved, with its certified lower bound. The algorithms then return the lightest design Ascend reaches from the uncritical design and a greedy one, with fewer schemata on ties.

Competitors. We compare eight key designs. The key-only design of Prop. 3.12 keeps R with its minimal keys, and 3NF synthesis enters with iConf [71] and with the classical synthesis [8] in the implementation against which iConf was evaluated. The classical BCNF decomposition [60] splits by the first violating FD, and a greedy BCNF decomposition by the violating FD whose split leaves the fewest minimal FDs unenforced. Hybrid [65] decomposes the schemata of a 3NF synthesis along the non-key FDs that permit more redundancy than their keys, and the last two designs are those of Repair and Drop. Every design declares the minimal keys of its

[1] https://hpi.de/naumann/projects/repeatability/data-profiling/fds.html

Table 2: Data Sets (Attributes $|R|$, Minimal FDs $|\Sigma_a|$, Critical Schemata of iConf), Dependency Loss of the Designs with the Certified Lower Bounds of Drop and Hit, Schemata, and Running Times (unit weights; bold: best, *: not proved minimal, †: unsound, ‡: Hybrid without cardinality constraints from data, dash: no records or not run)

						dependency loss									bound		schemata		time (s)	
								3NF		BCNF										
Data set	records	$\|R\|$	$\|\Sigma_a\|$	keys	crit.	ν	$\|O\|$	iConf	classical	classical	greedy	Hybrid	Repair	Drop	Drop	Hit	3NF	Repair	Repair	Drop
abalone	4,177	9	137	29	5	31	11	11	26	95	92	24	**8**	**8**	8	7	23	15	5.3	4.9
breast	691	11	46	2	2	46	3	3	3	38	37†	3	**3**	**3**	3	3	38	38	0.9	0.1
bridges	108	13	142	5	16	108	28	19	22	103	99†	20	**11**	**11**	11	11	57	58	11	0.0
echo	132	12	526	72	12	125	29	63	52	463	392	84	**11**	**11**	11	11	79	61	26	1,887
adult	32,537	15	78	2	0	78	0	0	0	68	67	0	**0**	**0**	0	0	46	42	0.0	0.0
letter	18,668	17	61	1	0	61	0	0	0	60	60	0	**0**	**0**	0	0	62	62	0.0	0.0
ncvoter	1,000	18	757	69	28	402	88	89	88	704†	668†	266	**48***	**48***	20	24	172	158	2,820	2,727
weather	262,920	18	918	2	117	894	192	147	149	869	858	234	**84***	**84***	53	77	407	453	5,289	4,091
lineitem	–	16	3,687	370	38	788	145	516	528	3,487	3,375†	550‡	**69***	–	25	33	594	450	7,272	–
uniprot	–	29	3,702	10	214	3,579	342	810	816	3,552	3,526	810‡	**136***	–	61	91	1,675	1,796	6,796	–
hepatitis	155	20	8,250	348	567	7,220	2,007	1,554	1,841	8,211†	7,995	2,578	**992***	–	192	403	1,677	1,962	7,306	–

schemata, and only the key-only design and the last two are BCNF designs in general, so the loss of the others also counts FDs that their keys imply only across schemata, which favours them. The original relation with every FD enforced is the reference.

6.1 RQ1: Dependency Loss in Practice

Tab. 2 shows that dependency loss is the rule for the designs that practice and the classical algorithm produce. The closure Σ^+ is enforceable on adult and letter only, where iConf is in BCNF and Repair and Drop find a design with loss zero within a second. On the nine others, the key-only design loses every non-key minimal FD, 7,220 of the 8,250 minimal FDs of hepatitis. Classical BCNF decomposition leaves more than half of the minimal FDs unenforced on all nine sets and on adult and letter, where loss zero is possible, and it returns unsound designs on two sets. Choosing every split greedily saves at most a sixth of this loss, since every split separates the attributes of the FDs that span both parts, and Hybrid loses more than the keys of iConf on seven of the nine sets.

The keys of 3NF synthesis, in contrast, already enforce most of the minimal FDs. Those of iConf leave at most 22% unenforced, and those of the classical synthesis more on six of the nine sets. On eight of the nine sets, the BCNF design of Repair leaves fewer FDs unenforced than the keys of either 3NF synthesis, by a factor of 7.5 on lineitem. The recovered FDs cost few schemata, at most 17% more than iConf needs.

Discovered FDs include accidental ones, so Tab. 9 in Supp. A adds four sets of meaningful FDs: those of HoloClean [50] on food inspections and hospitals, the rules of Royal Mail [52] on the UK Price Paid Data [24], and the historical dimensions of TPC-DS [59]. Each contains cycles such as *City*, *State*, *Address* $\rightarrow$ *Zip* and *Zip* $\rightarrow$ *City*, *State*, and Repair loses one FD per cycle, the minimum for BCNF designs. The keys of 3NF synthesis lose the other direction and every FD derived through it, but tie with Repair on TPC-DS.

6.2 RQ2: Efficiency and Optimality

The critical schemata themselves are easy (Tab. 2): they have at most eleven attributes, and the local searches prove their losses minimal, almost all within a second. Hence, Repair returns within a second on adult and letter and within half a minute on breast, abalone, bridges and echo, and what costs hours is the interaction of the local designs. Where the fallback does not finish, Repair takes 0.8 to 2.0 hours, as Drop and the ascents use their budgets, and its design lacks the optimality of Thm. 5.9 (asterisks in Tab. 2).

Local optima do not add up in practice either: on echo, the assembled design loses 150 minimal FDs while the local losses add up to 12, and Ascend brings the loss down to the proved minimum of 11 (Tab. 5 in Supp. A). Where Γ^+ is not enforceable, the ascents lower the loss of the uncritical design of (R, Γ^+) by more than 40%, and every design of Repair with positive loss comes from an ascent.

The weakened FDs that optimal designs need in theory were not needed where optimality is proved. Drop proves the loss minimal on six of its eight FD sets with at most 1,000 minimal FDs (no asterisk in Tab. 2), and every proved optimal design is plain.

Repair attains the proved minimum on abalone, breast, bridges and echo, and undercuts the obstruction bound of Thm. 3.22 on every set with positive loss but breast (columns $|O|$, Repair and bound of Tab. 2). On the five largest sets, however, the bounds of Drop are 1.6× to 5.2× smaller than the loss of Repair.

Hit narrows these gaps to a factor of 1.1 to 2.5 within an hour (column Hit of Tab. 2). Its bounds rise fast and then stall on plateaus of equally light sets: on weather, the bound reaches 77 within a second, and the remaining hour rejects 1,553 sets of this weight one by one. A plateau also stops Hit at 7 on abalone, where Drop proves the minimum of 8 in five seconds, while on echo Hit certifies the loss of Repair in 0.3 seconds, where Drop needs half an hour. Under the violation weights of RQ4, Hit proves six of the eleven weighted designs of Repair optimal, and it leaves gaps of at most 16% on echo, weather and TPC-DS and of about a factor of two on ncvoter and hepatitis. Hence, the loss of Repair is at most 2.5× the minimum on every FD set and weighting of our experiments.

6.3 RQ3: Operational Costs

Fig. 2a and Tab. 7 in Supp. A show what the designs of RQ1 cost to maintain.

Operational set up. RQ3 and RQ4 use the data sets with records in Tab. 2 and Tab. 9 in Supp. A, cleaned until every rule held. Every

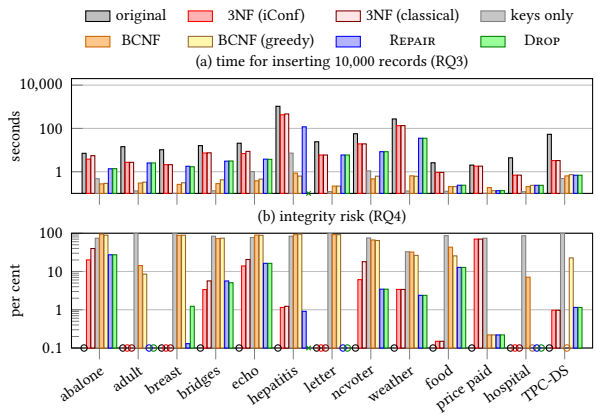


Figure 2: Insertion Time and Integrity Risk of the Designs (circle: below 0.1%, cross: not computed)

Table 3: Keys of iConf (iC) against Repair: Loss, Time (s) for Inserting 10,000 Records with Foreign Keys, Time (ms) for Reconstructing a Record from a Key, Integrity Risk on the Universal Relation, and Harmful Updates that Repair Misses on the Stored Tables

	loss		+FK (s)		query (ms)		risk (%)		stored (%)	
Data set	iC	Repair	iC	Repair	iC	Repair	iC	Repair	viol.	anom.
abalone	11	8	2.34	1.93	0.08	0.05	20.1	27.4	20.3	41.8
adult	0	0	4.88	4.29	0.22	0.21	<0.1	<0.1	0.0	13.8
breast	3	3	2.73	2.56	0.12	0.12	<0.1	0.1	0.0	52.6
bridges	19	11	4.47	4.49	0.22	0.23	3.4	5.7	1.1	28.8
echo	63	11	7.72	8.01	0.33	0.27	14.0	16.3	9.5	31.6
hepatitis	1,554	992	168	194	16.8	20.6	1.2	0.9	0.1	7.7
letter	0	0	9.13	9.13	0.39	0.39	<0.1	<0.1	0.0	92.1
ncvoter	89	48	18.5	16.6	0.63	0.57	6.1	3.5	0.7	32.9
weather	147	84	88.2	86.1	3.29	4.50	3.4	2.4	0.0	9.3
food	6	1	3.04	0.37	0.06	0.03	0.2	12.8	12.6	36.6
price paid	3	1	4.40	0.16	0.08	0.02	70.3	0.2	0.2	49.4
hospital	6	1	2.29	0.33	0.04	0.02	<0.1	<0.1	0.0	65.7
TPC-DS	2	2	26.5	0.85	0.44	0.07	1.0	1.2	1.0	87.7

design has a table per schema with the projection of the data, a `UNIQUE` constraint per declared key, and, on the original relation and the critical schemata of 3NF synthesis, an exclusion constraint per non-key FD of a minimal cover. Each synthetic record copies a stored record on a closed set of attributes and carries fresh values elsewhere, so no insertion is rejected while the constraints find tuples to compare with. We time the insertion of 10,000 such records in one transaction, five times on fresh tables, within two hours per design. We repeat these insertions with a foreign key from every table to every other table whose declared key lies in its schema, checked at the commit, and time the reconstruction of a record from a minimal key of R through the natural join of all tables.

Every exclusion constraint costs 0.24 to 0.69 seconds per 10,000 records, on every data set (Tab. 7 in Supp. A). Hence, the original relation costs 15× to over 2,000× the time of the key-only design, since every constraint probes its GiST index for the tuples that agree with the new tuple on the left-hand side, while a key costs one B-tree probe.

3NF synthesis pays the same price on its critical schemata, and the designs of Repair insert the same records at least 1.9× as fast as either 3NF synthesis on nine of the eleven sets where it has critical schemata: the non-key FDs that 3NF synthesis keeps at the logical level become its overhead at the operational level.

Keys are not free either: the designs of Repair and Drop pay 0.04 to 0.1 seconds per schema and 10,000 records, but still a fourth of the time of the original relation or less on every set. Fig. 3 in Supp. A shows that, on the six largest sets with positive loss, the designs of Repair lose fewer FDs than the keys of 3NF synthesis and take at most 9% longer, while the designs that enforce every FD take 1.8× to 9× as long.

Tab. 3 and Tab. 10 in Supp. A show that foreign keys and queries charge these schemata again, so the designs of Repair cost about as much as the keys of 3NF synthesis on the discovered sets and far less on the meaningful ones. Every foreign key adds 12 to 53 ms per 10,000 records. Hence, the 2,045 foreign keys of 3NF synthesis on TPC-DS multiply its insertion time by nine, while the nine of Repair add a fourth. A point query follows one index per schema, so it takes under a millisecond with up to 200 schemata and 17 to 21 ms with the 1,600 of hepatitis. The exception is the BCNF decomposition of price paid, which reaches the postcode only through address attributes that none of its declared keys covers, so that every lookup scans a table.

6.4 RQ4: Integrity Risk

Integrity set up. For every data set we drew 10,000 random updates, each replacing the value of a random attribute in a random stored record by the value of that attribute in another random record. An update is *harmful* if the updated relation violates some FD of Σ, and a design *catches* it if the updated relation violates one of its declared key dependencies, that is, an enforced FD. The *integrity risk* of a design is the share of harmful updates it does not catch, averaged over five seeds. The seven key designs of RQ3 leave the non-key FDs of 3NF synthesis to the application, as in Sec. 1, and the original relation has risk zero by construction.

Fig. 2b shows that the designs that lose most FDs in RQ1 miss most harmful updates. The key-only design misses more than half of them on 12 of the 13 sets, and the two BCNF decompositions miss 64% to 95% on seven.

The designs that lose few FDs miss few updates. The keys of both 3NF syntheses and the designs of Repair and Drop miss less than 7% of the harmful updates on every discovered set but abalone and echo (and ncvoter for the classical synthesis). On abalone, only about thirty updates are harmful (Tab. 7 in Supp. A). Among these designs, however, the number of unenforced FDs no longer predicts the risk. Repair loses at most as many FDs as the keys of iConf, but it misses more updates on four of the seven discovered sets with positive loss. On echo, Repair loses 11 FDs and misses 16%, while the keys of iConf lose 63 and miss 14%. Indeed, one FD with a two-attribute left-hand side accounts for three in four of the updates that Repair misses on echo, so the risk behaves like a weighted loss in which an FD weighs as much as the updates that violate it.

This suggests *violation weights*, which count the updates that violate an FD, and which turn the loss into an upper bound on the harmful updates that a design misses by Prop. 3.13. We gave every minimal FD the weight one plus the number of updates that violate it among 50,000 random updates drawn with five other seeds, ran

Repair again under the budgets of RQ2, and evaluated its designs on the updates of Fig. 2b. Tab. 8 in Supp. A shows that the weighted designs miss fewer harmful updates than those for unit weights on six of the seven discovered sets, and on echo Repair now misses 2.5% instead of 16%. They keep the loss on abalone, breast and bridges and trade light FDs for heavy ones elsewhere. With the violation shares of the evaluation updates as p, the bound λ_p ranks the designs by risk better than the unit loss on eight of the thirteen sets and worse only on breast and bridges, but it is loose by over 1,000× for Repair on hepatitis, where updates violate many FDs at once. The gain of the weighted designs survives drift (last two columns of Tab. 8 in Supp. A): when the updated attribute follows a Zipf distribution at a total variation of up to 0.7 from the updates behind the weights, the weighted designs miss fewer on the same six sets.

The meaningful FDs confirm the role of the weight and show its limit (Fig. 2b, last rows of Tab. 8 in Supp. A). On price paid, Repair misses 0.2% of the harmful updates and the keys of iConf 70%, since updates violate the FDs from Postcode that iConf loses far more often than the address rule that Repair loses. On TPC-DS, both miss about 1%, all violations of *i_item_sk* → *i_item_id*, which Repair enforces under the weights instead of the other direction of the item cycle, and then misses none. On food, however, Repair misses 13% and the keys of iConf 0.2%, since almost every update of an address violates the address rule that Repair loses, while the key {*Address*, *Zip*} of iConf enforces *Zip* → *City*, *State* wherever an address recurs.

An unenforced FD inside one schema can still be enforced by an exclusion constraint at the price of RQ3, while one across schemata needs a join. Most unenforced FDs of Repair lie inside one schema on seven of the eleven sets with positive loss, but most of those of the BCNF decompositions and Hybrid span several schemata on six of the seven discovered sets (Tab. 6 in Supp. A).

So far, every update changed one record of the universal relation. A database, however, updates the rows of its tables, and an SQL `UPDATE` of a row shared by several records changes all of them, an update anomaly that a key of the table catches only if the updated attribute belongs to the key. Tab. 11 in Supp. A and the last columns of Tab. 3 split the harmful updates that no key catches on the stored tables into those that leave a violation and those that change other records instead. The violations follow the loss: an update that changes no other record is caught on the tables whenever it is caught on the universal relation, and the designs of Repair leave at most 1.1% on every discovered set but abalone and echo. The anomalies follow the schemata: Repair lets 8% to 92% of the harmful updates change other records, within three points of the keys of iConf on ten of the thirteen sets, and on TPC-DS such an update changes about 120,000 sales. Only the key-only design has no anomalies, and it misses a third to all of the harmful updates.

In summary, the risk follows the loss across the designs and the weight of the unenforced FDs among those with small loss, while decomposition adds anomalies that depend on the schemata, and most FDs that Repair leaves unenforced need no join.

7 RELATED WORK

Fundamentally, our work bridges the lossless, dependency-preserving decompositions into BCNF of Osborn [46] and Köhler [27], which exist only for some schemata, with the lossless decompositions into BCNF of Tsou et al. [60], which preserve an arbitrary subset of the FDs, and level zero of k-loss BCNF coincides with the former. Normal forms between 3NF and BCNF [35, 67] enforce all non-key FDs they admit, attribute splitting [40] changes the schema rather than the enforced constraints, and Kolahi et al. [30, 31] quantify the redundancy that dependency preservation costs, while we quantify the dependencies that the absence of redundancy costs. Neither domain-key normal form [19], the design-stage pruning of constraints [7] nor the reflections on BCNF [9, 11, 34] count the FDs that keys leave unenforced. Soundness formalizes the observation that the losslessness of a decomposition is only as good as the constraints its schemata enforce [9, 51], and Thm. 3.24 characterizes for key designs the independence that independent and constant-time-maintainable schemata study [14, 22, 23, 53, 63].

Algorithmically, the closest work is the Hybrid algorithm of Wei et al. [65], which enforces the FDs it loses by assertions over joins, while our algorithms minimize the weight of the lost FDs and leave them to the application. The objectives of Zhang et al. [68, 71] assume full dependency preservation, certainty degrees [36] can serve as weights, Normalize [49] and RelaxRD [18] choose the FDs to decompose on heuristically or for storage, and approximate acyclic schemata [26] and constraint repair [10, 15] fit schemata or FDs to data, so none of them measures the FDs that a design leaves unenforced.

Operationally, non-key FDs enforced by triggers are the bottleneck of integrity maintenance [69, 70], and lost FDs enforced by assertions cost more than not decomposing [65]. Our experiments quantify the third option, which leaves the fewest FDs unenforced, and our work is the first to minimize the FDs that a lossless BCNF design leaves outside the guarantee of its keys.

8 CONCLUSION AND FUTURE WORK

We have introduced BCNF designs, which enforce FDs through the keys of their schemata only, and their dependency loss, the number of minimal FDs they leave unenforced. Three requirements single out the loss, violation weights turn it into a bound on integrity risk, and a polynomial-time test decides when keys alone keep the stored relations globally consistent. Loss zero recovers the classical dependency-preserving BCNF decompositions, membership in k-loss BCNF is NP-hard for every fixed k, and under the atomic cover, deciding loss one is coNP-hard and the minimum cannot be approximated within a factor below two. Drop and Hit minimize the loss, Ascend lowers it for any design, and Repair treats only the critical schemata of 3NF synthesis. In our experiments, Repair leaves fewer FDs unenforced than the keys of 3NF synthesis on all sets with positive loss but two, within 2.5× of the minimum and at a fraction of the cost of enforcing every FD, and violation weights lower its integrity risk further.

Future work will settle the exact complexity of MinLoss, which lies between DP and Σ_2^p for plain designs under the atomic cover, minimize the loss among independent designs, parameterize designs jointly by loss and redundancy [65], and study data models where only keys are enforced natively, such as property graphs [56]. In summary, counting the FDs that keys leave unenforced turns the choice between 3NF and BCNF into a matter of degree.

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A FURTHER RESULTS

This Supplementary Material collects the results that Secs. 3, 4, 5 and 6 refer to.

Unsound BCNF decompositions. The following proposition formalizes the observation from Ex. 3.5.

Proposition A.1 (Unsound BCNF decompositions). *There is a schema (R, Σ) and a lossless decomposition $\mathcal{D}$ of (R, Σ) into BCNF, computed by the classical decomposition algorithm, such that the key design $(\mathcal{D}, \kappa)$ that declares the minimal keys of $(S, \Sigma[S])$ on every $S \in \mathcal{D}$ is not sound.*

Inheritance. The following lemma shows that an uncritical FD stays uncritical when other FDs are left unenforced, which Thm. 3.22 and Alg. 4 rest on.

Lemma A.2 (Inheritance). *Let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set that contains every uncritical minimal FD of Σ. Then every uncritical minimal FD of Σ is an uncritical minimal FD of $\mathcal{E}$.*

Verification. The hardness of Thm. 4.5 rests on the representation of a design by its declared keys. A design is locally enforced iff, for every minimal FD $X \to A$ of $\mathcal{E}$ and every $S \in \mathcal{D}$ with $XA \subseteq S$, the set X contains some $K \in \kappa(S)$, which takes time polynomial in the size of the atomic cover of $\mathcal{E}$.

Shapes of weakened FDs. The following proposition shows that the enumeration of line 5 of Alg. 1 can be confined neither to one weakened form per unenforced FD nor to the left-hand sides of minimal FDs.

Proposition A.3 (Shapes of weakened FDs). *There is a schema (R, Σ) with $\lambda(R, \Sigma) = 1$ such that every BCNF design of (R, Σ) with loss one enforces two weakened forms of the same unenforced minimal FD. There is a schema (R, Σ) with $\lambda(R, \Sigma) = 2$ such that every BCNF design of (R, Σ) with loss two enforces a weakened form $X' \to A$ of an unenforced minimal FD where X' is not contained in the left-hand side of any minimal FD of Σ.*

The closures of the left-hand sides of critical minimal FDs do not confine the weakened FDs either. For $\Psi = \Sigma_a - \Delta$, call a weakened FD $X' \to A$ *local* iff $X' \subseteq Y^+_\Psi$ for some minimal FD $Y \to B$ of Ψ^+ that is critical for Ψ^+. Only the critical minimal FDs of Ψ^+ obstruct the enforceability of Ψ^+, and local weakened FDs suffice in the two schemata of Prop. A.3, but the following proposition shows that they do not suffice in general.

Proposition A.4 (Local weakened FDs). *There is a schema (R, Σ) with a set $\Delta \subseteq \Sigma_a$ and weights w such that $|\Delta| = \lambda(R, \Sigma)$, every BCNF design of (R, Σ) with loss $\lambda_w(R, \Sigma)$ leaves exactly Δ unenforced, and every BCNF design of (R, Σ) that leaves exactly Δ unenforced enforces a weakened FD that is not local.*

Hence, line 5 of Alg. 1 cannot be confined to local weakened FDs. The search of Lem. A.16 needs no such restriction, since it measures closures against the current set $\mathcal{E}$ rather than Ψ. Under unit weights, the schema of the proof has loss three as well, but a plain design attains it by leaving $BDE \to A$, $DF \to C$ and $AF \to C$ unenforced, so the proposition needs the weights. Under unit weights, whether such a design always exists is open. Local weakened FDs attain the minimum loss on every random schema of our machine checks that needs weakened FDs (**Machine checks** below). The weighted counterexample does not lift to unit weights by the natural amplification: making a heavy FD h expensive to lose by adding FDs that imply h also provides derivations of h from uncritical FDs in the test of Thm. 3.18, which removes the need for the non-local weakened FD.

Tightness of the cover condition. The cover condition of Lem. 5.3 cannot be strengthened to confine Δ to the critical minimal FDs and the FDs that violate their schemata.

Proposition A.5 (Tightness of the cover condition). *There is a schema (R, Σ) such that every BCNF design of (R, Σ) with minimum loss leaves a minimal FD of Σ unenforced that is uncritical and violates the schema of no critical minimal FD of Σ.*

The optimal designs of this schema are not plain, since every plain design leaves at least four minimal FDs unenforced, while the design of the proof leaves three.

Isolated critical schemata. The following proposition identifies a case in which Alg. 2 is optimal.

Proposition A.6 (Isolated critical schemata). *If the critical schemata of $\mathcal{D}_0$ share no attribute with each other or with any other schema of $\mathcal{D}_0$, except for a schema that is a minimal key of (R, Σ), then line 9 of Alg. 2 succeeds and the returned design has loss $\lambda_w(R, \Sigma)$.*

In this case, (R, Σ) is the attribute-disjoint union of the critical schemata and of the remaining attributes, on which the other schemata form a design with loss zero, so the local optima add up to the global optimum by Lem. 4.2, and a key schema only joins the parts.

Branches of Repair. Tab. 5 lists for every FD set of Sec. 6 with positive loss the loss of every design that Alg. 2 examines before it returns the lightest. When the assembled set Γ^+ is enforceable, these are the design of Γ, the uncritical design (column O) of Thm. 3.22 and the ascents of Alg. 4 from both. Otherwise, the fallback examines the uncritical design of (R, Γ^+) in place of the design of Γ, a greedy enforceable subset of Γ^+, the global Drop with the remaining budget, which includes its own ascents, and the ascent from the uncritical design of (R, Γ^+). A dash marks a step that its budget cut or that Drop had already run. The assembled design alone is never the lightest: on abalone and echo the ascent from it recovers the minimal FDs that no schema of iConf embeds, on the six sets of the fallback the time-limited Drop or the ascent from the uncritical design of (R, Γ^+) wins, and on breast the uncritical design of (R, Σ) is optimal as well.

The price of plainness. The following proposition bounds the price of confining the search of MinLoss to plain designs and shows that the bound is attained.

Proposition A.7 (Price of plainness). *Let $\lambda^{\mathrm{p}}_w(R, \Sigma)$ denote the minimum loss of the plain BCNF designs of (R, Σ), and let O be the set of obstructions of (R, Σ). Then $\lambda_w(R, \Sigma) \le \lambda^{\mathrm{p}}_w(R, \Sigma) \le w(O)$. For every $m \ge 2$ there is a schema (R_m, Σ_m) with $m + 3$ attributes such that $\lambda(R_m, \Sigma_m) = 1$ and $\lambda^{\mathrm{p}}(R_m, \Sigma_m) = |O| = \binom{m}{\lfloor m/2 \rfloor}$, and every BCNF design of (R_m, Σ_m) with loss one enforces $\binom{m}{\lfloor m/2 \rfloor}$ weakened forms of the same unenforced minimal FD.*

The upper bound holds since the uncritical design is plain. As $\lambda(R, \Sigma) \ge 1$ whenever $O \neq \emptyset$, plain designs lose at most $|O|$ times the minimum under unit weights, and (R_m, Σ_m) attains this factor.

Table 4: Notation

Symbol	Meaning	Symbol	Meaning
Σ, Σ^+, Σ_a	FD set, its closure, its atomic cover of minimal FDs	X^+_Σ	attribute closure of X under Σ
$(\mathcal{D}, \kappa)$	key design with decomposition $\mathcal{D}$ and declared keys $\kappa(S)$ for every $S \in \mathcal{D}$ (Def. 3.1)	$\mathcal{E}$	enforced FDs of a design, the closure of its key dependencies
λ_w, λ	dependency loss with weights w and with unit weights (Def. 3.10)	$\nu(R, \Sigma)$	number of non-key minimal FDs of Σ (Prop. 3.12)
critical	a schema of a 3NF decomposition that is not in BCNF (Sec. 2), or a minimal FD $X \to A$ of $\mathcal{E}$ such that $(XA, \mathcal{E}[XA])$ is not in BCNF (Def. 3.17)	$\mathcal{U}$, O	uncritical minimal FDs of Σ, obstructions $\Sigma_a - \mathcal{U}^+$ (Def. 3.21)
Ψ	uncritical set, which generates an enforceable FD set (Cor. 3.19)	Δ, W	minimal FDs of Σ left unenforced, weakened FDs enforced instead (Alg. 1)
$C(\Delta)$, $Q(g, \Delta)$	critical minimal FDs whose schemata Δ relieves, requirement set of g (Lem. 5.3, Prop. 5.4)	$\mathcal{D}_0$, Γ	3NF decomposition and key dependencies of the local designs in Repair (Alg. 2)

Table 5: Losses of the Designs that Repair Examines, Whether the Assembled Set Γ^+ Is Enforceable, and the Design It Returns (dash: not run)

Data set	Γ^+ enf.	Γ^+	O	greedy	Ascend(Γ^+)	Ascend(O)	Drop	returned
abalone	yes	22	11	–	8	8	–	Ascend(Γ^+)
breast	yes	5	3	–	3	3	–	Ascend(Γ^+)
bridges	no	30	28	30	11	11	11	Ascend(Γ^+)
echo	yes	150	29	–	11	11	–	Ascend(Γ^+)
ncvoter	no	248	88	241	65	–	48	Drop
weather	no	195	192	166	91	–	84	Drop
lineitem	no	424	145	423	70	–	69	Drop
uniprot	no	856	342	780	438	–	136	Drop
hepatitis	no	2,463	2,007	–	1,361	–	992	Drop

Table 6: Unenforced Minimal FDs of the Eight Key Designs that Span Several Schemata (the Rest of the Loss of Tabs. 2 and 9 Lies Inside One Schema), and the Sets on which the Design Is Independent

Data set	key-only	BCNF	greedy	iConf	classical	Hybrid	Repair	Drop
abalone	0	95	92	2	13	17	1	1
breast	0	38	37	1	1	1	1	2
bridges	0	103	99	7	9	13	2	3
echo	0	463	392	52	39	81	1	1
ncvoter	0	704	668	59	51	248	19	19
weather	0	869	858	54	55	209	13	13
hepatitis	0	8,211	7,995	961	1,187	2,230	550	–
food	0	4	4	4	4	–	1	1
price paid	0	2	1	0	0	–	1	1
hospital	0	4	1	4	4	–	1	1
TPC-DS	0	180	51	0	0	–	0	0
independent	11/11	11/11	10/11	2/11	2/11	0/7	3/11	3/10

Its FDs are $B \to A$ and, for every set Y of $\lfloor m/2 \rfloor$ attributes among $C_1, \ldots, C_m$, the FDs $BY \to E$ and $AYE \to B$, so that $m = 2$ yields the first schema of Prop. A.3. An optimal design gives up $B \to A$ and enforces $BY \to A$ on every schema $ABYE$, while a plain design gives up $AYE \to B$ or $BY \to E$ for every Y. Hence, the restriction of Drop to plain designs in Thm. 5.5 can lose a factor exponential in the number of attributes, although at most linear in the size of the atomic cover, as $O \subseteq \Sigma_a$.

Polynomial covers. The schemata (R_m, Σ_m) are their own atomic covers, so their size is exponential in the number of attributes as well, and whether a cover of polynomial size can force an exponential price of plainness is open. The natural attempt fails in an instructive way. For $m \geq 1$, let Σ'_m consist of $B \to A$, of $P_i \to Q_i$ and $Q_i \to P_i$ for $i = 1, \ldots, m$, and of $BP \to E$ and

Table 7: Exclusion Constraints (excl.), Mean Time (s) for Inserting 10,000 Records (keys: Key-Only Design, BCNF: Classical BCNF Decomposition, Other Designs in Fig. 2a), and Harmful Updates among 10,000 Random Updates

Data set	original		3NF (iConf)		3NF (classical)		keys	BCNF	Repair	harmful
	excl.	s	excl.	s	excl.	s	s	s	s	updates
abalone	25	7.10	9	3.81	16	5.53	0.48	0.28	1.37	30
adult	46	14.3	0	2.72	0	2.72	0.13	0.30	2.54	1,546
breast	40	10.4	2	2.12	2	2.12	0.09	0.26	1.79	1,655
bridges	65	16.1	19	7.34	20	7.48	0.13	0.28	3.08	5,825
echo	80	20.8	13	6.90	20	8.69	1.02	0.38	3.81	3,871
hepatitis	2,273	1,054	1,094	427	1,242	465	7.25	0.87	119	9,071
letter	61	24.2	0	5.92	0	5.92	0.12	0.22	5.92	244
ncvoter	216	56.9	46	19.4	50	19.4	1.10	0.47	8.43	3,745
weather	516	274	222	134	226	135	0.13	0.65	35.2	4,963
food	7	2.60	2	0.93	2	0.93	0.12	0.21	0.24	4,652
price paid	4	2.02	3	1.81	3	1.81	0.10	0.19	0.13	2,684
hospital	16	4.39	2	0.70	2	0.70	0.12	0.21	0.24	8,744
TPC-DS	77	53.5	2	3.28	2	3.28	0.49	0.64	0.69	7,684

Table 8: Unenforced Minimal FDs and Integrity Risk of Repair under Unit Weights and under Violation Weights w, without and with Drift

Data set	λ		risk (%)		drift (%)	
	unit	w	unit	w	unit	w
abalone	8	8	27.4	15.0	27.2	15.1
breast	3	3	0.1	0.3	0.1	0.2
bridges	11	11	5.7	5.2	5.4	4.3
echo	11	15	16.3	2.5	10.9	2.3
hepatitis	992	1,070	0.9	0.8	0.8	0.7
ncvoter	48	58	3.4	1.7	5.2	1.7
weather	84	145	2.4	1.5	2.2	1.3
food	1	1	12.8	12.8	16.0	16.0
price paid	1	1	0.2	0.2	0.2	0.2
hospital	1	1	0.0	0.0	0.0	0.0
TPC-DS	2	5	1.2	0.0	0.2	0.0

Table 9: Dependency Loss on Meaningful FDs (bold: best, †: unsound)

					3NF		BCNF			
Data set	records	$\lvert R\rvert$	$\lvert\Sigma_a\rvert$	ν	iConf	classical	classical	greedy	Repair	Drop
food	276,716	17	27	11	6	6	4	4	**1**	**1**
price paid	926,485	16	19	4	3	3	2†	1	**1**	**1**
hospital	1,000	19	26	23	6	6	4	1	**1**	**1**
TPC-DS	2,685,687	95	373	316	2	2	180†	51†	**2**	**2**

Table 10: Mean Time (s) for Inserting 10,000 Records without and with Foreign Keys (+FK), and Mean Time (ms) for Reconstructing a Record from a Key of R (keys: Key-Only Design)

Data set	keys			3NF keys (iConf)			BCNF			Repair		
	s	+FK s	ms	s	+FK s	ms	s	+FK s	ms	s	+FK s	ms
abalone	0.48	0.47	0.01	1.58	2.34	0.08	0.28	0.34	0.02	1.37	1.93	0.05
adult	0.13	0.13	0.02	2.72	4.88	0.22	0.30	0.47	0.03	2.54	4.29	0.21
breast	0.09	0.09	0.01	1.79	2.73	0.12	0.26	0.51	0.02	1.79	2.56	0.12
bridges	0.13	0.13	0.02	2.93	4.47	0.22	0.28	0.35	0.03	3.08	4.49	0.23
echo	1.02	0.99	0.02	3.97	7.72	0.33	0.38	0.52	0.03	3.81	8.01	0.27
hepatitis	7.25	7.06	0.02	109	168	16.8	0.87	1.16	0.09	119	194	20.6
letter	0.12	0.12	0.02	5.92	9.13	0.39	0.22	0.25	0.03	5.92	9.13	0.39
ncvoter	1.10	1.08	0.01	8.73	18.5	0.63	0.47	0.63	0.03	8.43	16.6	0.57
weather	0.13	0.14	0.02	34.2	88.2	3.29	0.65	1.12	0.06	35.2	86.1	4.50
food	0.12	0.13	0.01	0.55	3.04	0.06	0.21	0.29	0.02	0.24	0.37	0.03
price paid	0.10	0.10	0.02	0.52	4.40	0.08	0.19	0.21	15.6	0.13	0.16	0.02
hospital	0.12	0.13	0.01	0.47	2.29	0.04	0.21	0.28	0.02	0.24	0.33	0.02
TPC-DS	0.49	0.48	0.03	2.80	26.5	0.44	0.64	0.76	0.63	0.69	0.85	0.07

Table 11: Integrity Risk (%) on the Stored Tables: Harmful Updates that No Key Catches and that Leave a Violation (viol.) or Change Other Records (anom.)

Data set	key-only	iConf		BCNF		Repair	
		viol.	anom.	viol.	anom.	viol.	anom.
abalone	73.9	19.7	41.8	83.4	16.1	20.3	41.8
adult	100.0	0.0	13.8	14.3	0.7	0.0	13.8
breast	98.4	0.0	52.6	83.7	9.8	0.0	52.6
bridges	83.6	1.1	28.9	61.0	14.3	1.1	28.8
echo	77.6	9.6	31.6	83.3	9.3	9.5	31.6
hepatitis	83.8	0.1	7.8	84.5	9.3	0.1	7.7
letter	100.0	0.0	92.1	84.7	15.2	0.0	92.1
ncvoter	75.1	2.4	32.5	65.1	2.6	0.7	32.9
weather	32.8	0.0	10.0	31.9	0.9	0.0	9.3
food	86.9	0.2	49.0	25.2	36.6	12.6	36.6
price paid	74.3	70.3	3.0	0.2	49.4	0.2	49.4
hospital	86.0	0.0	72.8	7.0	64.1	0.0	65.7
TPC-DS	100.0	1.0	87.7	0.0	90.5	1.0	87.7

$APE \to B$, where $P = P_1 \cdots P_m$, a cover of $2m + 3$ FDs over $2m + 3$ attributes. Its atomic cover contains $BM \to E$ and $AME \to B$ for every set M with exactly one of P_i and Q_i for every i, so it has $2^{m+1} + 2m + 1$ elements, and the 2^m FDs $AME \to B$ are the obstructions, since $B \to A$ violates their schemata. The loss is one: leaving $B \to A$ unenforced and enforcing $BP \to A$ makes every BM a key of $AMEB$, and every minimal FD of the enforced set is then uncritical. The obstruction bound $w(O) = 2^m$ for plain designs, however, is far from tight. The plain design that leaves $B \to A$, $BP \to E$ and the m FDs $P_i \to Q_i$ unenforced has loss $m + 2$: nothing derives $BP \to E$ once no Q_i is reachable from BP, so $APE \to B$ is uncritical, and it implies every $AME \to B$ through the FDs $Q_i \to P_i$. In general, the obstructions that a polynomial cover generates are equivalent modulo such exchanges of attributes, and a plain design repairs an entire class by isolating one representative. Whether the price of plainness is always bounded by a polynomial in the size of the cover is open; an exhaustive search confirms the loss one and the plain loss $\min(2^m, m + 2)$ of Σ'_m for $m \leq 3$. Among 2,652 random schemata with covers of at most eight FDs, the plain loss never exceeded three quarters of the size of the cover times the loss; the largest ratio, three, already occurs for the cover $\{BEF \to A, AE \to D, D \to F, A \to B\}$ with loss one, and attribute-disjoint copies keep this ratio constant.

Realizability and upper bounds. A set of unenforced minimal FDs is no certificate of membership in k-loss BCNF that can be checked in polynomial time either, even when the atomic cover is given.

Theorem A.8 (Realizability). *Given (R, Σ) with $\Sigma = \Sigma_a$ and a set $\Delta \subseteq \Sigma_a$, deciding whether some BCNF design of (R, Σ) leaves exactly Δ unenforced is coNP-hard, even when $|\Delta| = 1$, and so is deciding whether some plain BCNF design (Def. 5.1) of (R, Σ) does.*

The proof reduces from the unsatisfiability of a CNF formula. The literal attributes together with an attribute T determine the attribute C_j of every clause that they satisfy, the clause attributes determine an attribute Z, and the FD $T \to Z$ keeps the atomic cover small. Once $T \to Z$ is unenforced, the schema of the FD from the literal attributes and Z to T exhibits a non-key FD exactly when a consistent set of literals satisfies every clause. Hence, the tests in line 9 of Alg. 1 and in line 7 of Alg. 3 are coNP-hard, which is consistent with their running time polynomial in the size of the atomic cover of $\mathcal{E}$, since this atomic cover may be exponentially larger than Σ_a. The same construction shows that a single unit of loss separates polynomial time from coNP-hardness.

Theorem A.9 (Fixed loss). *Let $\Sigma = \Sigma_a$ and let w be the unit weights. For every fixed $k \geq 1$, deciding whether $\lambda(R, \Sigma) \leq k$ is coNP-hard, and for k as part of the input it is DP-hard. Both statements hold for all BCNF designs and for plain BCNF designs alike.*

Membership in 0-loss BCNF is decidable in polynomial time under the atomic cover (Sec. 4), so the hardness of MinLoss sets in at loss one, and no algorithm decides k-loss BCNF for a fixed $k \geq 1$ in polynomial time unless P equals NP. The class DP consists of the intersections of a problem in NP with a problem in coNP [47], so DP-hardness strengthens the NP-hardness of Thm. 4.3 and the coNP-hardness above, and MinLoss under the atomic cover lies outside NP and coNP unless NP equals coNP. The gap between loss one and loss two also bounds approximation and parameterization.

Corollary A.10 (Approximation and obstructions). *Let $\Sigma = \Sigma_a$ and let w be the unit weights. Unless P equals NP, no polynomial-time algorithm computes for every (R, Σ) a BCNF design, or a plain BCNF design, whose loss is below $2 \cdot \lambda(R, \Sigma)$, or at most $\lambda(R, \Sigma) + c$ for a constant c. Deciding whether $\lambda(R, \Sigma) \leq 1$ is coNP-hard already for schemata with two obstructions, so it is neither fixed-parameter tractable nor in XP for the number of obstructions, unless P equals NP.*

The following theorem gives upper bounds.

Theorem A.11 (Upper bounds). *Deciding whether $\lambda_w(R, \Sigma) \leq k$ for a given schema (R, Σ), weights w and number k is in NEXPTIME. When $\Sigma = \Sigma_a$, deciding whether some plain BCNF design (Def. 5.1) of (R, Σ) has loss at most k is in Σ_2^p.*

For plain designs, a certificate consists of the set Δ and of at most $|\Sigma_a| \cdot |R|$ uncritical minimal FDs of $(\Sigma_a - \Delta)^+$ that imply $\Sigma_a - \Delta$, and a coNP oracle verifies that they are uncritical. With the atomic cover given, MinLoss for plain designs therefore lies between DP (Thm. A.9) and Σ_2^p, and whether it is Σ_2^p-complete is open: a hardness proof would have to encode an existential choice, but every path from a literal to a clause in the construction of

Thm. A.8 is a minimal FD of its own, which a design keeps or drops independently of the others. For unrestricted designs, the same certificate must include the weakened FDs, and membership in Σ_2^p follows if every enforceable FD set has an equivalent one that is generated by polynomially many uncritical minimal FDs. Prop. A.7 shows that a number of weakened forms linear in $|\Sigma_a|$ may be necessary, and whether polynomially many always suffice is open.

Repair distance. The *repair distance* $\delta_\Sigma(\mathcal{E}) = \min\{|\mathcal{F}| \mid \mathcal{F} \subseteq \Sigma^+,\ (\mathcal{E} \cup \mathcal{F})^+ = \Sigma^+\}$ satisfies (R1) but violates (R2), since it charges the unenforced rules by the effort of restoring them rather than by the rules at risk.

Proposition A.12 (Repair distance). *For every key design with enforced FDs $\mathcal{E}$, $\delta_\Sigma(\mathcal{E}) \le \lambda(\mathcal{D}, \kappa)$. For every $n \ge 1$ there is a schema with a BCNF design such that $\delta_\Sigma(\mathcal{E}) = 1$ and $\lambda(\mathcal{D}, \kappa) = n + 1$, and the join of some state that satisfies the keys violates each of the $n + 1$ unenforced minimal FDs.*

Decomposition by sources.

Prop. A.6 and Lem. 4.2 require parts of a schema that share no attribute. Real schemata share attributes that nothing determines, such as an identifier, and the following result shows that such attributes do not couple the parts. We call an attribute a *source* of (R, Σ) if it is the right-hand side of no minimal FD of Σ, so that $X \to A \in \Sigma^+$ implies $A \in X$ for every source A. The *parts* of (R, Σ) are the connected components of the graph on the non-sources that joins B and A whenever $B \in X$ for some minimal FD $X \to A$ of Σ. For a part P, let R_P consist of P and the sources in the left-hand sides of the minimal FDs with right-hand side in P, and let Σ_P be the set of these minimal FDs, which are FDs over R_P. A BCNF design with enforced FDs $\mathcal{E}$ is *part-respecting* if every minimal FD $X \to A$ of $\mathcal{E}$ satisfies $XA \subseteq R_P$ for the part P that contains A. Every plain design is part-respecting, since its enforced FDs are generated by minimal FDs of Σ.

Theorem A.13 (Decomposition by sources). *Let $P_1, \dots, P_r$ be the parts of (R, Σ), and let $R_i = R_{P_i}$ and $\Sigma_i = \Sigma_{P_i}$. Then $\lambda_w(R, \Sigma) \le \sum_{i=1}^r \lambda_w(R_i, \Sigma_i)$, and the sum is the minimum loss of the part-respecting BCNF designs of (R, Σ). Moreover, $\lambda_w^{\mathrm{p}}(R, \Sigma) = \sum_{i=1}^r \lambda_w^{\mathrm{p}}(R_i, \Sigma_i)$.*

Gluing designs for the parts yields a part-respecting design whose loss is the sum, and conversely, the enforced FDs of a part-respecting design restrict to an enforceable FD set of every part. Lem. 4.2 is the case without shared sources, and the attribute-disjoint schemata in Prop. A.6 are unions of parts and sources. For example, customers with a billing and a shipping address, $C \to S_1T_1$, $S_1T_1 \to Z_1$, $Z_1 \to T_1$, $C \to S_2T_2$, $S_2T_2 \to Z_2$ and $Z_2 \to T_2$, have the source C and the two parts $S_1T_1Z_1$ and $S_2T_2Z_2$, each with loss one, so the loss of the schema is two by Thm. A.13 and Prop. 5.4, and each address is designed on its own.

Corollary A.14 (Parameterized tractability). *Given $\Sigma = \Sigma_a$, the minimum loss of the plain BCNF designs and the minimum loss of the part-respecting BCNF designs of (R, Σ), together with designs that attain them, can be computed in time $f(m) \cdot (|R| \cdot |\Sigma_a|)^{O(1)}$, where m is the largest number of attributes of a schema R_i and f is a computable function.*

Indeed, the parts arise from a graph search over Σ_a, and Drop, restricted to plain designs or not, solves every part (R_i, Σ_i) in time bounded by a function of $|R_i|$. In the example above, the parts have three attributes plus the source C, however many addresses a customer has. We conjecture that the restriction to part-respecting designs loses nothing, that is, that equality holds in Thm. A.13. The difficulty is that a minimal FD $X' \to A$ of an enforceable FD set may contain attributes of other parts, which determine nothing in the part of A and therefore act as tags: they withhold A from X' without them, and grant it in every schema that contains them. Neither of the two natural constructions of a design for a part is sound in general. The contraction of $\mathcal{E}$, which treats the attributes of the other parts as present, is the enforced FD set of the projection of the design onto R_P; it preserves soundness and the loss, but it may add an unenforced minimal FD and thereby a violation, so that the projected design is no longer locally enforced. The restriction $\mathcal{E}[R_P]$ preserves local enforcement, but a derivation of an FD over R_P may pass through attributes of other parts that the shared sources determine, so that the restriction may lose the uncritical FDs that derive it. The conjecture follows from the following statement, which we call *uncrossing*: for every enforceable FD set $\mathcal{E}$, the uncritical minimal FDs $X \to A$ of $\mathcal{E}$ with $XA \subseteq R_P$ for the part $P \ni A$ generate an enforceable FD set $\mathcal{E}^\circ$ with $\mathcal{E}^\circ \cap \Sigma_a = \mathcal{E} \cap \Sigma_a$. Indeed, $\mathcal{E}^\circ$ is generated within the parts and is therefore part-respecting, and for an $\mathcal{E}$ of minimum loss it has the same loss, so the minimum loss is attained by a part-respecting design and equals the sum by Thm. A.13. Uncrossing in turn follows from the claim that these FDs imply every FD $X \to A \in \mathcal{E}$ with $XA \subseteq R_P$: then $\mathcal{E}^\circ$ and $\mathcal{E}$ agree on every such schema, every generator of $\mathcal{E}^\circ$ remains uncritical, and Cor. 3.19 applies. A counterexample to the claim would have to derive its first new attribute A through an uncritical minimal FD $W_PT \to A$ of $\mathcal{E}$ with attributes T of other parts, and by the minimality of this FD and the BCNF of its schema, W_P determines no attribute of T; what remains open is to exclude that other attributes of the counterexample determine T. Exhaustive tests support uncrossing: it held for all 45,064 enforceable FD sets of minimum loss with minimal FDs across parts in 956 random schemata, and for 6,000 random enforceable FD sets with such FDs that are not of minimum loss. Consistently, the minimum loss equalled the sum over the parts on all 9,552 random source-separated schemata of our machine checks, among them 2,000 weighted schemata whose sources determine attributes of both parts and 1,498 glued from the gadget of Prop. A.3 along shared sources, 1,332 of which need weakened FDs, and every inclusion-minimal enforceable FD set of minimum loss was part-respecting, although 92% of all those of minimum loss were not.

Axioms of the loss. The requirements (R1) to (R3) of Sec. 3 read formally as follows. A *loss measure* μ assigns to every FD set Σ over R and every closed FD set $\mathcal{E} \subseteq \Sigma^+$ a value $\mu_\Sigma(\mathcal{E}) \ge 0$. Let $F(\Sigma)$ denote the set of non-trivial FDs $X \to A$ in Σ^+ with a single attribute A.

(R1) *Representation independence:* $\mu_\Sigma = \mu_{\Sigma'}$ whenever $\Sigma^+ = \Sigma'^+$.
(R2) *Rule additivity:* there is a cost function $c_\Sigma : F(\Sigma) \to \mathbb{R}_{\ge 0}$ with $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in F(\Sigma) - \mathcal{E}} c_\Sigma(\varphi)$ for every $\mathcal{E}$, that is, every FD that a design leaves unenforced is a rule at risk (Prop. 3.9), and the costs of these rules add up.
(R3) *No double counting:* $c_\Sigma(X \to A) = 0$ whenever $Y \to A \in \Sigma^+$ for some $Y \subset X$.

Requirement (R3) holds for every cost that measures violations, since a relation that violates $X \to A$ also violates $Y \to A$ for every $Y \subset X$.

Proposition A.15 (Axioms). *A loss measure satisfies (R1) to (R3) iff $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in \Sigma_a - \mathcal{E}} w(\varphi)$ for weights $w : \Sigma_a \to \mathbb{R}_{\geq 0}$ that depend only on Σ^+. It satisfies in addition $c_\Sigma(\varphi) = 1$ for every minimal FD φ of Σ iff $\mu_\Sigma(\mathcal{E}) = |\Sigma_a - \mathcal{E}|$.*

Goal-directed weakening. The implementation of line 5 of Alg. 1 grows W by a goal-directed search, which the following lemma makes complete.

Lemma A.16 (Goal-directed weakening). *Let $\Delta \subseteq \Sigma_a$, let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set that contains $\Sigma_a - \Delta$ and no element of Δ, and let $g = Y \to B$ be a critical minimal FD of $\mathcal{E}$ that the uncritical minimal FDs of $\mathcal{E}$ do not imply. Then every enforceable FD set $\mathcal{E}' \supseteq \mathcal{E}$ with $\mathcal{E}' \cap \Delta = \emptyset$ contains a minimal FD $X' \to A \notin \mathcal{E}$ of $\mathcal{E}'$ such that $X \subset X' \subseteq Y^+_{\mathcal{E}}$ for some $X \to A \in \Delta$.*

Hence, the search picks such an FD g and branches only over the weakenings $X' \to A \notin \mathcal{E}$ of the elements $X \to A$ of Δ with $X' \subseteq Y^+_{\mathcal{E}}$. Neither the enumeration of W nor the cover condition can be confined further (Prop. A.3, Prop. A.4 and Prop. A.5).

The algorithms on the running example.

Example A.17 (Drop). For (R, Σ) from Ex. 1.1, the budget $b = 0$ fails since Σ^+ is not enforceable by Ex. 3.20. At budget $b = 1$, line 3 discards the sets $\{C \to S\}$, $\{C \to T\}$ and $\{C \to Z\}$, since none of their elements violates STZ and the uncritical minimal FDs $C \to S$, $C \to T$, $C \to Z$ and $Z \to T$ do not imply $ST \to Z$. The two remaining sets $\{ST \to Z\}$ and $\{Z \to T\}$ both succeed with $W = \emptyset$, and under unit weights the order of enumeration decides which of them is examined first. The set $\Delta = \{ST \to Z\}$ leaves $\Psi = \{C \to S, C \to T, C \to Z, Z \to T\}$, whose decomposition $\{CSZ, ZT\}$ is in BCNF, and the algorithm returns $\mathcal{D}_2$. The set $\Delta = \{Z \to T\}$ yields the plain design $\mathcal{D}'_4$, which is examined first whenever the weight of $Z \to T$ is smaller than that of $ST \to Z$. □

The weights thus decide between designs of equal loss.

Example A.18 (Ascend). For (R, Σ) from Ex. 1.1 and $\mathcal{E} = \mathcal{U}^+$, the algorithm starts from $\Psi = \{C \to S, C \to T, C \to Z, Z \to T\}$ with the single candidate $ST \to Z$. Its addition fails since $Z \to T$ violates STZ, and its exchange for $Z \to T$ keeps the loss at one under unit weights, so the climb accepts it and returns a design with the enforced FDs of the plain design $\mathcal{D}'_4$ of Ex. A.17. Had $Z \to T$ weighed more, it would reject the exchange and return the uncritical design, which is then optimal. □

Pseudocode of Ascend and the loss–time trade-off. Alg. 4 and Fig. 3 complement Sec. 5 and Sec. 6.3.

Machine checks. The artifact repository checks the results of the paper by exhaustive search on small schemata. An exact solver reproduces the minimum losses of the schemata in Thm. 5.2 and in Props. A.3, A.4 and A.5, and those of Prop. A.7 for $m \leq 4$. Drop agrees with an exhaustive enumeration of the closed FD sets on 374,052 random schemata with five and six attributes, Hit returns a design with minimum loss on 4,964,061 random schemata with five or six attributes, or a lower bound and a loss that bracket the minimum where a cap stops it, and the criterion of Thm. 3.24 agrees with the chase on 5,400 random designs. The reduction of Thm. A.8 yields the stated atomic cover on 60 random formulas, and some BCNF design leaves exactly $T \to Z$ unenforced iff the formula is unsatisfiable, on 20 small formulas. Under unit weights, local weakened FDs attain the minimum loss on the 91 of 3,382 random schemata whose minimum loss needs weakened FDs, and on the 16 such variants among 64 of the schema of Prop. A.4 in which FDs gain one of two further attributes on the left-hand side.

Algorithm 2 Repair(R, Σ, w)

Require: Relation schema R, FD set Σ over R, weights w on Σ_a
Ensure: A BCNF design of (R, Σ)
1: $\mathcal{D}_0 \leftarrow$ iConf(R, Σ) ▷ *lossless, dependency-preserving 3NF decomposition, BCNF whenever possible*
2: $\Gamma \leftarrow \emptyset$ ▷ *key dependencies of the local designs*
3: **for all** $S \in \mathcal{D}_0$ **do**
4: **if** $(S, \Sigma[S])$ is in BCNF **then**
5: add $K \to S$ to Γ for every minimal key K of $(S, \Sigma[S])$
6: **else**
7: $(\mathcal{D}_S, \kappa_S) \leftarrow$ Drop$(S, \Sigma[S], w)$ ▷ *local optimum for the critical schema*
8: add $K \to S'$ to Γ for every $S' \in \mathcal{D}_S$ and $K \in \kappa_S(S')$
9: $\mathcal{D} \leftarrow$ FaithfulBCNF(R, Γ) ▷ *as in Alg. 1*
10: **if** $\mathcal{D} \neq \bot$ **then**
11: **return** the lighter of Ascend(R, Σ, w, Γ^+) and Ascend$(R, \Sigma, w, \mathcal{U}^+)$
12: **return** Drop(R, Σ, w) ▷ *fallback*

Algorithm 3 Hit(R, Σ, w)

Require: Relation schema R, FD set Σ over R, weights w on Σ_a
Ensure: A BCNF design of (R, Σ) with minimum loss
1: compute Σ_a and $\mathcal{U}$, and let $\mathcal{K} \leftarrow \{(\emptyset, Q(g, \emptyset)) \mid g \in O\}$
2: **loop**
3: $\Delta \leftarrow$ an optimal solution of the master problem for $\mathcal{K}$ ▷ $w(\Delta) \leq \lambda_w(R, \Sigma)$
4: $\Psi \leftarrow \Sigma_a - \Delta$
5: **if** Ψ implies some $f \in \Delta$ **then**
6: add $(\{f\}, D_f)$ to $\mathcal{K}$ for these f, where $D_f \subseteq \Psi$ implies f ▷ *Lem. 5.6.2*
7: **else if** some enforceable $\mathcal{E} \supseteq \Psi$ with $\mathcal{E} \cap \Delta = \emptyset$ exists **then**
8: **return** the design of Cor. 3.19 for the uncritical minimal FDs of $\mathcal{E}$
9: **else**
10: add $(\Delta \cap \mathcal{N}(g), Q(g, \Delta))$ for every $g \in \Sigma_a$ not implied by $\mathcal{U} \cup C(\Delta) \cup \Delta$, and $(\Delta, \Sigma_a - \Delta)$, to $\mathcal{K}$

Algorithm 4 Ascend$(R, \Sigma, w, \mathcal{E})$

Require: Relation schema R, FD set Σ over R, weights w on Σ_a, an FD set $\mathcal{E} \subseteq \Sigma^+$ that is enforceable for (R, Σ), bounds s and k, a time budget
Ensure: A BCNF design of (R, Σ) with loss at most $w(\Sigma_a - \mathcal{E})$
1: $\Psi \leftarrow$ the uncritical minimal FDs of $\mathcal{E}$, and $\Psi^* \leftarrow \Psi$ ▷ $\Psi^+ = \mathcal{E}$
2: **repeat**
3: **repeat** ▷ *climb*
4: **for all** non-tabu $g = Z \to B \in \Sigma_a - \Psi^+$ in decreasing order of $w(g)$ **do**
5: **if** $\Psi \cup \{g\}$ is uncritical **then**
6: $\Psi \leftarrow \Psi \cup \{g\}$
7: **else**
8: $V \leftarrow$ the $X \to A \in \Psi$ with $XA \subseteq ZB$ and X not a key of ZB under $\Psi \cup \{g\}$ ▷ *the violators of the schema of g*
9: **if** $\Psi' = (\Psi - V) \cup \{g\}$ is uncritical and loses no more than Ψ **then**
10: $\Psi \leftarrow \Psi'$ ▷ *at equal loss at most s times in a row*
11: **until** Ψ is unchanged
12: $\Psi^* \leftarrow \Psi$ if it loses no more than Ψ^*, and $\Psi \leftarrow$ a random uncritical kick of Ψ^*
13: **until** k kicks in a row leave $w(\Sigma_a - \Psi^{*+})$ unchanged, or the budget is exhausted
14: **return** the design of Cor. 3.19 for Ψ^*

B PROOFS

Proof of Lemma 3.3. Since $\mathcal{E} \subseteq \Sigma^+$, the FD $X \to A$ is in Σ^+ and $A \notin X$. Hence, there is a minimal $Y \subseteq X$ with $Y \to A \in \Sigma^+$, which is a minimal FD of Σ. □

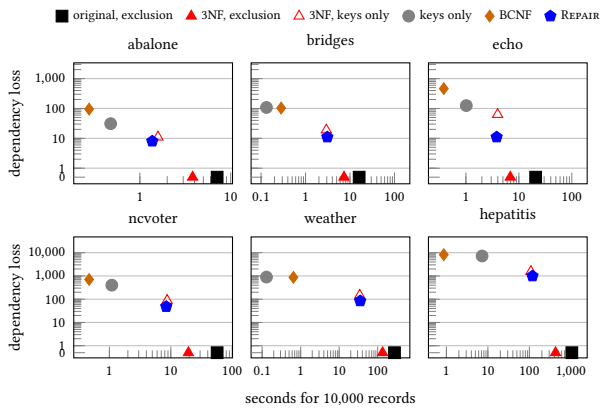


Figure 3: Time for Inserting 10,000 Records against the Dependency Loss of the Designs

Proof of Lemma 3.8. Dependency preservation holds since $\mathcal{E}$ is the closure of key dependencies, each of which is embedded in some schema of $\mathcal{D}$. If $(\mathcal{D}, \kappa)$ is locally enforced, then every non-trivial FD $X \to Y \in \mathcal{E}[S]$ has a left-hand side that contains some $K \in \kappa(S)$, so X is a key of S for $\mathcal{E}[S]$. For the converse, let $X \to Y \in \mathcal{E}[S]$ be non-trivial. Since $(S, \mathcal{E}[S])$ is in BCNF, X is a key of S for $\mathcal{E}[S]$, so X contains a minimal key of $(S, \mathcal{E}[S])$, which belongs to $\kappa(S)$ by assumption. □

Proof of Proposition 3.9. For statement 1, local enforcement gives $\mathcal{E}[S] = \{K \to S \mid K \in \kappa(S)\}^+[S]$, and r_S satisfies these key dependencies, so it satisfies $\mathcal{E}[S]$. By Lem. 3.8, $(S, \mathcal{E}[S])$ is in BCNF, and no relation over a BCNF schema that satisfies its FDs contains a redundant data value [62]. For statement 2, let t_1 and t_2 be tuples of the join that agree on X, and let $X \to Y \in \mathcal{E}$. The closure $X^+_{\mathcal{E}}$ arises from X by repeatedly adding a schema $S \in \mathcal{D}$ that has a key $K \in \kappa(S)$ within the attributes reached so far. If t_1 and t_2 agree on these attributes, then $t_1[S]$ and $t_2[S]$ are tuples of r_S that agree on its key K, so t_1 and t_2 agree on S as well. By induction, they agree on $X^+_{\mathcal{E}} \supseteq Y$. Statement 3 is the definition of soundness. For the last statement, the join of a state that satisfies the keys satisfies every FD of $\mathcal{E}$ by statement 2. Conversely, if $\varphi \notin \mathcal{E}$, then some relation r over R satisfies $\mathcal{E}$ and violates φ, since $\mathcal{E}$ is closed. Its projections form a state that satisfies the keys, and their join contains r, so it violates φ. □

Proof of Proposition A.15. The minimal FDs of Σ are the elements of $F(\Sigma)$ whose left-hand side has no proper subset that determines the right-hand side, so (R3) leaves non-zero costs only on Σ_a, and (R2) becomes $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in \Sigma_a - \mathcal{E}} c_\Sigma(\varphi)$. By (R1) and since Σ_a is determined by Σ^+, the weights $w = c_\Sigma$ restricted to Σ_a depend only on Σ^+. Conversely, the measure $\sum_{\varphi \in \Sigma_a - \mathcal{E}} w(\varphi)$ satisfies (R2) with $c_\Sigma = w$ on Σ_a and $c_\Sigma = 0$ elsewhere, (R3) by construction, and (R1) since Σ_a and w depend only on Σ^+. The second statement is the case $w = 1$. □

Proof of Proposition A.12. The set $\Sigma_a - \mathcal{E}$ restores Σ^+, since Σ_a is a cover of Σ, so $\delta_\Sigma(\mathcal{E}) \le |\Sigma_a - \mathcal{E}|$. For the second statement, take the schema (R_n, Σ_n) of the proof of Prop. 3.16 and a BCNF design of the classical algorithm, which enforces $\mathcal{E} = \{Z \to TP_1 \cdots P_n\}^+$ and has loss $n + 1$, as $ST \to Z$ and every $ST \to P_i$ are unenforced. Adding $ST \to Z$ restores Σ^+_n, and $\mathcal{E} \neq \Sigma^+_n$, so $\delta_{\Sigma_n}(\mathcal{E}) = 1$. The last claim is the last statement of Prop. 3.9. □

Proof of Proposition 3.13. A relation over R satisfies a declared key dependency $K \to S$ iff its projection onto S does, and it satisfies all of them iff it satisfies their closure $\mathcal{E}$. If r' violates Σ, then it violates some minimal FD of Σ, since Σ_a is a cover of Σ. If, in addition, r' satisfies $\mathcal{E}$, then every minimal FD of Σ that r' violates lies outside $\mathcal{E}$, so the event is contained in the union of the events that r' violates φ for $\varphi \in \Sigma_a - \mathcal{E}$, whose probability is at most $\sum_{\varphi \in \Sigma_a - \mathcal{E}} p(\varphi) = \lambda_p(\mathcal{D}, \kappa)$. The expected number of violated minimal FDs outside $\mathcal{E}$ is the same sum by linearity of expectation. For a plain design, $\mathcal{E}$ is generated by $\Sigma_a \cap \mathcal{E}$, so a relation that violates $\mathcal{E}$ violates a minimal FD of Σ in $\mathcal{E}$. If, in addition, r' never violates two minimal FDs of Σ, then the events for $\varphi \in \Sigma_a - \mathcal{E}$ are disjoint, and an r' that violates such a φ satisfies $\mathcal{E}$ and violates Σ, so the union bound is an equality. □

Proof of Theorem 3.24. A relation over R satisfies $\mathcal{E}$ iff it satisfies the declared key dependencies, so global consistency asks for a relation r that satisfies them with $r_S \subseteq r[S]$ for all S.

Necessity. Let $S \neq T$, $K \in \kappa(T)$ with $K \subseteq S^+_{\setminus T}$, and $B \in (T - K) \cap S^+_{\setminus T}$. Let every relation r_U with $U \neq T$ consist of the tuple that is 0 on U, and let r_T consist of the tuple that is 1 on B and 0 elsewhere on T. Every relation has one tuple, so the state satisfies the keys. Suppose r satisfies the key dependencies and contains a tuple u with $u[S] = 0$. Induction along the closure computation of $S^+_{\setminus T}$ shows $u[S^+_{\setminus T}] = 0$: whenever u is 0 on a key K' of $U \neq T$, it agrees with the tuple of r_U on K' and hence on U. Then $u[K] = 0$ agrees with the tuple of r_T on K, so $u[B] = 1$, which contradicts $u[B] = 0$. Hence, the state is not globally consistent.

Sufficiency. Suppose the condition holds, and let a state satisfy the keys. For a stored tuple $t \in r_S$, we extend t to a partial tuple $e(t)$ by lookups. Initially, $e(t) = t$ and only S is *used*. While some schema T that is not used has a key $K \subseteq \mathrm{dom}(e(t))$, mark T as used, and for every $B \in T - K$ set $e(t)[B]$ to $s[B]$ if r_T contains a tuple s with $s[K] = e(t)[K]$, and to the term $\nu_{T,K,B}(e(t)[K])$ otherwise. Since only schemata other than T were used before, $\mathrm{dom}(e(t)) \subseteq S^+_{\setminus T}$, so $T - K$ is disjoint from $\mathrm{dom}(e(t))$ by the condition, and no value is overwritten. If two keys K, K' of T lie in $\mathrm{dom}(e(t))$, then $T \cap \mathrm{dom}(e(t)) \subseteq K \cap K'$ by the condition, so $K = K'$, and the lookup is determined. Finally, $e(t)$ is completed by terms $\nu_{t,B}$ that are unique to t and B, and $r = \{e(t) \mid t \text{ stored}\}$ satisfies $r_S \subseteq r[S]$ for all S, where terms are read as fresh constants. A term $\nu_{T,K,B}(\cdot)$ arises only from a lookup in T by K, which happens at most once in every extension.

It remains to show that r satisfies every declared key $K \to T$. Let $e(t_1)[K] = e(t_2)[K]$. No attribute of K carries a term $\nu_{t,B}$, unless $t_1 = t_2$, so $K \subseteq \mathrm{dom}(e(t_i))$ before completion, and T is used in both extensions, either as the schema of t_i or by a lookup with some key K_i; in both cases $T \subseteq \mathrm{dom}(e(t_i))$. We say that $e(t_i)$ *hits* $s \in r_T$ if $t_i = s$ or the lookup in T found s; then $e(t_i)[T] = s$. If both hit tuples s_1, s_2, then $s_1[K] = s_2[K]$, so $s_1 = s_2$ by the key K of r_T, and $e(t_1)[T] = e(t_2)[T]$. If $e(t_1)$ hits s_1 and $e(t_2)$ hits no tuple, then $e(t_2)$ carries terms $\nu_{T,K_2,\cdot}$ on $T - K_2$, so $K \subseteq K_2$,

since $e(t_1)[K] = s_1[K]$ consists of constants; as K and K_2 lie in $\mathrm{dom}(e(t_2))$ when T is looked up, $K = K_2$, and s_1 would have been found, a contradiction. If neither hits a tuple, then the terms on $K - K_1$ and $K - K_2$ arise from the lookups in T, and the values of K known before these lookups contain no term $v_{T,\cdot,\cdot}$. Hence, $K - K_1$ is empty iff $K - K_2$ is, and in this case $K = K_1 = K_2$ as before; otherwise the terms on $K - K_1 = K - K_2$ coincide, which forces $K_1 = K_2$ and $e(t_1)[K_1] = e(t_2)[K_2]$. In both cases the lookups produce the same values on T, so $e(t_1)[T] = e(t_2)[T]$. Thus, r witnesses global consistency. The condition is tested with $O(|\mathcal{D}|^2)$ closure computations. □

Proof of Proposition 3.12. The enforced FDs of $(\{R\}, \kappa)$ are the closure $\mathcal{E}$ of the key dependencies of Σ. For every $X \subseteq R$, $X^+_{\mathcal{E}} = R$ if X contains a minimal key and $X^+_{\mathcal{E}} = X$ otherwise, so every non-trivial FD in $\mathcal{E}$ has a key as left-hand side. Hence, the design is locally enforced, it is sound as $\mathcal{D} = \{R\}$, and $\Sigma_a - \mathcal{E}$ consists of the non-key minimal FDs. For the second statement, let $X \to A$ be a non-key minimal FD of Σ and suppose $X \to A \in \mathcal{E}'$ for the enforced FDs $\mathcal{E}'$ of $(\{R\}, \kappa')$. Local enforcement requires that X contains some $K \in \kappa'(R)$, so $X \to R \in \Sigma^+$, which contradicts that X is not a key. □

Proof of Theorem 3.15. Let $(\mathcal{D}, \kappa)$ be a BCNF design with $\lambda(\mathcal{D}, \kappa) = 0$ and enforced FDs $\mathcal{E}$. Then $\Sigma_a \subseteq \mathcal{E} \subseteq \Sigma^+$, so $\mathcal{E} = \Sigma^+$ as Σ_a is a cover of Σ. Then $\mathcal{D}$ is lossless for Σ by soundness, and dependency-preserving and in BCNF for Σ by Lem. 3.8. Conversely, let $\mathcal{D}$ be a lossless, dependency-preserving decomposition of (R, Σ) into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \Sigma[S])$ for every $S \in \mathcal{D}$. Since $(S, \Sigma[S])$ is in BCNF, every FD in $\Sigma[S]$ is implied by the key dependencies of S, so $\mathcal{E}(\mathcal{D}, \kappa)$ is the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$, which is Σ^+ by dependency preservation. Hence, $(\mathcal{D}, \kappa)$ is sound, locally enforced, and has loss zero. □

Proof of Proposition 3.16. Let $R_n = \{S, T, Z, P_1, \ldots, P_n\}$ and $\Sigma_n = \{ST \to Z, Z \to T, Z \to P_1, \ldots, Z \to P_n\}$. The minimal keys are ST and SZ, and the minimal FDs are $ST \to Z$, $ST \to P_i$, $Z \to T$ and $Z \to P_i$ for $i = 1, \ldots, n$. The non-key minimal FDs are $Z \to T$ and $Z \to P_i$, so the design $(\{R_n\}, \kappa)$ has loss $n+1$. A set that contains neither Z nor ST determines only itself, and a set that contains SZ or ST is a key, so every violating FD $X \to W$ of a schema has $Z \in X$ and $S \notin X$. A step of the classical algorithm on such an FD splits a schema S' into XW and $S' - W$, which both contain Z, and only $S' - W$ contains S, since no set without S determines S. As long as the schema that contains S also contains some attribute B of $TP_1 \cdots P_n$, the FD $Z \to B$ violates it, since Z does not determine S. Hence, every run ends with the schema SZ with key SZ and schemata that contain Z but not S, whose only minimal key is Z. Their keys imply $Z \to T$ and every $Z \to P_i$, while the closure of ST under them is ST, so they imply neither $ST \to Z$ nor any $ST \to P_i$, and the loss is $n + 1$. The design $\{STZ, ZP_1 \cdots P_n\}$ with keys ST and SZ of STZ and key Z of $ZP_1 \cdots P_n$ is sound since $STZ \cap ZP_1 \cdots P_n = Z$ determines $P_1 \cdots P_n$, it is locally enforced, and it enforces every minimal FD except $Z \to T$. Finally, $ST \to Z$ is the only minimal FD of Σ_n with right-hand side Z, and it is critical since $Z \to T$ is a non-key FD of $(STZ, \Sigma_n[STZ])$, so Σ^+_n is not enforceable by Thm. 3.18 and $\lambda(R_n, \Sigma_n) \geq 1$. □

Proof of Theorem 3.18. Statement 1 implies statement 2 by Lem. 3.8 and soundness. For the converse, let $\mathcal{D}$ be a lossless, dependency-preserving decomposition of $(R, \mathcal{E})$ into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \mathcal{E}[S])$ for every $S \in \mathcal{D}$. As in the proof of Thm. 3.15, $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) with $\mathcal{E}(\mathcal{D}, \kappa) = \mathcal{E}$, it is sound since $\mathcal{D}$ is lossless for $\mathcal{E}$, and it is locally enforced by Lem. 3.8. Next, let $\mathcal{D}$ be as before, let $S \in \mathcal{D}$, let K be a minimal key of $(S, \mathcal{E}[S])$, and let $A \in S - K$. Some minimal $X \subseteq K$ satisfies $X \to A \in \mathcal{E}$. Every non-trivial FD $Y \to B \in \mathcal{E}[XA]$ belongs to $\mathcal{E}[S]$, so Y is a key of S for $\mathcal{E}$ since $(S, \mathcal{E}[S])$ is in BCNF, and therefore Y is a key of XA for $\mathcal{E}$. Hence, $X \to A$ is uncritical. The FDs obtained in this way imply every key dependency $K \to S$ of $\mathcal{D}$, and these imply $\mathcal{E}$ since $\mathcal{D}$ is dependency-preserving and in BCNF for $\mathcal{E}$. Finally, let the uncritical minimal FDs U of $\mathcal{E}$ form a cover of $\mathcal{E}$, and let K be a minimal key of $(R, \mathcal{E})$. The decomposition $\{XA \mid X \to A \in U\} \cup \{K\}$ is in BCNF for $\mathcal{E}$, since $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. It is dependency-preserving for $\mathcal{E}$ since U is a cover of $\mathcal{E}$, and it is lossless for $\mathcal{E}$ since it is dependency-preserving and contains a key [12]. □

Proof of Corollary 3.19. Let $\mathcal{E} = \Psi^+$, let K be a minimal key of $(R, \mathcal{E})$, and let $\mathcal{D}$ consist of the schemata XA of the elements $X \to A$ of Ψ and of K. Every $(XA, \mathcal{E}[XA])$ is in BCNF since Ψ is uncritical, and $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. Hence, $\mathcal{D}$ is in BCNF for $\mathcal{E}$, it is dependency-preserving for $\mathcal{E}$ since Ψ is a cover of $\mathcal{E}$ whose elements are embedded in its schemata, and it is lossless for $\mathcal{E}$ since it contains a key [12]. As in the proof of Thm. 3.18, declaring the minimal keys of $(S, \mathcal{E}[S])$ on every $S \in \mathcal{D}$ yields a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$, so $\mathcal{E}$ is enforceable. □

Proof of Lemma A.2. Let $X \to A$ be an uncritical minimal FD of Σ, and let $\mathcal{U} \subseteq \mathcal{E} \subseteq \Sigma^+$ with $\mathcal{E}$ closed. Since $\mathcal{E} \subseteq \Sigma^+$ contains $X \to A$, the FD $X \to A$ is a minimal FD of $\mathcal{E}$. Let $Y \to B \in \mathcal{E}[XA]$ be non-trivial, and let $Z \in XA - Y$. Since $Y \to B \in \Sigma^+[XA]$ and $X \to A$ is uncritical for Σ^+, Y is a key of XA for Σ, so some minimal FD $Y' \to Z$ of Σ satisfies $Y' \subseteq Y$. If $Y' \to Z$ were critical for Σ^+, some non-trivial FD $V \to W \in \Sigma^+[Y'Z]$ would have a left-hand side V that is not a key of $Y'Z$ for Σ, but $V \to W \in \Sigma^+[XA]$ since $Y'Z \subseteq XA$, so V is a key of XA for Σ and hence of $Y'Z$, a contradiction. Consequently, $Y' \to Z \in \mathcal{U} \subseteq \mathcal{E}$ and $Y \to Z \in \mathcal{E}$ for every $Z \in XA - Y$, so Y is a key of XA for $\mathcal{E}$. Hence, $(XA, \mathcal{E}[XA])$ is in BCNF, that is, $X \to A$ is uncritical for $\mathcal{E}$. □

Proof of Theorem 3.22. Let $\mathcal{E} = \mathcal{U}^+$. By Lem. A.2, every element of $\mathcal{U}$ is an uncritical minimal FD of $\mathcal{E}$, and $\mathcal{U}$ is a cover of $\mathcal{E}$. Hence, the uncritical minimal FDs of $\mathcal{E}$ form a cover of $\mathcal{E}$, and $\mathcal{E}$ is enforceable for (R, Σ) by Thm. 3.18. A design with enforced FDs $\mathcal{E}$ has loss $w(\Sigma_a - \mathcal{E}) = w(\mathcal{O})$, so $\lambda_w(R, \Sigma) \leq w(\mathcal{O})$. □

Proof of Proposition A.1. Ex. 3.5 provides the schema and the decomposition $\mathcal{D}_3$. □

Proof of Proposition 3.23. The key dependency $K \to K$ is trivial, so the enforced FDs remain $\mathcal{E}$, and every $S \in \mathcal{D}$ remains locally enforced. The projection $\mathcal{E}[K]$ contains no non-trivial FD, since an FD $Y \to B \in \mathcal{E}$ with $YB \subseteq K$ and $B \notin Y$ would make $K - B$ a key of $(R, \mathcal{E})$, so the design is locally enforced on K as well. It is sound since $\mathcal{D} \cup \{K\}$ is dependency-preserving for $\mathcal{E}$, whose

generating key dependencies are embedded in its schemata, and contains the key K of $(R, \mathcal{E})$ [12]. For the second statement, let $\mathcal{E}$ be the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$. Then $\Sigma[S] \subseteq \mathcal{E} \subseteq \Sigma^+$ gives $\mathcal{E}[S] = \Sigma[S]$, which is in BCNF and therefore implied by the key dependencies of the minimal keys of $(S, \Sigma[S])$. Hence, declaring these keys yields a locally enforced key design with enforced FDs $\mathcal{E}$, and the first statement applies. □

Proof of Theorem 4.1. For $k = 0$ the statement is Thm. 3.15 together with the NP-hardness of deciding the existence of a lossless, dependency-preserving decomposition into BCNF [5]. This NP-hardness persists under the assumption of Sec. 2, since deleting the set C of constant attributes from R and from the FDs of Σ takes polynomial time and preserves the existence of such a decomposition. Indeed, a schema that contains an attribute of C and an attribute outside C violates BCNF by $\emptyset \to C$, so the schemata disjoint from C form such a decomposition of $R - C$, and conversely every such decomposition of $R - C$ extends by the schema C. For $k > 0$ we reduce from the case $k = 0$. Let (R, Σ) be an instance, and let $(R_1, \Sigma_1), \ldots, (R_k, \Sigma_k)$ be k copies of the schema $(\{S, T, Z\}, \{ST \to Z, Z \to T\})$ over pairwise disjoint sets of fresh attributes. Each copy has loss one, since its key-only design leaves only $Z \to T$ unenforced (Prop. 3.12), while its closure is not enforceable, as $ST \to Z$ is critical and the only minimal FD with right-hand side Z, and the copies share no attribute with R and introduce no constant attribute, so the union satisfies the assumption of Sec. 2. By Lem. 4.2, applied k times, $\lambda(R \cup R_1 \cup \cdots \cup R_k, \Sigma \cup \Sigma_1 \cup \cdots \cup \Sigma_k) = \lambda(R, \Sigma) + k$, so (R, Σ) is in 0-loss BCNF iff the union is in k-loss BCNF. □

Proof of Lemma 4.2. Let $R = R_1 \cup R_2$ and $\Sigma = \Sigma_1 \cup \Sigma_2$. For $X \subseteq R$, the attribute closure X^+_Σ is the union of $(X \cap R_1)^+_{\Sigma_1}$ and $(X \cap R_2)^+_{\Sigma_2}$. Hence, the minimal FDs of Σ are those of Σ_1 together with those of Σ_2, and the closure of a set $X \subseteq R_i$ under any subset of Σ^+ is contained in R_i, since an FD $Y \to B \in \Sigma^+$ with $Y \subseteq R_i$ has $B \in Y^+_\Sigma = Y^+_{\Sigma_i} \subseteq R_i$, as $\emptyset^+_\Sigma = \emptyset$ by the assumption of Sec. 2. We use the characterization of enforceable FD sets in Thm. 3.18. Let $\mathcal{E}$ be enforceable for (R, Σ), and let $\mathcal{E}_i = \mathcal{E}[R_i]$ for $i = 1, 2$. Then $\mathcal{E}_i$ is a closed subset of Σ_i^+, its minimal FDs are the minimal FDs of $\mathcal{E}$ over R_i, and such an FD $X \to A$ is critical for $\mathcal{E}_i$ iff it is critical for $\mathcal{E}$, since both properties depend on $\mathcal{E}[XA] = \mathcal{E}_i[XA]$ only. Every FD $X \to A \in \mathcal{E}_i$ is implied by the uncritical minimal FDs of $\mathcal{E}$, and its derivation uses FDs over R_i only, since the closure of X stays within R_i. Hence, $\mathcal{E}_i$ is enforceable for (R_i, Σ_i), and $\lambda(R_1, \Sigma_1) + \lambda(R_2, \Sigma_2) \leq |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2| = |\Sigma_a - \mathcal{E}|$. Conversely, let $\mathcal{E}_i$ be enforceable for (R_i, Σ_i) with uncritical minimal FDs U_i, and let $\mathcal{E} = (\mathcal{E}_1 \cup \mathcal{E}_2)^+$. The minimal FDs of $\mathcal{E}$ are those of $\mathcal{E}_1$ together with those of $\mathcal{E}_2$, each of them is critical for $\mathcal{E}$ iff it is critical for its $\mathcal{E}_i$, and $U_1 \cup U_2$ is a cover of $\mathcal{E}$. Hence, $\mathcal{E}$ is enforceable for (R, Σ) with $|\Sigma_a - \mathcal{E}| = |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2|$. □

Proof of Corollary 4.4. Let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\rho \cdot \lambda(R, \Sigma)$. The design returned for (R, Σ) has loss zero iff (R, Σ) is in 0-loss BCNF, and it has loss zero iff its declared key dependencies imply every FD of Σ, which attribute closures decide in polynomial time. Hence, $\mathcal{A}$ decides 0-loss BCNF, which is NP-hard by Thm. 4.1. Now let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\lambda(R, \Sigma) + c$. Given (R, Σ), let (R', Σ') consist of $c + 1$ copies of (R, Σ) over pairwise disjoint sets of attributes, and let $\mathcal{E}$ be the enforced FDs of the design that $\mathcal{A}$ returns for (R', Σ'). If (R, Σ) is in 0-loss BCNF, then $\lambda(R', \Sigma') = 0$ by Lem. 4.2, so $\mathcal{E}$ misses at most c of the minimal FDs of Σ', which are the minimal FDs of the copies, and the declared key dependencies imply every FD of some copy. Otherwise, the restriction of $\mathcal{E}$ to every copy is enforceable for that copy by the proof of Lem. 4.2, so it misses a minimal FD of the copy, and the declared key dependencies imply the FD set of no copy. Attribute closures decide which case applies in polynomial time, so $\mathcal{A}$ decides 0-loss BCNF. □

Proof of Theorem 4.3. We reduce the maximum independent set problem, which is NP-complete [21], to the decision whether $\lambda(R, \Sigma) \leq k$. Let $G = (V, E)$ be a graph with n vertices and m edges, and let α be the size of a maximum independent set of G. Let R consist of attributes P_v and Q_v for every vertex v, and of attributes G_e, C_e, D^1_e, D^2_e and D^3_e for every edge e. Let Σ consist of the *vertex FD* $a_v = P_v \to Q_v$ for every vertex v and, for every edge $e = uv$, of the *edge FD* $b_e = Q_u Q_v G_e \to C_e$, the *blockers* $s^u_e = P_u G_e \to C_e$ and $s^v_e = P_v G_e \to C_e$, and the *victims* $h^i_e = C_e P_u P_v D^i_e \to G_e$ for $i = 1, 2, 3$. The schema (R, Σ) has $2n + 5m$ attributes and $n + 6m$ FDs, so it is computed in polynomial time. We show that $\Sigma = \Sigma_a$ and $\lambda(R, \Sigma) = 2m + n - \alpha$, so that G has an independent set of size α' iff $\lambda(R, \Sigma) \leq 2m + n - \alpha'$.

Atomic cover. The attributes P_v and D^i_e occur on no right-hand side of Σ, and Q_v occurs on the right-hand side of a_v only, so a_v is the only minimal FD with right-hand side Q_v. The attributes C_e and G_e occur only in the six FDs of the edge e, each of which has one of them on its left-hand side and the other on its right-hand side, so a set $X \subseteq R$ determines neither unless it contains one of them. Hence, X determines G_e iff $G_e \in X$ or X contains C_e, P_u, P_v and some D^i_e, and X determines C_e iff $C_e \in X$ or X contains G_e and P_u, or G_e and P_v, or G_e and one of P_u and Q_u and one of P_v and Q_v. Consequently, the victims are the minimal FDs with right-hand side G_e, the edge FD and the blockers are those with right-hand side C_e, and $\Sigma = \Sigma_a$. In particular, $P_u P_v G_e \to C_e$ is implied by Σ but not minimal, so it does not count towards the loss.

Lower bound. Let $\mathcal{E}$ be enforceable for (R, Σ), let Ψ be the set of its uncritical minimal FDs, which is a cover of $\mathcal{E}$ by Thm. 3.18, and let S be the set of vertices v with $a_v \in \mathcal{E}$. We first show that no victim h^i_e belongs to $\mathcal{E}$ if $\mathcal{E}$ contains one of s^u_e, s^v_e and $P_u P_v G_e \to C_e$. Suppose that $h^i_e \in \mathcal{E} = \Psi^+$. Then Ψ contains an FD $Z \to G_e$ with $Z \subseteq (C_e P_u P_v D^i_e)^+_{\mathcal{E}}$, and $Z \to G_e \in \Sigma^+$ shows $Z \supseteq C_e P_u P_v D^j_e$ for some j, where $j = i$ since $(C_e P_u P_v D^i_e)^+_\Sigma = C_e P_u Q_u P_v Q_v D^i_e G_e$. Each of the three FDs has its attributes in $Z G_e$ and a left-hand side whose closure under Σ, and hence under $\mathcal{E}$, does not contain D^i_e. Hence, if one of them belongs to $\mathcal{E}$, then $(Z G_e, \mathcal{E}[Z G_e])$ is not in BCNF, which contradicts the uncriticality of $Z \to G_e$. Now consider an edge $e = uv$ and its six FDs. If $s^u_e \in \mathcal{E}$ or $s^v_e \in \mathcal{E}$, then the three victims are not in $\mathcal{E}$. Otherwise, s^u_e and s^v_e are not in $\mathcal{E}$, and if $u, v \in S$, then either $b_e \notin \mathcal{E}$ or $P_u P_v G_e \to C_e \in \mathcal{E}$, which again excludes the victims. In every case, $\mathcal{E}$ misses at least two of the six FDs, and at least three if $u, v \in S$. Together with the $n - |S|$ vertex FDs outside $\mathcal{E}$, this gives $|\Sigma_a - \mathcal{E}| \geq n - |S| + 2m + e(S)$ where $e(S)$ is the number of edges with both vertices in S. Removing one vertex of every such edge from S leaves an independent set of size at least $|S| - e(S)$, so $|\Sigma_a - \mathcal{E}| \geq 2m + n - \alpha$.

Upper bound. Let I be an independent set of G, let Δ consist of the blockers of all edges and of the vertex FDs a_v with $v \notin I$, and let $\mathcal{E} = (\Sigma - \Delta)^+$. Then $\mathcal{E}$ contains no element of Δ, since Q_v occurs on no right-hand side of $\Sigma - \Delta$ for $v \notin I$, and $(P_u G_e)^+_{\mathcal{E}} \subseteq P_u Q_u G_e$ because b_e requires Q_v as well. Every element of $\Sigma - \Delta$ is uncritical for $\mathcal{E}$. This is clear for the vertex FDs, whose schemata have two attributes, and for the edge FD b_e, since the only non-trivial FDs of $\mathcal{E}$ on $Q_u Q_v G_e C_e$ have a left-hand side that contains $Q_u Q_v G_e$. For a victim h^i_e, the non-trivial FDs of $\mathcal{E}$ on its schema $C_e P_u P_v D^i_e G_e$ have a left-hand side that contains $C_e P_u P_v D^i_e$: the attributes P_u, P_v and D^i_e occur on no right-hand side of $\Sigma - \Delta$, so G_e is determined only by sets that contain $C_e P_u P_v D^i_e$, and C_e only through b_e, which requires Q_u and Q_v and thus a_u and a_v in $\mathcal{E}$, while I contains at most one of u and v. By Thm. 3.18, $\mathcal{E}$ is enforceable, and its loss is $|\Delta| = 2m + n - |I|$. Hence, $\lambda(R, \Sigma) = 2m + n - \alpha$. □

Proof of the last statement of Corollary 4.4. The statement is the first part of Cor. A.10. □

Proof of Corollary A.10. Let φ be a CNF formula with at least two clauses, each with at least two literals, which does not affect the coNP-hardness of unsatisfiability, and let (R, Σ) be its schema from the proof of Thm. A.8. By the proof of Thm. A.9, $\lambda(R, \Sigma) = 1$ if φ is unsatisfiable and $\lambda(R, \Sigma) \geq 2$ otherwise, for all and for plain designs. Since $\Sigma = \Sigma_a$, the loss of a given design is computed in polynomial time by testing whether its declared key dependencies imply each minimal FD. An algorithm whose designs have loss below $2 \cdot \lambda(R, \Sigma)$ therefore returns a design of loss one iff φ is unsatisfiable. An algorithm with additive error c, applied to $c + 1$ attribute-disjoint copies of (R, Σ), whose loss is $c + 1$ or at least $2c + 2$ by Lem. 4.2, returns a design of loss at most $2c + 1$ iff φ is unsatisfiable.

For the obstructions, $f = LZ \to T$ and $LC \to T$ are the only minimal FDs of Σ with right-hand side T, and both are critical for Σ^+, by the argument of the proof of Thm. A.9 applied to $\mathcal{E} = \Sigma^+$, so the uncritical minimal FDs do not imply them. Every other minimal FD of Σ in the list of the proof of Thm. A.8 is uncritical. The schemata TZ and CZ have no violator, since T determines only Z and no proper subset of C determines Z, as φ has at least two clauses. The remaining schemata contain T but not Z, or Z but not T and not all of C, and in each of them a subset determines a further attribute of the schema only if it contains the left-hand side of the FD. Hence, the obstructions are exactly f and $LC \to T$. An algorithm that decides $\lambda(R, \Sigma) \leq 1$ in time $h(|O|) \cdot n^{O(1)}$ or $n^{h(|O|)}$ for a function h would therefore decide unsatisfiability in polynomial time. □

Proof of Theorem A.8. We reduce from the unsatisfiability of a formula φ in conjunctive normal form with variables $x_1, \ldots, x_n$, $n \geq 1$, and non-empty clauses $c_1, \ldots, c_m$. Let R consist of the *literal attributes* P_i and Q_i for x_i and $\neg x_i$, collected in L, the *clause attributes* $C_1, \ldots, C_m$, collected in C, and the attributes T and Z, and let $X = LZ$. Let Σ consist of $f = X \to T$, of $P_i Q_i T \to X$ for every i, of $T\ell \to C_j$ for every literal attribute ℓ of a literal in c_j, of $C \to Z$, and of $T \to Z$, and let $\Delta = \{T \to Z\}$.

Atomic cover. Without T, only $C \to Z$ and f apply, and the literal attributes are determined only by the FDs $P_i Q_i T \to X$. A case distinction over the right-hand sides then shows that Σ_a consists of $X \to T$, $LC \to T$, $T \to Z$, $C \to Z$, the FDs $T\ell \to C_j$ above, $P_i Q_i T \to C_j$ for every clause c_j without a literal of x_i, $X \to C_j$ for every j, and $P_i Q_i T \to B$ for every i and $B \in L - P_i Q_i$. Hence, Σ_a has $O(n^2 + nm + |\varphi|)$ elements and is computed in polynomial time, and we take it as the input. In particular, every minimal FD of Σ with right-hand side T has a left-hand side that contains X or LC.

Satisfiable formulas. Let $\mathcal{E}$ be the enforced FDs of a BCNF design with $\Sigma_a - \mathcal{E} = \Delta$, and let $S = XT$ be the schema of f. Let H be the set of literal attributes of the true literals of a satisfying assignment. Then $\mathcal{E} \supseteq \Sigma_a - \Delta$ implies $TH \to Z \in \mathcal{E}[S]$, while $(TH)^+_\Sigma = THCZ$ does not contain S, so $(S, \mathcal{E}[S])$ is not in BCNF. Since $\mathcal{E} \subseteq \Sigma^+$ contains f and $LC \to T$, these two FDs are the only minimal FDs of $\mathcal{E}$ with right-hand side T, and f is critical for $\mathcal{E}$. The FD $LC \to T$ is critical for $\mathcal{E}$ as well, since $T\ell \to C_1 \in \mathcal{E}[LCT]$ for a literal attribute ℓ of c_1, while $(T\ell)^+_\Sigma$ contains no literal attribute other than ℓ. Hence, no uncritical minimal FD of $\mathcal{E}$ has right-hand side T, the uncritical minimal FDs of $\mathcal{E}$ do not imply f, and $\mathcal{E}$ is not enforceable by Thm. 3.18, a contradiction. So no BCNF design, plain or not, leaves exactly Δ unenforced.

Unsatisfiable formulas. Let Ψ' consist of f, of the FDs $P_i Q_i T \to B$ in Σ_a, of the FDs $T\ell \to C_j$, and of $C \to Z$, and let $\mathcal{E} = \Psi'^+$. Then Ψ' implies every element of $\Sigma_a - \Delta$, since $LC \to T$ follows from $C \to Z$ and f, and the FDs $P_i Q_i T \to C_j$ and $X \to C_j$ follow through T and the FDs $T\ell \to C_j$, while $T^+_{\mathcal{E}} = T$, so $\mathcal{E} \cap \Sigma_a = \Sigma_a - \Delta$. Every element of Ψ' is uncritical for $\mathcal{E}$. On S, a subset without T determines no further attribute of S unless it contains X, and a subset Y with $T \in Y$ is a key of S if it contains some $P_i Q_i$, and otherwise determines within S at most Z, which it determines iff its literal attributes, which are consistent, satisfy every clause, which is impossible. On $P_i Q_i TB$, every subset without $P_i Q_i T$ determines no further attribute of the schema, since the literal attributes are determined only through some $P_k Q_k T$. On $T\ell C_j$ and on CZ, only the declared left-hand sides and their supersets determine further attributes of the schema. Hence, Ψ' is an uncritical set, and by Cor. 3.19, $\mathcal{E}$ is enforced by a BCNF design, which is plain and leaves exactly Δ unenforced. □

Proof of Theorem A.9. Let (R, Σ) be the schema of a CNF formula φ from the proof of Thm. A.8, where every clause has at least two literals, which does not affect the coNP-hardness of unsatisfiability, so that $\Sigma = \Sigma_a$. We show that $\lambda(R, \Sigma) = 1$ if φ is unsatisfiable and $\lambda(R, \Sigma) \geq 2$ otherwise, for all BCNF designs and for plain ones. By Thm. A.8, some BCNF design, and then a plain one, leaves exactly $\{T \to Z\}$ unenforced iff φ is unsatisfiable, so it suffices to show that no BCNF design leaves at most one minimal FD unenforced while it enforces $T \to Z$. Let $\mathcal{E}$ be the enforced FDs of such a design. By Lem. 3.3, every minimal FD of $\mathcal{E}$ with right-hand side T has a left-hand side that contains LZ or LC, since $f = LZ \to T$ and $LC \to T$ are the minimal FDs of Σ with right-hand side T. An FD of the first kind is critical for $\mathcal{E}$, since its schema contains T and Z, and $T \to Z \in \mathcal{E}$ is a minimal FD of $\mathcal{E}$ whose left-hand side is not a key of the schema, as $T^+_\Sigma = TZ$. An FD of the second kind is critical for $\mathcal{E}$ as well: its schema contains T, the clause attribute C_1 and the literal attributes of c_1, at most one of the FDs $T\ell \to C_1$ for the literals ℓ of c_1 is unenforced, and $(T\ell)^+_\Sigma$ contains no literal attribute other than ℓ, so some $T\ell \to C_1 \in \mathcal{E}$ is a minimal FD of $\mathcal{E}$ inside the schema with a left-hand side that is not a key. Hence, no uncritical minimal FD of $\mathcal{E}$ has right-hand side T, while $\mathcal{E}$ contains

f or $LC \to T$, which the uncritical minimal FDs of $\mathcal{E}$ therefore do not imply, and $\mathcal{E}$ is not enforceable by Thm. 3.18. Consequently, $\lambda(R, \Sigma) \leq 1$ iff φ is unsatisfiable, in both settings, which proves the case $k = 1$. For $k > 1$, add $k - 1$ attribute-disjoint copies of the schema $(\{S, T, Z\}, \{ST \to Z, Z \to T\})$, each of loss one for all and for plain designs, and apply Lem. 4.2, whose proof applies verbatim to plain designs.

For DP-hardness we reduce from SAT–UNSAT, which asks for CNF formulas φ_1 and φ_2 whether φ_1 is satisfiable and φ_2 is unsatisfiable, and which is DP-complete [47]. Let G be the graph with a vertex for every literal occurrence of φ_1 and an edge between two occurrences in the same clause and between occurrences of complementary literals. An independent set of G contains at most one vertex per clause, and G has an independent set of size m_1, the number of clauses, iff φ_1 is satisfiable. Let (R_1, Σ_1) be the schema of G from the proof of Thm. 4.3, whose minimum loss is $2|E(G)| + |V(G)| - \alpha(G)$ for all and for plain designs, since its optimal designs are plain, and let $a = 2|E(G)| + |V(G)| - m_1$, so that $\lambda(R_1, \Sigma_1) \geq a$ with equality iff φ_1 is satisfiable. Let (R_2, Σ_2) be the schema of φ_2 from the first part over disjoint attributes, so that $\lambda(R_2, \Sigma_2) = 1$ if φ_2 is unsatisfiable and $\lambda(R_2, \Sigma_2) \geq 2$ otherwise. Both schemata are their own atomic covers and have no constant attributes, so by Lem. 4.2 the loss of their union is at most $a + 1$ iff φ_1 is satisfiable and φ_2 is unsatisfiable. □

Proof of Theorem A.11. For the first statement, a nondeterministic algorithm guesses a closed FD set $\mathcal{E}$ by the table of the closures $X^+_{\mathcal{E}}$ for all $X \subseteq R$, of size $2^{|R|} \cdot |R|$. In time polynomial in $2^{|R|}$ and the size of the input, it verifies that the table is a closure operator, that $X^+_{\mathcal{E}} \subseteq X^+_{\Sigma}$ for every X, so that $\mathcal{E} \subseteq \Sigma^+$, computes Σ_a and $w(\Sigma_a - \mathcal{E})$, and tests the condition of Thm. 3.18 by computing the minimal FDs of $\mathcal{E}$, testing every schema XA for BCNF on its subsets, and comparing closures under the uncritical ones. For the second statement, some plain BCNF design has loss at most k iff there are a set $\Delta \subseteq \Sigma_a$ with $w(\Delta) \leq k$ and a set U of FDs such that, for $\Psi = \Sigma_a - \Delta$ and $\mathcal{E} = \Psi^+$, (i) $\mathcal{E} \cap \Delta = \emptyset$, (ii) every element of U is a minimal FD of $\mathcal{E}$, (iii) U implies Ψ, and (iv) every element $X \to A$ of U is uncritical for $\mathcal{E}$. Indeed, (ii) and (iii) give $U^+ = \mathcal{E}$, so (iv) and Thm. 3.18 make $\mathcal{E}$ enforceable, and (i) gives $\Sigma_a \cap \mathcal{E} = \Psi$, so the design has loss $w(\Delta)$ and is plain. Conversely, for a plain design with $\Sigma_a - \mathcal{E} = \Delta$, the uncritical minimal FDs of $\mathcal{E}$ imply every $g \in \Psi$, and the closure computation for g applies at most $|R|$ of them, so some such U has at most $|\Psi| \cdot |R|$ elements. Conditions (i) to (iii) take polynomial time with closures under Ψ, and (iv) states that for every $Y \subseteq XA$ the set $Y^+_{\Psi} \cap XA$ is Y or XA, a coNP condition. Hence, the problem is in Σ^p_2. □

Proof of Theorem 4.5. Soundness is losslessness of $\mathcal{D}$ for the cover $\{K \to S \mid S \in \mathcal{D}, K \in \kappa(S)\}$ of $\mathcal{E}$, which the chase decides in polynomial time [1]. A witness against local enforcement is a schema $S \in \mathcal{D}$ and an FD $X \to A$ with $XA \subseteq S$ such that $X \to A \in \mathcal{E}$, which the attribute closure verifies in linear time, and $K \not\subseteq X$ for every $K \in \kappa(S)$, so local enforcement is in coNP. For the hardness, let φ be a propositional formula in conjunctive normal form with variables $x_1, \ldots, x_n$ and clauses $C_1, \ldots, C_m$, and let R consist of attributes P_i and Q_i for the literals x_i and $\neg x_i$ of every variable, an attribute C_j for every clause, and an attribute Z. Let $\mathcal{D}$ consist of the schema $S_0 = P_1 Q_1 \cdots P_n Q_n Z$ with $\kappa(S_0) = \{P_i Q_i \mid 1 \leq i \leq n\}$, of a schema LC_j with $\kappa(LC_j) = \{L\}$ for every clause C_j and every attribute L of a literal in C_j, and of the schema $S_1 = C_1 \cdots C_m Z$ with $\kappa(S_1) = \{C_1 \cdots C_m\}$, and let Σ be the set of these key dependencies. Then $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) with $\mathcal{E} = \Sigma^+$, its size is polynomial in the size of φ, and it is sound since S_0 contains the key $P_1 Q_1$ of (R, Σ). No attribute C_j or Z occurs in a left-hand side of Σ other than $C_1 \cdots C_m$, so the only non-trivial FDs of $\mathcal{E}$ on a schema LC_j or on S_1 are implied by its declared key, and these schemata are locally enforced. Let $X \subseteq S_0$ contain no declared key of S_0, that is, at most one of P_i and Q_i for every i, so that the literals of X are consistent. The closure of X under Σ consists of X, of the attributes C_j of the clauses that contain a literal of X, and of Z iff every clause contains a literal of X. Hence, $\mathcal{E}$ contains a non-trivial FD on S_0 with left-hand side X iff $Z \notin X$ and the literals of X satisfy φ. Consequently, $(\mathcal{D}, \kappa)$ is locally enforced iff φ is unsatisfiable. □

Proof of Theorem 5.2. For the first statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{D \to C, BE \to C, BD \to A, ABC \to D\}$. The minimal keys of (R, Σ) are ABE and BDE, and Σ_a consists of $BD \to A$, $D \to C$, $BE \to C$, $ABC \to D$ and $ABE \to D$. The design $\mathcal{D} = \{ABCD, BCE, BDE\}$ with $\kappa(ABCD) = \{ABC, BD\}$, $\kappa(BCE) = \{BE\}$ and $\kappa(BDE) = \{BDE\}$ enforces $\mathcal{E} = \{ABC \to D, BD \to AC, BE \to C\}^+$. It is sound since BDE is a key of $(R, \mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $\mathcal{E}$ contains $ABE \to D$ but not $D \to C$. The loss is minimum because Σ^+ is not enforceable: $ABC \to D$ and $ABE \to D$ are critical for Σ^+, since $D \to C$ is a non-key FD of $(ABCD, \Sigma[ABCD])$ and $BD \to A$ is a non-key FD of $(ABDE, \Sigma[ABDE])$, and the uncritical minimal FDs $BD \to A$, $D \to C$ and $BE \to C$ do not imply $ABC \to D$. A plain design with loss one enforces $\Psi = \Sigma_a - \{\varphi\}$ for some minimal FD φ with $\varphi \notin \Psi^+$ and Ψ^+ enforceable. For $\varphi = ABE \to D$, the FDs $BE \to C$ and $ABC \to D$ imply φ. For $\varphi = D \to C$, the FD $ABC \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $BD \to A$ and $BE \to C$ of Ψ^+ do not imply it. For $\varphi = ABC \to D$, the FD $ABE \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABDE, \Psi^+[ABDE])$, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BE \to C$, both $ABC \to D$ and $ABE \to D$ are critical for Ψ^+ by the same two non-key FDs, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BD \to A$, the FD $ABC \to D$ is critical for Ψ^+ since $D \to C$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $ABE \to D$, $D \to C$ and $BE \to C$ of Ψ^+ do not imply it. Hence, every plain design has a loss of at least two.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to E, CE \to A, CE \to B, ADE \to B, ABD \to C\}$. The minimal keys of (R, Σ) are ABD, ADE, BCD and CDE, and Σ_a consists of $B \to E$, $BC \to A$, $CE \to A$, $CE \to B$, $ADE \to B$, $ABD \to C$ and $ADE \to C$. The design $\mathcal{D} = \{ACE, BCE, ABDE, ABCD\}$ with the single keys CE, CE, ADE and ABD enforces $\mathcal{E} = \{CE \to AB, ADE \to B, ABD \to C\}^+$, which contains $ADE \to C$ but neither $B \to E$ nor $BC \to A$. It is sound since ADE is a key of $(R, \mathcal{E})$ contained in $ABDE$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has the declared key as left-hand side, and its loss is two. The

loss is minimum. Indeed, each of $BC \to A$, $CE \to A$, $ADE \to B$, $ABD \to C$ and $ADE \to C$ is implied by the other minimal FDs, so a design that leaves only one of them unenforced does not exist. A design that leaves only $B \to E$ unenforced enforces $BC \to A$ and $CE \to A$, which remain non-key FDs of every schema that contains $ABCD$ or $ACDE$, so every minimal FD with right-hand side C is critical and $ABD \to C$ cannot be enforced. A design that enforces $B \to E$, in particular one with loss zero or one that leaves only $CE \to B$ unenforced, has the non-key FD $B \to E$ on every schema that contains BE properly. Every minimal FD of its enforced FDs with right-hand side B has E on its left-hand side and is therefore critical, so by Thm. 3.18 the design enforces neither $CE \to B$ nor $ADE \to B$. Hence, $\lambda(R, \Sigma) = 2$. Now let $(\mathcal{D}', \kappa')$ be a key-complete BCNF design with enforced FDs $\mathcal{E}'$, and suppose its loss is at most two. We use that C and E determine nothing but themselves, that CE determines $ABCE$ only, and that B determines BE only. Suppose first that $B \to E \in \mathcal{E}'$. Then no schema of $\mathcal{D}'$ contains BE properly, since B is not a key of such a schema. The key dependencies of $\mathcal{D}'$ derive B from CE or from ADE only through a schema that contains BE properly, so $CE \to B$ and $ADE \to B$ are both unenforced. They derive C from ABD only through a schema that contains $ABCD$ or $ACDE$, on which $BC \to A$ or $CE \to A$ would violate local enforcement, so a third minimal FD is unenforced. Suppose next that $B \to E \notin \mathcal{E}'$ and $CE \to B \in \mathcal{E}'$. Then some schema $S^* \in \{BCE, ABCE\}$ with $CE \in \kappa'(S^*)$ belongs to $\mathcal{D}'$. Since BC is a minimal key of $(S^*, \Sigma[S^*])$, key-completeness gives $BC \in \kappa'(S^*)$ and $BC \to E \in \mathcal{E}'$. If $CE \to A \notin \mathcal{E}'$, then $BC \to A \notin \mathcal{E}'$ either, as $BC \to A$ and $CE \to B$ would imply $CE \to A$, and three minimal FDs are unenforced. If $CE \to A \in \mathcal{E}'$, then $BC \to A \in \mathcal{E}'$, and C is derived from ABD only through a schema that contains $ABCD$ or $ACDE$, which $BC \to A$ or $CE \to A$ prevent, so $ABD \to C$ and $ADE \to C$ are unenforced. Suppose finally that neither $B \to E$ nor $CE \to B$ is in $\mathcal{E}'$. Then $BC \to A$ and $CE \to A$ must be enforced, and C is again derived from ABD only through a schema that contains $ABCD$ or $ACDE$, so $ABD \to C$ is unenforced. In all cases the loss is at least three. □

Proof of Proposition A.7. Every plain BCNF design is a BCNF design, which gives the first inequality. The uncritical design of Thm. 3.22 enforces $\mathcal{E} = \mathcal{U}^+$, and $\mathcal{U} \subseteq \Sigma_a \cap \mathcal{E}$ gives $(\Sigma_a \cap \mathcal{E})^+ = \mathcal{E}$, so it is a plain BCNF design with loss $w(O)$, which gives the second inequality.

For the second statement, let $\mathcal{V} = \{C_1, \dots, C_m\}$, let $\mathcal{Y}$ be the set of subsets of $\mathcal{V}$ with $\lfloor m/2 \rfloor$ elements, which is an antichain with $|\mathcal{Y}| = \binom{m}{\lfloor m/2 \rfloor} \geq 2$ whose elements are non-empty proper subsets of $\mathcal{V}$, let $R_m = \{A, B, E\} \cup \mathcal{V}$, and let $\Sigma_m = \{B \to A\} \cup \{BY \to E, AYE \to B \mid Y \in \mathcal{Y}\}$. No attribute of $\mathcal{V}$ occurs on a right-hand side, A occurs only on the right-hand side of $B \to A$, E only on those of the FDs $BY \to E$, and B only on those of the FDs $AYE \to B$. Hence, a set $X \subseteq R_m$ determines A iff it contains A or B, it determines E iff it contains E or some BY with $Y \in \mathcal{Y}$, and it determines B iff it contains B or some AYE with $Y \in \mathcal{Y}$. Since $\mathcal{Y}$ is an antichain, $\Sigma_a = \Sigma_m$, $B^+_{\Sigma_m} = AB$, $(BY)^+_{\Sigma_m} = (AYE)^+_{\Sigma_m} = ABYE$ for every $Y \in \mathcal{Y}$, and no subset of $\mathcal{V}$ determines A. The FD $B \to A$ is uncritical since its schema has two attributes, and so is $BY \to E$, since every non-trivial FD of Σ^+_m on BYE has a left-hand side that contains BY' for some $Y' \in \mathcal{Y}$ with $Y' \subseteq Y$, that is, $Y' = Y$. The FD $AYE \to B$ is critical, since $B \to A$ is a non-key FD of $(ABYE, \Sigma_m[ABYE])$, and the uncritical minimal FDs do not imply it, since the FDs $AY'E \to B$ are the only minimal FDs with right-hand side B. Hence, the obstructions of (R_m, Σ_m) are the $|\mathcal{Y}|$ FDs $AYE \to B$.

We first show that every enforceable $\mathcal{E}$ that contains $AYE \to B$ has it as an uncritical minimal FD. By Lem. 3.3, every minimal FD of $\mathcal{E}$ with right-hand side B has the form $X \to B$ with $AY'E \subseteq X$ for some $Y' \in \mathcal{Y}$, so $AYE \to B$ is a minimal FD of $\mathcal{E}$, as $\mathcal{Y}$ is an antichain. By Thm. 3.18, the uncritical minimal FDs of $\mathcal{E}$ imply it, so the closure of AYE under them applies an uncritical minimal FD $X \to B$ of $\mathcal{E}$ with $X \subseteq (AYE)^+_{\mathcal{E}} \subseteq ABYE$. Since $B \notin X$ and X contains some $AY'E$, we obtain $Y' \subseteq Y$, hence $Y' = Y$ and $X = AYE$.

Next, let $\mathcal{E}$ be enforceable. If $B \to A \in \mathcal{E}$, then $\mathcal{E}$ contains no FD $AYE \to B$, since $B \to A$ would be a non-key FD of $(ABYE, \mathcal{E}[ABYE])$ by $B^+_{\mathcal{E}} \subseteq B^+_{\Sigma_m} = AB$, which contradicts the uncriticality of $AYE \to B$. Hence, every BCNF design that enforces $B \to A$ has a loss of at least $|\mathcal{Y}| \geq 2$, so a design with loss one enforces every minimal FD of Σ_m except $B \to A$. For such a design and every $Y \in \mathcal{Y}$, the FD $AYE \to B$ is uncritical for $\mathcal{E}$, and $BY \to E$ is a non-trivial FD of $\mathcal{E}[ABYE]$, so BY is a key of $ABYE$ for $\mathcal{E}$ and $BY \to A \in \mathcal{E}$. This FD is a minimal FD of $\mathcal{E}$, since $B \to A \notin \mathcal{E}$, no subset of $\mathcal{V}$ determines A, and a set BY'' with $Y'' \subset Y$ contains no BY' with $Y' \in \mathcal{Y}$, so its closure under $\mathcal{E}$ does not contain A. Hence, the design enforces the $|\mathcal{Y}|$ weakened forms $BY \to A$ of $B \to A$. In particular, Σ^+_m is not enforceable and $\lambda(R_m, \Sigma_m) \geq 1$.

The design with the schemata $ABYE$ for $Y \in \mathcal{Y}$, each declaring the keys BY and AYE, and the schema $K = B\mathcal{V}$ declaring the key K has loss one. Its enforced FDs $\mathcal{E} = \{BY \to AE, AYE \to B \mid Y \in \mathcal{Y}\}^+$ contain every minimal FD of Σ_m except $B \to A$. A subset of $ABYE$ that contains neither BY nor AYE is closed under $\mathcal{E}$, since every other declared key contains an attribute of $\mathcal{V} - Y$, and the closure of a subset of K adds at most A and E, which lie outside K. Hence, the design is locally enforced, and it is sound since it is dependency-preserving for $\mathcal{E}$ and contains the key K of $(R_m, \mathcal{E})$ [12]. Thus, $\lambda(R_m, \Sigma_m) = 1$.

Finally, let $\mathcal{E}$ be the enforced FDs of a plain BCNF design with $B \to A \notin \mathcal{E}$. Then $\mathcal{E}$ is generated by minimal FDs of Σ_m other than $B \to A$, none of which has A as right-hand side, so a set X determines A under $\mathcal{E}$ only if $A \in X$. If $\mathcal{E}$ contained both $BY \to E$ and $AYE \to B$ for some $Y \in \mathcal{Y}$, then $AYE \to B$ would be uncritical for $\mathcal{E}$, so BY would be a key of $ABYE$ for $\mathcal{E}$ and determine A, a contradiction. Hence, the design leaves $B \to A$ and one of $BY \to E$ and $AYE \to B$ for every $Y \in \mathcal{Y}$ unenforced, and its loss is at least $|\mathcal{Y}| + 1$. Together with the case $B \to A \in \mathcal{E}$, every plain BCNF design has a loss of at least $|\mathcal{Y}|$, and $\lambda^{\mathrm{p}}(R_m, \Sigma_m) = |O| = |\mathcal{Y}|$ by the first statement. □

Proof of Lemma 5.3. Let $\mathcal{U}_{\mathcal{E}}$ denote the uncritical minimal FDs of $\mathcal{E}$, which form a cover of $\mathcal{E}$ by Thm. 3.18. Since $\Sigma_a \subseteq \mathcal{E} \cup \Delta$, the set $\mathcal{U}_{\mathcal{E}} \cup \Delta$ is a cover of Σ, so it suffices to show that $\mathcal{U} \cup \mathcal{C}(\Delta) \cup \Delta$ implies every $Y \to B \in \mathcal{U}_{\mathcal{E}}$. Firstly, let $Y \to B$ be a minimal FD of Σ that is critical for Σ^+, so that some minimal FD of Σ violates YB. If some such FD $Z \to C$ belonged to $\mathcal{E}$, then $Z \to C \in \mathcal{E}[YB]$ where Z is not a key of YB for $\mathcal{E} \subseteq \Sigma^+$, so $(YB, \mathcal{E}[YB])$ would not be in BCNF, which contradicts $Y \to B \in \mathcal{U}_{\mathcal{E}}$. Hence, every minimal FD of Σ that violates YB belongs to Δ, that is, $Y \to B \in \mathcal{C}(\Delta)$. Secondly, let $Y \to B$ not be a minimal FD of Σ. Then $Y' \to B$

is a minimal FD of Σ for some $Y' \subset Y$, and $Y' \to B \notin \mathcal{E}$ since $Y \to B$ is minimal for $\mathcal{E}$. Hence, $Y' \to B \in \Delta$ implies $Y \to B$. Every remaining element of $\mathcal{U}_\mathcal{E}$ is a minimal FD of Σ that is uncritical for Σ^+, that is, an element of $\mathcal{U}$. For the second claim, suppose that Δ contains neither a critical minimal FD of Σ nor an FD that violates the schema of one. Then $C(\Delta) = \emptyset$ and $\Delta \subseteq \mathcal{U}$, so $\mathcal{U}$ is a cover of Σ and Σ^+ is enforceable by Thm. 3.18. □

Proof of Proposition 5.4. Let $(\mathcal{D}, \kappa)$ be a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$ such that $\Delta' = \Sigma_a - \mathcal{E}$ contains Δ, and let $g_i = Y \to B$. By Lem. 5.3, g_i is implied by $\mathcal{U} \cup C(\Delta') \cup \Delta'$, and the attribute closure of Y under this set applies only FDs whose left-hand sides are contained in Y^+_Σ. Since $\mathcal{U} \cup C(\Delta) \cup \Delta$ does not imply g_i, some FD f applied in the closure is not in this set, and $f \in C(\Delta') \cup \Delta'$. If $f \in \Delta'$, then f belongs to $Q(g_i, \Delta)$ by the first clause of its definition. Otherwise, $f = X \to A$ is a critical minimal FD of Σ with $X \subseteq Y^+_\Sigma$ and $f \in C(\Delta') - C(\Delta)$, so every minimal FD of Σ that violates XA belongs to Δ', and some such FD v is not in Δ since $f \notin C(\Delta)$. Hence, v belongs to $Q(g_i, \Delta)$ by the second clause, and $v \in \Delta'$. In both cases Δ' meets $Q(g_i, \Delta)$, which is impossible if the requirement set is empty. Since the requirement sets are pairwise disjoint and disjoint from Δ, the set Δ' contains Δ and one distinct element of every $Q(g_i, \Delta)$, which gives the bound on $w(\Delta')$. □

Proof of Theorem 5.5. A returned pair $(\mathcal{D}, \kappa)$ is a key design since every minimal key of $(S, \mathcal{E}[S])$ is a key of S for Σ. It is sound and locally enforced by Thm. 3.18 applied to $\mathcal{E}$, and its loss is $w(\Delta)$ by line 7. For optimality, let $\mathcal{E}^*$ be the enforced FDs of a BCNF design with minimum loss, and let $\Delta^* = \Sigma_a - \mathcal{E}^*$. By Lem. 3.3, every minimal FD of $\mathcal{E}^*$ either belongs to $\Sigma_a - \Delta^*$ or is a weakening of some minimal FD $X \to A$ of Σ with $X \to A \notin \mathcal{E}^*$, that is, an element of $\mathcal{W}(\Delta^*)$. Hence, $\mathcal{E}^* = (\Psi^* \cup W^*)^+$ for $\Psi^* = \Sigma_a - \Delta^*$ and the set W^* of minimal FDs of $\mathcal{E}^*$ in $\mathcal{W}(\Delta^*)$, and $\mathcal{E}^* \cap \Delta^* = \emptyset$. By Lem. 5.3, Δ^* passes the test in line 3. Since $\mathcal{E}^*$ is enforceable, the combination of Δ^* and W^* causes a return, unless the algorithm returns earlier with a loss of at most $w(\Delta^*)$. The restriction to $W = \emptyset$ examines the sets Ψ^+ with $\Psi \subseteq \Sigma_a$ and $\Psi^+ \cap \Sigma_a = \Psi$ whose complement $\Sigma_a - \Psi$ passes line 3, and these include the enforced FDs of every plain design by Lem. 5.3. □

Proof of Lemma A.16. Let $\mathcal{U}_\mathcal{E}$ and $\mathcal{U}_{\mathcal{E}'}$ denote the uncritical minimal FDs of $\mathcal{E}$ and $\mathcal{E}'$. By Thm. 3.18, $\mathcal{U}_{\mathcal{E}'}$ implies g, so the computation of the closure of Y under $\mathcal{U}_{\mathcal{E}'}$ derives B. Since $\mathcal{U}_\mathcal{E}$ does not imply g, this computation applies some FD of $\mathcal{U}_{\mathcal{E}'} - \mathcal{U}_\mathcal{E}$, and we let $f = V \to C$ be the first one. The FDs applied before f belong to $\mathcal{U}_\mathcal{E} \subseteq \mathcal{E}$, so $V \subseteq Y^+_\mathcal{E}$. Firstly, let $f \notin \mathcal{E}$. By Lem. 3.3, some minimal FD $X \to C$ of Σ satisfies $X \subseteq V$. It belongs to Δ, since otherwise $X \to C \in \mathcal{E}$ and hence $f \in \mathcal{E}$, and $X \neq V$, since $f \in \mathcal{E}'$ and $\mathcal{E}' \cap \Delta = \emptyset$. Hence, f is the required FD. Secondly, let $f \in \mathcal{E}$. Then f is a minimal FD of $\mathcal{E}$, since some $V' \subset V$ with $V' \to C \in \mathcal{E} \subseteq \mathcal{E}'$ would contradict the minimality of f in $\mathcal{E}'$, and f is critical for $\mathcal{E}$ since $f \notin \mathcal{U}_\mathcal{E}$. So some non-trivial FD $Z \to D \in \mathcal{E}[VC]$ has a left-hand side Z that is not a key of VC for $\mathcal{E}$, while Z is a key of VC for $\mathcal{E}'$ since f is uncritical for $\mathcal{E}'$. Hence, $Z \to a \in \mathcal{E}' - \mathcal{E}$ for some $a \in VC$, and a minimal $Z' \subseteq Z$ with $Z' \to a \in \mathcal{E}'$ yields a minimal FD $Z' \to a$ of $\mathcal{E}'$ that is not in $\mathcal{E}$. As before, Lem. 3.3 provides some $X \to a \in \Delta$ with $X \subset Z'$, and $Z' \subseteq VC \subseteq Y^+_\mathcal{E}$ since $V \subseteq Y^+_\mathcal{E}$ and $f \in \mathcal{E}$. □

Proof of Lemma 5.6. Let $\mathcal{E}$ be the enforced FDs of a BCNF design and $\Delta' = \Sigma_a - \mathcal{E}$. For statement 1, if $\Delta \subseteq \Delta'$, then Δ' meets $Q(g, \Delta)$ by Prop. 5.4. For statement 2, if $f \in \Delta'$, then $D \notin \mathcal{E}$, since $\mathcal{E}$ is closed under implication and $f \notin \mathcal{E}$, so D meets Δ'. For statement 3, $\Delta' \neq \Delta$ by assumption, so $\Delta \not\subseteq \Delta'$ or Δ' contains an element of $\Sigma_a - \Delta$. For statement 4, let $\Delta_g = \Delta \cap \mathcal{N}(g)$. Every element of $Q(g, \Delta)$ or of $Q(g, \Delta_g)$ is a critical minimal FD with left-hand side contained in Y^+_Σ or a minimal FD that violates the schema of such an FD, so it belongs to $\mathcal{N}(g)$, where Δ and Δ_g agree. Whether such a critical FD $X \to A$ belongs to $C(\Delta)$ depends only on the minimal FDs that violate XA, which belong to $\mathcal{N}(g)$ as well, so $X \to A \in C(\Delta)$ iff $X \to A \in C(\Delta_g)$. Hence, $Q(g, \Delta) = Q(g, \Delta_g)$. Since $\Delta_g \subseteq \Delta$ and C is monotone, $\mathcal{U} \cup C(\Delta_g) \cup \Delta_g$ does not imply g either, so statement 1 for Δ_g shows that $(\Delta_g, Q(g, \Delta))$ is valid. □

Proof of Theorem 5.7. Every cut in $\mathcal{K}$ is valid. The initial cuts are of the first kind of Lem. 5.6 with $\Delta = \emptyset$, since the obstructions are the minimal FDs not implied by $\mathcal{U}$ and $C(\emptyset) = \emptyset$. The cuts added after line 5 are of the second kind. The cuts added in line 10 are of the fourth kind and, since line 7 failed and every BCNF design that leaves exactly Δ unenforced enforces an enforceable $\mathcal{E} \supseteq \Psi$ with $\mathcal{E} \cap \Delta = \emptyset$, of the third kind. Hence, the set $\Sigma_a - \mathcal{E}^*$ of a BCNF design with minimum loss satisfies $\mathcal{K}$ in every iteration, and $w(\Delta) \leq \lambda_w(R, \Sigma)$ in line 3. Every iteration that does not return adds a cut that Δ violates: a cut $(\{f\}, D_f)$ with $f \in \Delta$ and $D_f \cap \Delta = \emptyset$, or the cut $(\Delta, \Sigma_a - \Delta)$. Hence, no set Δ is computed twice, and the algorithm terminates after at most $2^{|\Sigma_a|}$ iterations, since the master problem remains feasible. When line 7 succeeds, $\mathcal{E}$ is enforceable with $\Sigma_a \cap \mathcal{E} = \Psi$, so Cor. 3.19 and Thm. 3.18 yield a BCNF design with enforced FDs $\mathcal{E}$ and loss $w(\Delta) \leq \lambda_w(R, \Sigma)$, which is therefore minimum. The search in line 7 is complete by Lem. A.16, as in the proof of Thm. 5.5. For the restriction to $\mathcal{E} = \Psi^+$, the cuts of the first two kinds are valid for plain designs as for all BCNF designs, and the cuts of the third kind are valid for plain designs, since a plain BCNF design that leaves exactly Δ unenforced enforces Ψ^+. The same argument then shows termination, the lower bounds and the minimality among plain designs. □

Proof of Proposition A.3. For the first statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to A, BC \to E, BD \to E, ACE \to B, ADE \to B\}$, whose minimal FDs are the elements of Σ. The design $\{ABCE, ABDE, BCD\}$ with $\kappa(ABCE) = \{ACE, BC\}$, $\kappa(ABDE) = \{ADE, BD\}$ and $\kappa(BCD) = \{BCD\}$ enforces $\mathcal{E} = \{ACE \to B, ADE \to B, BC \to AE, BD \to AE\}^+$. It is sound since BCD is a key of $(R, \mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $B \to A$ is the only minimal FD outside $\mathcal{E}$. Every BCNF design with loss one leaves $B \to A$ unenforced. Indeed, if one of the other four minimal FDs is the only unenforced one, then $B \to A$ is a non-key FD of $ABCE$ and of $ABDE$, so every FD with right-hand side B is critical and $ACE \to B$ or $ADE \to B$ cannot be enforced. With $B \to A$ unenforced, $ACE \to B$ and $ADE \to B$ are the only FDs with right-hand side B, so both must be uncritical, which requires that BC is a key of $ABCE$ and that BD is a key of $ABDE$. Hence, $BC \to A$ and $BD \to A$ are enforced, and

both are minimal FDs of the enforced set since neither C nor D determines A.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to D, DE \to A, AE \to B, BCE \to D\}$, whose minimal FDs are the elements of Σ together with $DE \to B$ and $BCE \to A$. The design $\{ABCE, ABD, ADE, BDE\}$ with the single keys BCE, AB, DE and DE enforces $\mathcal{E} = \{BCE \to A, AB \to D, DE \to AB\}^+$, which contains $BCE \to D$ but neither $A \to D$ nor $AE \to B$. It is sound since BCE is a key of $(R, \mathcal{E})$, it is locally enforced, and its loss is two. An exhaustive enumeration of the closed FD sets between Σ^+ and the trivial FD set shows that every enforceable FD set with loss two leaves exactly $A \to D$ and $AE \to B$ unenforced. Such an FD set contains $BCE \to D$, which is critical since $DE \to B$ is a non-key FD of $BCDE$, so $BCE \to D$ is derived from $BCE \to A$ and some weakened form $X' \to D$ of $A \to D$ with $A \in X' \subseteq ABCE$. If $X' \subseteq AE$, then $X' \to D$ and $DE \to B$ imply the unenforced FD $AE \to B$. Hence, X' contains B or C, and no minimal FD of Σ has a left-hand side that contains AB or AC. □

Proof of Proposition A.4. Let $R = \{A, B, C, D, E, F\}$ and $\Sigma = \{DF \to C, ACE \to D, AE \to B, AF \to D, BDE \to A, DE \to F\}$. The atomic cover Σ_a consists of the elements of Σ and the minimal FDs $DE \to C$, $AF \to C$ and $ACE \to F$. Let $\Delta = \{AE \to B, AF \to C, AF \to D\}$, let $\Psi = \Sigma_a - \Delta$, and let w assign weight one to the elements of Δ and weight four to the other six minimal FDs. A closure computation shows that the elements of Ψ are the minimal FDs of Ψ^+, that only $ACE \to D$ among them is critical for Ψ^+, since $DE \to C$ violates $ACDE$, and that $ACE^+_\Psi = ACDEF$.

We first show that an enforceable FD set $\mathcal{E} \subseteq \Sigma^+$ contains neither both $AE \to B$ and $BDE \to A$, nor all of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. Since $\mathcal{E} \subseteq \Sigma^+$, the left-hand side of every non-trivial FD in $\mathcal{E}$ with right-hand side A contains BDE, with right-hand side D it contains ACE or AF, and with right-hand side F it contains DE or ACE. If $AE \to B$ and $BDE \to A$ are in $\mathcal{E}$, then $BDE \to A$ is the only minimal FD of $\mathcal{E}$ with right-hand side A, and it is critical for $\mathcal{E}$ since $AE \to B$ violates $ABDE$. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $BDE \to A$, and $\mathcal{E}$ is not enforceable by Thm. 3.18. If $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$ are in $\mathcal{E}$, then $ACE \to D$ and $ACE \to F$ are minimal FDs of $\mathcal{E}$ that are critical for $\mathcal{E}$, since $DE \to C$ violates $ACDE$ and $AF \to C$ violates $ACEF$. Every other minimal FD of $\mathcal{E}$ with right-hand side D has F in its left-hand side, and vice versa. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $ACE \to D$, and $\mathcal{E}$ is not enforceable.

Next, the design $\mathcal{D} = \{ABDE, ABCE, CDE, ABCF, CDF, ABDF, ACEF, DEF\}$ with $\kappa(ABDE) = \{ADE, BDE\}$, $\kappa(ABCE) = \{ACE\}$, $\kappa(ABCF) = \kappa(ABDF) = \{ABF\}$ and the single keys DE, DF, ACE and DE of CDE, CDF, $ACEF$ and DEF enforces $\mathcal{E} = (\Psi \cup \{ACE \to B, ADE \to B, ABF \to C, ABF \to D\})^+$. It is sound since ACE is a key of $(R, \mathcal{E})$ contained in $ACEF$ and $\mathcal{D}$ is dependency-preserving for $\mathcal{E}$, and it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key in its left-hand side. It leaves exactly Δ unenforced, since AE and AF are closed for $\mathcal{E}$, so its loss is $w(\Delta) = 3$. Now let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) with loss less than four, so that its unenforced FDs form a subset Δ' of Δ. If $AE \to B \in \mathcal{E}'$, then $\mathcal{E}'$ contains $AE \to B$ and $BDE \to A$. If $AE \to B \notin \mathcal{E}'$ and $AF \to D \in \mathcal{E}'$, then $AF \to C \in \mathcal{E}'$ as well, so $\mathcal{E}'$ contains $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. If $AE \to B$ and $AF \to D$ are not in $\mathcal{E}'$ but $AF \to C$ is, then $\mathcal{E}'$ contains the same four FDs. In each case $\mathcal{E}'$ is not enforceable, so $\Delta' = \Delta$. Hence, $\lambda_w(R, \Sigma) = 3$, and every BCNF design with this loss leaves exactly Δ unenforced. Under unit weights, let Δ' be the set of unenforced FDs of a BCNF design with enforced FDs $\mathcal{E}'$, and suppose that $|\Delta'| \leq 2$. Then Δ' contains $AE \to B$ or $BDE \to A$, and the other element of Δ', if any, is none of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$, since each of these is implied by the elements of $\Sigma_a - \Delta'$. Hence, $\mathcal{E}'$ contains these four FDs and is not enforceable, so $\lambda(R, \Sigma) = 3 = |\Delta|$.

Finally, let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) that leaves exactly Δ unenforced. Then $ACE \to D$ is a minimal FD of $\mathcal{E}'$ that is critical for $\mathcal{E}'$, since $DE \to C \in \mathcal{E}'$. By Thm. 3.18, the uncritical minimal FDs of $\mathcal{E}'$ imply $ACE \to D$, so D is derived from ACE through some uncritical minimal FD $X \to D$ of $\mathcal{E}'$ with $X \neq ACE$. Then X does not contain ACE and is a proper superset of AF, since $X \to D$ is minimal and $AF \to D \notin \mathcal{E}'$. If $C \in X$, then $DF \to C$ violates XD, and if $E \in X$, then $DE \to F$ violates XD, since neither DF nor DE determines A for Σ. Hence, $X = ABF$, and $ABF \to D$ is not local since $B \notin ACE^+_\Psi$. □

Proof of Proposition A.5. Let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to B, A \to D, D \to A, B \to E, E \to B, BC \to A\}$. The atomic cover Σ_a consists of $A \to B$, $A \to D$, $A \to E$, $D \to A$, $D \to B$, $D \to E$, $B \to E$, $E \to B$, $BC \to A$, $BC \to D$, $CE \to A$ and $CE \to D$. The minimal FDs with two attributes are uncritical. The minimal FD $BC \to A$ is critical since $A \to B$ violates ABC, and $A \to B$ is the only minimal FD that violates ABC. Likewise, $BC \to D$, $CE \to A$ and $CE \to D$ are critical, and their schemata are violated by $D \to B$, $A \to E$ and $D \to E$ only. Let G consist of these four critical minimal FDs and the four FDs that violate their schemata. The uncritical minimal FDs $\mathcal{U} = \Sigma_a - \{BC \to A, BC \to D, CE \to A, CE \to D\}$ do not imply $BC \to A$ since $BC^+_\mathcal{U} = BCE$, so Σ^+ is not enforceable by Thm. 3.18.

Let $\mathcal{D} = \{ABD, ADE, BE, BCD, CDE\}$ with $\kappa(ABD) = \{A, BD\}$, $\kappa(ADE) = \{A, DE\}$, $\kappa(BE) = \{B, E\}$, $\kappa(BCD) = \{BC\}$ and $\kappa(CDE) = \{CE\}$, and let $\mathcal{E}$ be the closure of the declared key dependencies. Then $D^+_\mathcal{E} = D$, so $D \to A$, $D \to B$ and $D \to E$ are not in $\mathcal{E}$, while every other minimal FD of Σ is, as $BC \to A$ follows from $BC \to D$ and $BD \to A$, and $CE \to A$ follows from $CE \to D$ and $DE \to A$. The decomposition is dependency-preserving for $\mathcal{E}$ and contains the key BC of $(R, \mathcal{E})$, so it is lossless for $\mathcal{E}$ and the design is sound. It is locally enforced since $\mathcal{E}[ABD] = \{A \to BD, BD \to A\}^+$, $\mathcal{E}[ADE] = \{A \to DE, DE \to A\}^+$, $\mathcal{E}[BE] = \{B \to E, E \to B\}^+$, $\mathcal{E}[BCD] = \{BC \to D\}^+$ and $\mathcal{E}[CDE] = \{CE \to D\}^+$, as C and D determine nothing and B and E determine only each other. Hence, $\lambda(R, \Sigma) \leq 3$.

Let $\mathcal{E}'$ be enforceable for (R, Σ) with $\Delta = \Sigma_a - \mathcal{E}'$. Exchanging A with D or B with E maps Σ to a cover of itself, and the two exchanges act transitively on the critical minimal FDs and on the FDs that violate their schemata. By Lem. 5.3, Δ contains an element of G, so it suffices to consider $A \to B \in \Delta$ and $BC \to A \in \Delta$. If $A \to B \in \Delta$, then $\mathcal{E}'$ contains neither both $A \to D$ and $D \to B$ nor both $A \to E$ and $E \to B$, since $\mathcal{E}'$ is closed under implication and $A \to B \notin \mathcal{E}'$. If $BC \to A \in \Delta$, then $\mathcal{E}'$ contains neither both $BC \to D$ and $D \to A$ nor both $B \to E$ and $CE \to A$. In both cases $|\Delta| \geq 3$, so $\lambda(R, \Sigma) = 3$. Finally, let $|\Delta| = 3$ and $\Delta \subseteq G$. In the first

case, $\Delta = \{A \to B, D \to B, A \to E\}$ since $A \to D$ and $E \to B$ are not in G, but $D \to E$ and $E \to B$ belong to $\mathcal{E}'$ and imply $D \to B$. In the second case, $\Delta = \{BC \to A, BC \to D, CE \to A\}$ since $D \to A$ and $B \to E$ are not in G, but $CE \to D$ and $D \to A$ belong to $\mathcal{E}'$ and imply $CE \to A$. Both cases contradict $\Delta = \Sigma_a - \mathcal{E}'$, so every design with loss three leaves an element of $\Sigma_a - G$ unenforced, and every element of $\Sigma_a - G$ is uncritical and violates the schema of no critical minimal FD. □

Proof of Proposition 5.8. The uncritical minimal FDs of the enforceable set $\mathcal{E}$ form a cover of $\mathcal{E}$ by Thm. 3.18, and they form an uncritical set, since each of them is uncritical for $\mathcal{E}$. Every set Ψ that the algorithm maintains is therefore uncritical: the initial set by the previous sentence, every set of a climb by the tests in lines 5 and 9, and every kick by line 12. Line 5 adds an FD $g \in \Sigma_a - \Psi^+$, which enlarges Ψ^+ and enforces g, so $w(\Sigma_a - \Psi^+)$ decreases by at least $w(g)$, and line 9 changes Ψ only if $w(\Sigma_a - \Psi^+)$ does not increase, and at most s times in a row without decreasing it. Since the subsets of Σ^+ are finite in number, every climb terminates. Line 12 replaces Ψ^* only by a set of at most its loss, so $w(\Sigma_a - \Psi^{*+})$ never increases from $w(\Sigma_a - \mathcal{E})$, the loss of the initial set, and it decreases only finitely often, after which k kicks end the algorithm. Finally, Cor. 3.19 turns the uncritical set Ψ^* into a BCNF design of (R, Σ) with enforced FDs Ψ^{*+}, whose loss is $w(\Sigma_a - \Psi^{*+})$ by definition, at any time. □

Proof of Theorem A.13. No source A is the right-hand side of a non-trivial FD of Σ^+, since such an FD would yield a minimal FD of Σ with right-hand side A. Let $\mathcal{E} \subseteq \Sigma^+$ be closed, and suppose that every minimal FD $X \to A$ of $\mathcal{E}$ satisfies $XA \subseteq R_P$ for the part $P \ni A$. We first show that $Y^+_{\mathcal{E}} \cap P_i = (Y \cap R_i)^+_{\mathcal{E}} \cap P_i$ for every $Y \subseteq R$. The closure of Y under the minimal FDs of $\mathcal{E}$ adds an attribute of P_i only through a minimal FD whose left-hand side lies in R_i, whose attributes are sources, which belong to Y, or attributes of P_i, which by induction lie in $(Y \cap R_i)^+_{\mathcal{E}}$. Consequently, $\mathcal{E}_i = \mathcal{E}[R_i]$ satisfies $X^+_{\mathcal{E}_i} = X^+_{\mathcal{E}} \cap R_i$ for $X \subseteq R_i$, the minimal FDs of $\mathcal{E}$ with right-hand side in P_i are the minimal FDs of $\mathcal{E}_i$ with right-hand side in P_i, and for each of them, $X \to A$, we have $\mathcal{E}[XA] = \mathcal{E}_i[XA]$, so it is uncritical for $\mathcal{E}$ iff it is uncritical for $\mathcal{E}_i$. The same argument for Σ, whose minimal FDs satisfy the assumption by the definition of the parts, shows that $\Sigma^+_i = \Sigma^+[R_i]$ and that the minimal FDs of Σ with right-hand side in P_i are the minimal FDs of (Σ_i).

For the upper bound, let $\mathcal{E}_i$ be enforceable for (R_i, Σ_i) with loss $\lambda_w(R_i, \Sigma_i)$, and let U_i be its uncritical minimal FDs, which form an uncritical set that generates $\mathcal{E}_i$ by Thm. 3.18. Let $\mathcal{E} = (U_1 \cup \cdots \cup U_r)^+$. Every minimal FD of $\mathcal{E}$ with right-hand side in P_i has its left-hand side in R_i by the argument above applied to the generators, so $\mathcal{E}$ satisfies the assumption, $\mathcal{E}[R_i] = \mathcal{E}_i$, and every element of U_i is an uncritical minimal FD of $\mathcal{E}$. Hence, $\mathcal{E}$ is enforceable by Cor. 3.19, it is the set of enforced FDs of a part-respecting BCNF design, and a minimal FD of Σ with right-hand side in P_i belongs to $\mathcal{E}$ iff it belongs to $\mathcal{E}_i$, so the loss of the design is $\sum_i \lambda_w(R_i, \Sigma_i)$. If every $\mathcal{E}_i$ is plain, then $U_1 \cup \cdots \cup U_r$ may be replaced by the minimal FDs of Σ in $\mathcal{E}_1 \cup \cdots \cup \mathcal{E}_r$, and the design is plain.

For the lower bound, let $\mathcal{E}$ be the enforced FDs of a part-respecting BCNF design. Every FD $X \to A$ of $\mathcal{E}_i = \mathcal{E}[R_i]$ with $A \in P_i$ is implied by the uncritical minimal FDs of $\mathcal{E}$, and the closure of X under them adds attributes of P_i only through uncritical minimal FDs of $\mathcal{E}$ with right-hand side in P_i, which are uncritical minimal FDs of $\mathcal{E}_i$. Hence, $\mathcal{E}_i$ is enforceable for (R_i, Σ_i) by Thm. 3.18, and the loss of the design is the sum of the losses $w((\Sigma_i)_a - \mathcal{E}_i) \geq \lambda_w(R_i, \Sigma_i)$. If the design is plain, so is $\mathcal{E}_i$, since $\mathcal{E}_i$ is generated by the minimal FDs of Σ in $\mathcal{E}$ with right-hand side in P_i. Together with the upper bound, this proves both equalities. □

Proof of Corollary A.14. The sources and the parts are computed from Σ_a in polynomial time, and $(\Sigma_i)_a$ is the set of elements of Σ_a with right-hand side in P_i. Alg. 1, restricted to $W = \emptyset$ for plain designs, solves (R_i, Σ_i) in time bounded by a function of $|R_i|$, since $|(\Sigma_i)_a| \leq |R_i| \cdot 2^{|R_i|}$, and the glued design of the proof of Thm. A.13 is computed in polynomial time from the designs of the parts. □

Proof of Proposition A.6. Let $S_1, \ldots, S_m$ be the critical schemata of $\mathcal{D}_0$ and let $T = R - (S_1 \cup \cdots \cup S_m)$, so that $S_1, \ldots, S_m, T$ partition R and every schema of $\mathcal{D}_0$ other than a key schema lies inside one part. The projection of Σ onto a minimal key K of (R, Σ) is trivial, since an FD $Y \to B$ with $Y \subset K$ and $B \in K - Y$ would make $K - B$ a key. Since $\mathcal{D}_0$ is dependency-preserving for Σ, the set Σ^+ is therefore the closure of $\Sigma[S_1] \cup \cdots \cup \Sigma[S_m] \cup \Sigma[T]$, and the attribute closure of a set inside a part stays inside the part. By Lem. 4.2, applied m times, $\lambda_w(R, \Sigma) = \sum_{i=1}^m \lambda_w(S_i, \Sigma[S_i]) + \lambda_w(T, \Sigma[T])$, where the weighted version of the lemma holds with the same proof. The schemata of $\mathcal{D}_0$ inside T are in BCNF and their projections generate $\Sigma[T]$, and $K \cap T$ is a minimal key of $(T, \Sigma[T])$ since closures stay inside the parts, so these schemata and $K \cap T$ form a lossless, dependency-preserving decomposition of $(T, \Sigma[T])$ into BCNF, $\lambda_w(T, \Sigma[T]) = 0$ by Thm. 3.15, and the minimal keys of the schemata inside T generate $\Sigma[T]$. Alg. 2 adds to Γ these key dependencies, the trivial key dependency of a key schema if $\mathcal{D}_0$ has one, and, for every S_i, the key dependencies of a design of $(S_i, \Sigma[S_i])$ with loss $\lambda_w(S_i, \Sigma[S_i])$ by Thm. 5.5, whose enforced FDs $\mathcal{E}_i$ are enforceable for $(S_i, \Sigma[S_i])$. As in the proof of Lem. 4.2, the closure Γ^+ of the union of the enforceable sets $\mathcal{E}_1, \ldots, \mathcal{E}_m, \Sigma[T]$ over attribute-disjoint schemata is enforceable for (R, Σ), so line 9 succeeds by Thm. 3.18, and $w(\Sigma_a - \Gamma^+) = \sum_i w((\Sigma[S_i])_a - \mathcal{E}_i) = \lambda_w(R, \Sigma)$. The design of line 11 has loss at most $w(\Sigma_a - \Gamma^+)$ by Prop. 5.8 and at least $\lambda_w(R, \Sigma)$ by definition, which proves the claim. □

Proof of Theorem 5.9. Every element of Γ is a key dependency of a key design of some $(S, \Sigma[S])$, so $\Gamma \subseteq \Sigma^+$. If line 9 returns a lossless, dependency-preserving decomposition $\mathcal{D}$ of (R, Γ^+) into BCNF, then $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) since every minimal key of $(S, \Gamma^+[S])$ is a key of S for Σ, its enforced FDs are Γ^+ since $\mathcal{D}$ is dependency-preserving and in BCNF for Γ^+, and it is sound and locally enforced as in the proof of Thm. 5.5. Its loss is $w(\Sigma_a - \Gamma^+)$ by definition, and the uncritical design is a BCNF design of (R, Σ) with loss $w(\mathcal{O})$ by Thm. 3.22, so line 11 returns a BCNF design whose loss is at most the minimum of the two by Prop. 5.8. Otherwise, the claim follows from Thm. 5.5. □